Role-Based

AcademIQ Role-Based Thread Creation & Searchability Infrastructure

Master DevSpec Document

1. Executive Summary

This document defines the role-based thread creation, searchability, discoverability, linkage, permissions, and operational infrastructure across the AcademIQ ecosystem.

This document exists to ensure:

- structural consistency
- unified platform architecture
- role-adaptive workflows
- scalable discoverability
- standardized metadata
- linkage consistency
- institutional governance
- searchability consistency
- moderation consistency

across all platform roles.

This infrastructure extends the core:
“AcademIQ Thread-Based Learning & Teaching Infrastructure”

and applies role-specific operational flows while maintaining:

- ONE Channel consistency
- thread-first architecture
- Discovery Engine integration
- permission governance
- metadata structure
- universal searchability

2. Core Architectural Principle

AcademIQ shall operate using:
ONE unified thread infrastructure with role-adaptive workflows.

This means:

- all users operate through threads
- all threads remain searchable/discoverable

- all threads integrate into the same ecosystem infrastructure
- all thread systems utilize the same metadata architecture
- all thread systems integrate with Discovery Engine and Emma

while:

- creation flows
- permissions
- defaults
- recommendations
- visibility
- operational behavior

adapt dynamically based on user role.

3. Supported Platform Roles

The platform may support role-specific thread infrastructure for:

- Students
- Instructors
- Tutors
- Peer Tutors
- Advisors
- Counselors
- Institution Administrators
- Mentors
- Support Staff
- Future Parent Accounts

Each role should operate through:

- structured thread workflows
- guided thread creation
- permission-governed participation
- searchable thread infrastructure
- linkage systems
- Discovery Engine integration

4. Unified Thread Infrastructure

All role-based thread systems must inherit the same core infrastructure:

- ONE Channel integration
- thread-first communication
- permission-governed access
- searchable infrastructure
- Discovery Engine indexing
- metadata systems
- moderation systems
- analytics systems
- institutional governance
- linkage architecture
- Emma integration

This ensures:

- consistent UX
- scalable infrastructure
- operational continuity
- unified searchability
- ecosystem intelligence

5. Universal Thread Searchability Principle

All thread systems across all roles must remain:

- searchable
- discoverable
- linkable
- permission-governed
- metadata-driven

Threads should remain operationally accessible throughout:

- Library
- Discovery Engine
- universal search
- Student Control Center
- Instructor Control Center
- Tutor Control Center
- Advisor Portal
- Institution Portal
- Emma conversational search
- contextual recommendations

6. Standardized Metadata Infrastructure

All thread types must utilize standardized metadata infrastructure.

This ensures:

- scalable searchability
- Discovery Engine performance
- intelligent recommendations
- moderation consistency
- linkage consistency
- ecosystem intelligence

Examples of shared metadata:

- thread type
- subject
- course
- institution
- academic level
- grade/year
- thread visibility
- tags
- keywords
- engagement level
- role ownership
- permissions
- timestamps
- linkage identifiers

7. Student Thread Creation Infrastructure

Students should utilize guided thread creation workflows.

Student thread creation should prioritize:

- ease of use
- guided flows
- discoverability
- collaboration
- academic continuity

7.1 Student Thread Creation Flow

Step 1 — Select Thread Type

Students should select thread type using guided selectable cards, checkbox interfaces, or dynamic selection modules.

Examples:
☐ Study Circle
☐ Study Session
☐ Tutoring Request
☐ Academic Discussion
☐ Community Discussion
☐ Cohort
☐ Peer Collaboration
☐ Other

Selected thread type dynamically changes the remaining creation flow and available configuration options.

If:
☐ Other

Then:

- custom thread type input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

This ensures:

- platform flexibility
- future scalability
- metadata consistency
- structured discoverability
- cleaner recommendations
- searchability consistency

while preventing:

- fragmented thread categorization
- inconsistent metadata
- Discovery Engine degradation
- organizational chaos

Step 2 — Academic Context

Examples:

Subject:
☐ Calculus
☐ Biology
☐ Chemistry
☐ Physics

Academic Level:
☐ Elementary School
☐ Middle School
☐ High School
☐ College Undergraduate
☐ Graduate School
☐ Professional/Adult Learning

Grade / Year:
☐ Grades K–2
☐ Grades 3–5
☐ Grades 6–8
☐ Grades 9–12
☐ College Freshman
☐ College Sophomore
☐ College Junior
☐ College Senior
☐ Graduate Student

Additional Metadata:
☐ Course
☐ Institution
☐ Topic Tags
☐ Keywords

Step 3 — Audience & Visibility

Examples:
☐ Public
☐ Invite Only
☐ My Institution
☐ Course Participants
☐ Cohort Members

Step 4 — Participation Rules

Examples:

☐ Approval Required
☐ Participant Limits
☐ Grade Restrictions

8. Instructor Thread Creation Infrastructure

Instructor workflows should prioritize:

- course management
- office hours
- academic collaboration
- student engagement
- structured course support

8.1 Instructor Thread Creation Flow

Step 1 — Select Course(s)

Examples:
☐ Calculus I
☐ Calculus II
☐ Biology 101
☐ English Composition
☐ Other

If:
☐ Other

Then:

- custom course/thread input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 2 — Select Thread Type

Examples:
☐ Office Hours Thread
☐ Assignment Discussion Thread
☐ Study Session Thread
☐ Course Announcement Thread
☐ Research Collaboration Thread
☐ Other

If:
☐ Other

Then:

- custom thread type input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 3 — Audience & Permissions

Examples:
☐ Entire Course
☐ Specific Sections
☐ Invite Only
☐ Teaching Assistants Only
☐ Institution Faculty

Step 4 — Discovery & Visibility

Examples:
☑️ Searchable within Course
☑️ Discovery Engine Eligible
☐ Institution-Wide Visibility
☐ Public Visibility

9. Tutor Thread Creation Infrastructure

Tutor workflows should prioritize:

- tutoring discovery
- scheduling
- subject specialization
- student matching
- tutoring continuity

9.1 Tutor Thread Creation Flow

Step 1 — Subject Expertise

Examples:
☐ Calculus
☐ SAT Math

☐ Chemistry
☐ Physics
☐ Other

If:
☐ Other

Then:

- custom subject/thread input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 2 — Tutoring Type

Examples:
☐ 1:1 Tutoring
☐ Group Tutoring
☐ Exam Preparation
☐ Homework Support
☐ Other

If:
☐ Other

Then:

- custom tutoring type input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 3 — Tutoring Availability

Examples:
☐ Weekdays
☐ Weekends
☐ Recurring Availability
☐ Emergency/Urgent Support

Step 4 — Pricing & Session Rules

Examples:
☐ Hourly Pricing

☐ Session Limits
☐ Cancellation Policies
☐ Approval Required

10. Peer Tutor Thread Infrastructure

Peer Tutor infrastructure should remain institution-aware and moderation-driven.

Peer Tutors may require:

- GPA verification
- institution linkage
- subject approval
- moderation approval
- academic verification

10.1 Peer Tutor Thread Creation Flow

Step 1 — Institution Verification

Examples:
☐ Institution Linked
☐ GPA Verified
☐ Subject Approved
☐ Other

If:
☐ Other

Then:

- custom input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 2 — Tutoring Subject

Examples:
☐ Algebra
☐ Biology
☐ History
☐ Other

If:

☐ Other

Then:

- custom subject input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 3 — Audience Restrictions

Examples:
☐ Same Institution Only
☐ Same Academic Level Only
☐ Course Participants Only

11. Advisor & Counselor Thread Infrastructure

Advisor workflows should prioritize:

- support continuity
- institutional governance
- structured advising
- appointment coordination

11.1 Advisor Thread Creation Flow

Step 1 — Support Type

Examples:
☐ Academic Advising
☐ Financial Aid
☐ Career Guidance
☐ Wellness Support
☐ Other

If:
☐ Other

Then:

- custom support type input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear

- system attempts metadata mapping into existing Discovery Engine structures

Step 2 — Audience & Institution Scope

Examples:
☐ Institution Students
☐ Specific Programs
☐ Invite Only
☐ Cohort Support

Step 3 — Appointment & Communication Rules

Examples:
☐ Appointment Required
☐ Office Hours Enabled
☐ Follow-Up Threads Enabled

12. Institution Admin Thread Infrastructure

Institution Admin infrastructure should support:

- institution-wide governance
- cohort management
- moderation
- onboarding
- retention systems
- institution communication

12.1 Institution Admin Thread Creation Flow

Step 1 — Institution Scope

Examples:
☐ Entire Institution
☐ Department Specific
☐ Program Specific
☐ Cohort Specific
☐ Other

If:
☐ Other

Then:

- custom institution scope input field appears

- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 2 — Thread Purpose

Examples:
☐ Institution Announcement
☐ Cohort Collaboration
☐ Retention Initiative
☐ Academic Support Initiative
☐ Other

If:
☐ Other

Then:

- custom purpose input field appears
- suggested categories/tags appear
- AI-assisted category suggestions may appear
- system attempts metadata mapping into existing Discovery Engine structures

Step 3 — Governance Rules

Examples:
☐ Institution Moderation
☐ Restricted Participation
☐ Faculty Oversight
☐ Student Access Controls

13. Universal Search & Discoverability Infrastructure

All role-based thread systems must integrate with:

- Discovery Engine
- universal search
- Emma conversational search
- recommendation systems
- Library
- Control Centers

All threads should remain:

- searchable
- filterable
- discoverable
- permission-governed

13.1 Searchable Metadata Infrastructure

Thread searchability may support:

- subject
- course
- institution
- instructor
- tutor
- advisor
- thread type
- tags
- academic level
- grade/year
- topic
- cohort
- engagement
- timestamps
- activity
- popularity

13.2 Emma Conversational Search Infrastructure

Emma should support role-aware conversational search.

Examples:

- “Emma find my Calculus office hours thread.”
- “Emma open active SAT tutoring threads.”
- “Emma show Biology cohort discussions.”
- “Emma find yesterday’s study session.”

Emma should utilize:

- metadata
- timestamps
- linkage systems
- role permissions

- saved history
- engagement history
- Discovery Engine indexing

to retrieve and recommend relevant threads.

14. Linkage Infrastructure

All role-based thread systems must integrate with:

- subject linkage
- course linkage
- institution linkage
- cohort linkage
- community linkage
- tutoring linkage
- advisor linkage
- instructor linkage
- engagement linkage

This enables:

- intelligent recommendations
- scalable discoverability
- ecosystem continuity
- AI-powered retrieval
- personalized experiences

15. Strategic Infrastructure Principle

AcademIQ operates through:
ONE unified thread ecosystem.

Role-specific workflows should adapt:

- creation flows
- permissions
- governance
- recommendations
- visibility
- participation rules

while maintaining:

- unified infrastructure
- universal searchability
- standardized metadata
- Discovery Engine consistency
- ONE Channel integration
- Emma integration
- scalable linkage architecture

This architecture enables:

- scalable academic collaboration
- structured academic continuity
- intelligent discoverability
- AI-powered academic retrieval
- institutional governance
- role-adaptive workflows
- ecosystem-wide consistency

while maintaining a clean, structured, academically focused platform architecture.

Academic Support Network (ASN)

Academic Support Network (ASN)

The Academic Support Network (ASN) is AcademIQ's structured academic and student support ecosystem designed to connect students with trusted educators, tutors, advisors, mentors, therapists, peer academic support participants, and institution-linked professionals across the platform.

The ASN creates a tiered, identity-aware, verification-based support infrastructure that promotes:

- academic collaboration
- mentorship
- guided learning
- student engagement
- knowledge sharing
- community-driven support
- academic opportunity creation
- wellness and student support accessibility
- institutional accountability
- structured academic continuity

Unlike open marketplaces, AcademIQ's ASN is designed as a structured, permissioned, and trust-based ecosystem centered around:

- academic integrity
- accountability
- role transparency
- verification transparency
- credential visibility
- institutional trust
- verified participation
- structured communication
- student safety

Academic Support Network Structure

The ASN is divided into three primary support tiers:

1. Certified Academic & Professional Support
2. Verified Academic Support
3. Peer & Community Academic Support

Each role contains different permissions, verification standards, capabilities, visibility levels, moderation requirements, institutional authority, and platform responsibilities.

The ASN is designed to clearly distinguish between:

- verified users

- certified users
- licensed professionals
- institution-linked users
- peer-based support participants

This distinction exists to ensure complete transparency regarding qualifications, authority levels, institutional linkage, and platform trust status.

Minimum Qualification Standards

ASN qualification standards are designed to support platform trust, transparency, institutional confidence, student safety, and responsible academic support participation.

Additional qualification standards, restrictions, or verification requirements may be imposed based on:

- jurisdiction
- institution
- support category
- local laws and regulations
- institutional governance requirements
- platform moderation policies
- service type
- risk classification

Platform verification does not independently constitute:
- government accreditation
- institutional endorsement
- professional licensure
- legal authorization outside approved jurisdictions

Users are prohibited from:
- misrepresenting credentials
- falsely claiming certifications
- falsely claiming institutional affiliation
- impersonating licensed professionals
- misrepresenting professional authority

Tier 1 — Certified Academic & Professional Support

Overview

Tier 1 represents the highest level of academic, professional, institutional, and support authority within the ASN ecosystem.

These users are certified educators, licensed professionals, institution-affiliated personnel, credentialed advisors, therapists, certified tutors, and verified subject matter experts capable of providing structured academic and student support services.

These users receive enhanced visibility, trust status, platform authority, and institutional integration capabilities.

Tier 1 Roles

Certified Instructor

Overview

Certified Instructors represent the highest level of instructional authority and academic trust within the ASN ecosystem.

Examples

- Licensed teachers
- University professors
- Institution instructors
- Certified educators
- Subject matter experts
- Credentialed academic professionals

Minimum Requirements

Certified Instructors may be required to complete:

- Government-issued ID verification
- Email verification
- Selfie/liveness verification
- Bachelor’s degree from an accredited institution (or international equivalent)
- Teaching credential, license, certification, or demonstrable professional instructional qualification
- Education credential upload
- Optional institution affiliation verification
- Optional video verification

Capabilities

Certified Instructors may:

- Offer tutoring services
- Create and host study sessions
- Lead study circles
- Create educational content
- Host office hours
- Mentor students
- Build academic communities
- Create courses (if enabled)

- Offer premium academic services
- Participate in institution-linked ecosystems
- Receive enhanced platform visibility

Restrictions

Certified Instructors:

- Must comply with academic integrity policies
- Cannot complete assignments on behalf of students
- Cannot bypass institutional restrictions
- Must follow platform tutoring guidelines

Badge

✅ Certified Instructor
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Certified Tutor

Overview

Certified Tutors are professional or credentialed tutors who may not hold formal instructor positions but possess validated expertise, certifications, or professional tutoring qualifications.

Examples

- Certified independent tutors
- Professional academic coaches
- Exam preparation specialists
- Credentialed tutoring professionals

Minimum Requirements

Certified Tutors may be required to complete:

- Identity verification
- Subject competency validation
- Bachelor’s degree preferred from an accredited institution (or international equivalent)
- Demonstrable subject competency
- Optional tutoring certifications
- Community standards agreement
- Optional assessment review

Capabilities

Certified Tutors may:

- Offer advanced tutoring services
- Provide premium tutoring support
- Host structured tutoring sessions
- Participate in institutional support ecosystems
- Build tutoring programs
- Offer specialized exam preparation

Restrictions

Certified Tutors:

- Cannot misrepresent institutional authority
- Cannot participate in unauthorized grading workflows
- Must comply with tutoring conduct standards

Badge

🎓 Certified Tutor
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Certified Academic Advisor

Overview

Certified Academic Advisors provide structured academic guidance related to:

- academic planning
- course selection
- degree pathways
- institutional guidance
- academic progression

Minimum Requirements

Certified Academic Advisors may be required to complete:

- Identity verification
- Bachelor’s degree from an accredited institution (or international equivalent)
- Relevant academic advising, counseling, or education background
- Optional institution affiliation verification
- Optional academic advising certifications

Badge

📘 Certified Academic Advisor
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Certified Career Advisor

Overview

Certified Career Advisors provide structured career guidance related to:

- career planning
- internships
- graduate preparation
- professional development
- career pathways

Minimum Requirements

Certified Career Advisors may be required to complete:

- Identity verification
- Bachelor’s degree from an accredited institution (or international equivalent)
- Relevant industry, advising, or career support experience
- Optional coaching or advising certifications

Badge

💼 Certified Career Advisor
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Certified Financial Aid Advisor

Overview

Certified Financial Aid Advisors provide educational funding and financial aid guidance related to:

- FAFSA assistance
- scholarship guidance
- grant opportunities
- tuition planning
- educational budgeting
- financial literacy
- academic funding resources
- institution funding guidance

This role is designed to support students and families navigating educational funding opportunities and financial aid processes.

Important Limitation

Certified Financial Aid Advisors do NOT provide:

- investment advice
- securities advice
- wealth management services
- tax advisory services
- regulated financial planning services

Minimum Requirements

Certified Financial Aid Advisors may be required to complete:

- Identity verification
- Bachelor’s degree from an accredited institution (or international equivalent)
- Relevant financial aid, education, advising, or counseling background
- Optional financial aid certifications
- Optional institution affiliation verification

Capabilities

Certified Financial Aid Advisors may:

- Assist students with financial aid guidance
- Support scholarship discovery
- Provide educational funding resources
- Offer FAFSA-related support guidance
- Participate in institution-linked support ecosystems
- Conduct financial aid workshops and guidance sessions

Restrictions

Certified Financial Aid Advisors:

- Cannot provide regulated investment advice
- Cannot provide securities or wealth management services
- Cannot guarantee scholarship or aid outcomes
- Must comply with platform conduct and educational guidance policies

Badge

💰 Certified Financial Aid Advisor
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Licensed Therapist

Overview

Licensed Therapists represent licensed mental health and wellness professionals operating within approved platform support policies.

Important Distinction

Licensed Therapists are distinct from:
- peer wellness support
- mentorship
- general emotional support
- non-clinical coaching

Minimum Requirements

Licensed Therapists may be required to complete:

- Identity verification
- Bachelor’s degree minimum
- Master’s degree preferred or required depending on jurisdiction
- Active professional license in applicable jurisdiction
- License verification
- Jurisdiction verification
- Optional institution affiliation verification

Examples of Applicable Licenses

Examples may include, where applicable by jurisdiction:

- LMFT
- LCSW
- LPC
- Licensed Mental Health Counselor
- Licensed Clinical Counselor
- Psychologist licenses

Restrictions

Licensed Therapists:

- Must comply with applicable laws and regulations
- Must comply with platform conduct standards
- Cannot operate outside approved jurisdictional permissions
- May be subject to enhanced moderation and compliance review

Badge

🧠 Licensed Therapist
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Institution-Linked Professionals

Overview

Institution-linked ASN participants are users formally connected to educational institutions integrated into AcademIQ.

These users may include:

- institution instructors
- counselors
- advisors
- tutors
- support personnel
- mentors

Institution-linked users may receive:

- institution verification badges
- institution authority indicators
- institution-linked visibility enhancements

Institutions may impose:

- additional verification requirements
- role restrictions
- workflow permissions
- moderation controls
- communication policies

Tier 2 — Verified Academic Support

Overview

Tier 2 represents the verified academic support layer within the ASN.

These users may possess strong subject competency, mentorship ability, advising experience, or tutoring capabilities but may not possess formal certifications, licenses, or institutional educator authority.

This tier creates scalable and accessible academic support while maintaining accountability and trust.

Tier 2 Roles

Verified Tutor

Examples

- Graduate students
- Skilled college students
- Independent tutors
- Retired educators
- Subject specialists

Minimum Requirements

Verified Tutors may be required to complete:

- Government-issued ID verification
- Email verification
- Selfie/liveness verification
- Subject competency assessment
- Academic background review
- Community standards agreement

Capabilities

Verified Tutors may:

- Offer tutoring services
- Host study sessions
- Support study circles
- Provide concept-based support
- Participate in ONE Channel environments
- Build tutoring profiles and ratings

Restrictions

Verified Tutors:

- Do not possess institutional instructor authority
- Cannot grade official institution assignments
- Cannot misrepresent credentials
- Must comply with tutoring policies

Badge

🎓 Verified Tutor
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Verified Advisor

Overview

Verified Advisors may provide:

- academic guidance
- mentorship
- student support
- college preparation support
- general educational guidance

These users are verified but may not possess professional certifications or institutional authority.

Minimum Requirements

Verified Advisors may be required to complete:

- Identity verification
- Community standards agreement
- Educational or advising background review (if applicable)

Badge

📗 Verified Advisor
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Verified Mentor

Overview

Verified Mentors provide:

- student mentorship
- educational encouragement
- academic accountability support
- guidance-based collaboration

Minimum Requirements

Verified Mentors may be required to complete:

- Identity verification
- Community standards agreement
- Platform onboarding requirements

Badge

🌟 Verified Mentor
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Verified Academic Coach

Overview

Verified Academic Coaches may assist with:

- study habits
- organization
- productivity
- learning strategies
- academic planning support

Minimum Requirements

Verified Academic Coaches may be required to complete:

- Identity verification
- Community standards agreement
- Optional competency assessments

Badge

📖 Verified Academic Coach
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Tier 3 — Peer & Community Academic Support

Overview

Tier 3 represents the community-driven academic support layer of the ASN.

These users are primarily students and community participants providing collaborative learning support, mentorship, peer tutoring, study assistance, and cohort-based engagement.

This layer is designed to encourage:

- leadership
- collaboration
- engagement
- community growth
- peer-based learning

Tier 3 Roles

Peer Tutor

Examples

- High-performing students
- Cohort leaders
- Study group facilitators
- Peer mentors
- Student ambassadors

Eligibility Requirements

Peer Tutors may be required to:

- Be verified students
- Maintain good community standing
- Meet academic thresholds (if enabled)
- Complete onboarding requirements
- Accept tutoring conduct standards

Optional requirements may include:

- Instructor recommendations
- Cohort endorsements
- Competency assessments

Capabilities

Peer Tutors may:

- Offer peer tutoring
- Support study circles
- Host peer-led sessions
- Provide concept clarification
- Support cohort learning

Restrictions

Peer Tutors:

- Cannot present themselves as certified educators
- Cannot complete assignments for students
- Cannot participate in institutional grading workflows
- May operate under enhanced moderation initially

Badge

⭐ Peer Tutor
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Cohort Leader

Overview

Cohort Leaders are student leaders responsible for facilitating:

- cohort engagement
- communication
- collaboration
- study participation

Badge

👥 Cohort Leader
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Peer Mentor

Overview

Peer Mentors provide:

- student encouragement
- peer accountability
- onboarding assistance
- community engagement support

Badge

🌱 Peer Mentor
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Credential Transparency & Role Disclosure

The ASN is designed to provide complete role and credential transparency across the platform.

Users may be able to view:

- verification status
- certification status
- institution linkage

- support category
- badge classifications
- visibility status

Institutions may establish:

- additional credential requirements
- role restrictions
- institution-specific verification rules
- institutional moderation standards

Platform Safeguards & Academic Integrity

All ASN participants must comply with:

- platform conduct guidelines
- tutoring policies
- communication policies
- academic integrity standards
- identity verification requirements
- moderation policies
- institutional governance requirements (if applicable)

AcademIQ may monitor:

- tutoring activity
- session quality
- ratings and feedback
- suspicious activity
- policy violations
- credential misuse
- impersonation attempts
- institutional compliance concerns

AcademIQ reserves the right to:

- suspend ASN privileges
- revoke verification status
- revoke certification visibility
- restrict visibility
- remove users from ASN
- limit institutional access
- apply moderation restrictions

ONE Channel Integration

All tutoring, mentorship, advising, therapy support, study sessions, office hours, and academic support interactions may operate within ONE Channel environments to support:

- structured communication
- persistent discussions
- session continuity
- accountability
- moderation
- resource sharing
- searchable history
- workflow continuity
- collaboration infrastructure

Long-Term Vision

The Academic Support Network is designed to become a scalable global academic and student support ecosystem combining:

- professional instruction
- tutoring
- mentorship
- advising
- wellness support
- peer collaboration
- academic networking
- institutional collaboration
- opportunity creation
- community engagement

The ASN transforms AcademIQ from a traditional learning platform into a structured academic support ecosystem built around:

- identity
- trust
- transparency
- accountability
- collaboration
- engagement
- continuity
- academic growth
- institutional integration

Academic Support Thread Infrastructure System

AcademIQ Academic Support Thread Infrastructure System

FINAL DevSpec Document — Governance + Guided Workflow Edition

1. Executive Summary

The AcademIQ Academic Support Thread Infrastructure System defines the complete infrastructure for academic support, tutoring, study circles, study sessions, mentorship, advising, collaboration, scheduling, communication, continuity, and support governance throughout the AcademIQ ecosystem.

This infrastructure is powered by ONE Channel and establishes that academic support on AcademIQ is NOT:

- a standalone tutoring marketplace
- a disconnected booking system
- a generic messaging platform
- an isolated scheduling application
- a fragmented support environment

Instead, academic support on AcademIQ operates through:

- structured
- persistent
- searchable
- discoverable
- linkable
- subject-specific academic support threads

These academic support threads become centralized operational environments for:

- tutoring
- study circles
- study sessions
- mentorship
- advising
- academic collaboration
- office hours
- institution support programs
- communication
- scheduling
- academic continuity
- file sharing
- live sessions

- recordings
- summaries
- analytics
- moderation
- follow-up
- re-engagement
- long-term academic support

2. Core Architectural Principle

All academic support operations on AcademIQ must be thread-based.

Academic support shall NOT operate through:

- disconnected DMs
- isolated booking pages
- unmanaged external communication
- standalone support chats
- fragmented scheduling systems
- isolated collaboration tools

Every academic support interaction must operate through an academic support thread inside ONE Channel.

3. Academic Support Thread Architecture

Academic support threads must be:

- subject-specific
- structured
- persistent
- searchable
- discoverable
- linkable

Examples

- Calculus Support Thread
- Biology Study Circle
- SAT Math Prep Support Thread
- Organic Chemistry Study Session
- Engineering Mentorship Thread
- Institutional Academic Recovery Thread

All activities related to that support environment should remain inside the thread including:

- communication
- scheduling
- support requests
- files
- assignments
- recordings
- summaries
- collaboration
- continuity
- recommendations
- re-engagement history

This creates:

- academic continuity
- persistent support history
- structured collaboration
- scalable analytics
- recommendation intelligence
- long-term learning infrastructure

4. Academic Support Thread Object Infrastructure

Each academic support thread shall function as a persistent academic object within the AcademIQ ecosystem.

Academic support thread objects may contain:

- thread ID
- support type
- subject classification
- academic level
- grade/course linkage
- institution linkage
- instructor linkage
- tutor linkage
- student linkage
- cohort linkage
- study circle linkage
- study session linkage
- Discovery Engine metadata

- tags
- permissions
- moderation states
- visibility states
- session history
- analytics
- engagement history
- AI summaries
- recommendation signals
- academic continuity history

This infrastructure enables threads to become:

- searchable
- discoverable
- persistent
- analyzable
- recommendation-ready
- institution-ready
- AI-compatible

5. Academic Support Thread Discoverability

Academic support threads may be:

- saved to Library
- searchable through Discovery Engine
- recommended through linkage systems
- linked to courses
- linked to communities
- linked to study circles
- linked to study sessions
- linked to institution ecosystems
- linked to cohorts

Students should be able to revisit:

- Saved Threads
- Recent Threads
- Archived Threads
- Continuing Support Threads
- Re-engagement Threads

6. Academic Support Visibility Governance

Only subject-based academic support environments may support Public Discovery visibility.

Examples

- tutoring threads
- study circles
- study sessions
- subject collaboration threads
- MasterSessions
- subject-based academic communities

Sensitive support environments should NOT support Public Discovery.

Examples

- advising
- mentorship
- intervention support
- institutional recovery programs
- academic support cases
- personalized guidance threads

Sensitive support environments should instead support:

- Private
- Institution
- Shared with Approved Participant

This governance model protects:

- student privacy
- academic confidentiality
- institutional trust
- support integrity
- platform safety

7. Visibility Eligibility Infrastructure

Visibility settings should dynamically change based on support type.

Tutoring Thread

Available Visibility Modes:

- Public Discovery
- Hybrid
- Institution
- Shared
- Private

Study Circle

Available Visibility Modes:

- Public Discovery
- Hybrid
- Institution
- Shared
- Private

Advising Thread

Available Visibility Modes:

- Institution
- Shared with Approved Participant
- Private

Mentorship Thread

Available Visibility Modes:

- Institution
- Shared with Approved Participant
- Private

This infrastructure ensures:

- governance automation
- privacy enforcement
- cleaner user experience

- institutional safety
- reduced moderation risk

8. Shared Participant Infrastructure

If “Shared with Approved Participant” is selected, students must explicitly choose approved participants.

Possible Participant Selections

- Parent / Guardian
- Academic Advisor
- Mentor
- Counselor
- Instructor
- Financial Aid Advisor
- Career Services Representative
- Institution Support Staff
- Other

If “Other” is selected:

- custom participant role required
- user selection or email required
- permission level required

Permission Level Examples

- View Only
- Comment Only
- Participate in Thread
- Attend Session
- Receive Notifications
- Upload Documents

Shared access must require explicit student consent before any participant is added.

This infrastructure supports:

- privacy protection
- controlled collaboration
- institutional transparency

- family support workflows
- secure participation governance

9. Academic Support Thread Metadata Infrastructure

Academic support thread metadata may include:

- support type
- subject
- topic
- subtopic
- academic level
- grade
- institution
- course name
- course code
- course description
- major
- exam type
- support category
- urgency
- tags
- language
- country
- curriculum type
- linked study circle
- linked study session
- linked cohort

Metadata should support:

- Discovery Engine
- recommendation systems
- Emma
- analytics
- institutional reporting
- academic linkage systems

10. Academic Support Thread Creation Methods

Academic support threads may be initiated through:

A. Support Profiles

- tutor profile
- instructor profile
- advisor profile
- mentor profile

B. Course Environment

- course page
- assignment
- study session
- study circle
- academic thread

C. Discovery Engine

- support discovery
- recommendations
- search results
- academic linkage

D. Institutional Referrals

- instructor referral
- advisor referral
- institutional intervention
- support program assignment

E. Student-Initiated Support

- create support thread
- request collaboration
- request study support
- create study circle

11. Guided Academic Support Infrastructure

Academic support thread creation should operate through structured guided workflows.

All guided workflows throughout AcademIQ should support:

- checkboxes

- searchable fields
- dropdowns
- multi-select
- AI recommendations
- attachment support
- optional custom input
- “Other” fallback options
- linkage settings
- visibility settings
- discovery settings

12. Adaptive Guided Workflow Infrastructure

Guided workflows should support adaptive option population and behavioral prioritization.

The system should dynamically prioritize:

- frequently selected support types
- commonly selected subjects
- recurring provider preferences
- repeated linkage selections
- frequently used scheduling preferences
- institution-specific trends
- regional academic trends

Examples

- If a student selects “Tutoring” in 90% of support workflows, the system should prioritize Tutoring near the top of future guided workflows.
- If a student repeatedly selects Calculus support, Calculus-related options may appear earlier in subject recommendations.
- If a region consistently selects specific curriculum subjects, those subjects may dynamically surface higher in guided selections.

Dynamic prioritization may consider:

- personal usage behavior
- institution usage patterns
- regional trends
- curriculum trends
- Discovery Engine analytics
- engagement frequency
- recurring support relationships

The system should support:

- adaptive workflow ordering
- behavioral personalization
- intelligent workflow acceleration
- analytics-driven UX optimization
- recommendation optimization

This infrastructure should improve:

- workflow efficiency
- user experience
- personalization
- engagement
- recommendation intelligence
- platform analytics

This infrastructure transforms guided workflows from:

- static form infrastructure

into:

- intelligent adaptive workflow infrastructure

13. Guided Workflow — Support Type Selection

Students should select the academic support type.

Checkbox Examples

- Tutoring
- Study Circle
- Study Session
- Peer Collaboration
- Mentorship
- Academic Advising
- Homework Support
- Exam Preparation
- Concept Review

- Assignment Support
- Study Planning
- Long-Term Academic Support
- AP Course Support
- SAT/ACT Prep
- College-Level Support
- Graduate-Level Support
- Institutional Academic Support
- Concept Mastery Support
- Academic Recovery Support
- Study Strategy Support
- Learning Reinforcement
- Guided Concept Review
- Office Hours
- Community Support
- Other

If “Other” is selected:

- custom support type field required

14. Guided Workflow — Subject Selection

Students should select the subject area.

Checkbox Examples

- Algebra
- Geometry
- Calculus
- Statistics
- Biology
- Chemistry
- Physics
- Literature
- History
- Economics
- Computer Science
- Engineering
- SAT Prep
- ACT Prep
- GRE Prep
- MCAT Prep
- Other

If “Other” is selected:

- custom subject field required

Dynamic Subject Population Infrastructure

Subject selections should support dynamic regional population and adaptive taxonomy infrastructure.

The system should dynamically adjust subject availability based on:

- country
- region
- institution
- curriculum type
- academic system
- language
- user activity trends
- Discovery Engine trends

Examples

- Swahili → East Africa
- KCSE subjects → Kenya
- JEE Prep → India
- WAEC subjects → West Africa
- IB curriculum subjects → IB-linked institutions

If users consistently enter a subject using the “Other” field, the system may:

- flag the subject for review
- recommend taxonomy inclusion
- dynamically promote the subject into regional subject selections
- improve Discovery Engine indexing
- improve recommendation systems

The system should support:

- regional academic adaptability
- dynamic taxonomy evolution

- localization infrastructure
- curriculum-aware recommendations
- global scalability

This infrastructure transforms the subject system from:

- static dropdown infrastructure

into:

- intelligent evolving academic taxonomy infrastructure

15. Guided Workflow — Academic Level Selection

Students should select the academic level.

Checkbox Examples

- Elementary School
- Middle School
- High School
- AP Level
- IB Level
- Community College
- Undergraduate
- Graduate School
- Professional Certification
- Continuing Education
- Other

If “Other” is selected:

- custom academic level field required

16. Guided Workflow — Academic Linkage

Students should configure academic linkage settings.

Fields

- Course Name

- Course Code
- Institution
- Instructor Name
- Course Description

Checkbox Examples

- Link Existing Course
- Link to Study Session
- Link to Study Circle
- Link to Cohort
- Link to Community
- Link to Institution Program
- Create Standalone Support Thread
- Other

If “Other” is selected:

- custom linkage field required

17. Guided Workflow — Support Preferences

Students should configure support preferences.

Checkbox Examples

- Verified Tutor
- Certified Instructor
- Peer Tutor
- Institution-Approved Support
- Same Time Zone Preferred
- Same Language Preferred
- Video Sessions Preferred
- Audio Sessions Preferred
- Recurring Support
- Emergency Availability
- Affordable Pricing Priority
- Highest Rated Support Providers
- Other

If “Other” is selected:

- custom preference field required

18. Emma Integration

Students may request support recommendations from Emma directly inside academic support threads.

Example

“Emma, recommend Calculus support resources for derivatives.”

Emma may provide:

- tutor recommendations
- instructor recommendations
- study circles
- study sessions
- mentorship recommendations
- advisor recommendations
- ratings
- pricing
- verification status
- availability
- institution affiliation
- recommended resources
- recurring support suggestions
- suggested thread tags

Emma may also assist during guided thread creation.

19. Academic Support Request Template Infrastructure

Before support providers create outlines or booking recommendations, students must complete a structured academic support request template.

Academic Information

- subject
- topic
- subtopic
- academic level
- course name
- course code
- instructor name
- institution

- assignment/exam type

Support Request Information

- issue/concern
- learning objective
- support goals
- urgency level
- preferred support style
- preferred provider type

Scheduling Preferences

- preferred date
- preferred time
- preferred duration
- recurring support preference

Attachments & Resources

- assignments
- study guides
- notes
- screenshots
- documents
- practice problems

Checkbox Examples

- Homework Support
- Exam Preparation
- Concept Reinforcement
- Study Planning
- Long-Term Support
- Group Collaboration
- Other

If “Other” is selected:

- custom support request field required

This infrastructure improves:

- structured intake
- provider preparation
- recommendation quality
- analytics
- institutional workflows
- academic continuity

20. Academic Support Outline Workflow

Support providers may create support outlines including:

- objectives
- structure
- concepts to review
- estimated duration
- preparation requirements
- support strategy
- recommended resources

Students may:

- review outlines
- approve outlines
- request edits
- comment on outlines
- request revisions

21. Automated Provider Profile Review Infrastructure

Provider profile review should automatically appear during support booking workflows.

Before confirmation, students should automatically see:

- provider identity
- verification status
- institution affiliation
- ratings
- session count
- specialization
- response time

- support categories
- pricing
- availability
- badges
- reviews

Students may:

- Continue to Booking
- View Full Profile
- Compare Providers
- Save Provider
- Skip Profile Review

For recurring support relationships, students may also select:

- Never Show Again

If selected:

- future bookings with that provider may bypass automatic profile review
- students may still manually access profiles anytime

This infrastructure supports:

- transparency
- trust
- continuity
- reduced booking friction
- informed decision-making

22. Student Confirmation Review Infrastructure

Before booking finalization, students must review and approve:

- support outline
- recommended duration
- pricing
- provider selection
- schedule
- cancellation policy

- support expectations

Students may:

- Approve & Continue
- Request Edits
- Request Provider Change
- Modify Scheduling Preferences
- Request Clarification
- Cancel Request

Booking confirmation should only occur after explicit student approval.

23. Thread Priority Status

Students should be able to assign thread priority levels.

Examples

- Standard
- Urgent
- Emergency
- Exam Prep
- Assignment Due Soon
- Ongoing Support

Priority status should support:

- prioritization
- notification urgency
- recommendation systems
- escalation systems

24. Session Duration Infrastructure

Support sessions may operate in configurable increments.

Examples

- 30 minutes
- 1 hour
- 1 hour 30 minutes

- 2 hours
- 2 hours 30 minutes
- 3 hours

Final duration recommendations may be adjusted based on academic needs.

25. Pricing Infrastructure

Paid support providers must display pricing transparently throughout workflows.

Pricing visibility should exist on:

- profiles
- recommendation cards
- support threads
- confirmations
- summaries

Pricing models may include:

- hourly pricing
- session pricing
- package pricing
- subscription support
- institution-covered support

26. Cancellation Policy Infrastructure

Cancellation/refund policies must remain visible inside support threads.

Possible States

- refundable
- partially refundable
- non-refundable
- institution-covered
- reschedulable

27. Booking Infrastructure

All bookings and scheduling operations must occur inside academic support threads.

Booking actions may include:

- request booking
- confirm booking
- approve schedule
- modify duration
- reschedule
- cancel
- recurring support

28. Payment Confirmation Infrastructure

For paid support engagements, the system should generate confirmation messages inside support threads.

Confirmations may include:

- provider name
- support type
- duration
- schedule
- payment status
- cancellation policy
- support status

29. Academic Support Notifications Infrastructure

Notifications should exist:

1. inside platform notifications
2. inside academic support threads
3. inside thread timelines

Notification Examples

- support request submitted
- provider responded
- outline created
- outline updated
- confirmation pending
- booking confirmed
- payment confirmed
- session reminder
- session starting soon

- support completed
- follow-up requested
- continuation recommendation
- recurring support reminder

Reminder Examples

- 24-hour reminder
- 1-hour reminder
- session starting now
- incomplete confirmation reminder
- outline awaiting approval

This infrastructure ensures:

- operational continuity
- engagement persistence
- academic follow-through
- workflow visibility

30. Academic Support Timeline Infrastructure

Each support thread should maintain a visible event timeline.

Timeline events may include:

- support request submitted
- provider response received
- outline created
- booking confirmed
- payment completed
- session started
- support completed
- follow-up requested
- continuation requested

This creates:

- continuity
- auditability
- operational history
- academic memory

31. Academic Support Lifecycle States

Academic support threads may support lifecycle states including:

- Draft
- Pending Review
- Awaiting Approval
- Booking Pending
- Scheduled
- Active
- In Session
- Follow-Up Required
- Continuing Support
- Dormant
- Archived
- Escalated
- Institution Intervention

32. Live Session Infrastructure

Academic support sessions must launch directly from support threads.

Capabilities may include:

- video
- audio
- screen sharing
- collaborative whiteboard
- file review
- notes
- resource sharing

33. Session Lifecycle

Pre-session
→ active session
→ post-session summary
→ feedback
→ continuation/re-engagement

34. Session Artifacts

Support threads may store:

- recordings
- summaries
- notes
- assignments
- action plans
- recommendations
- shared files
- study materials

35. Ratings & Feedback Infrastructure

Students should rate academic support experiences after completion.

Profiles may display:

- rating score
- session count
- response time
- repeat engagement rate
- cancellation reliability
- student feedback
- verification status

36. Assist Integration

Assist may generate:

- summaries
- concept breakdowns
- preparation reminders
- study recommendations
- continuation suggestions

Assist must support:

- learning
- guidance
- reinforcement

Assist must NOT:

- complete student work

- bypass academic integrity
- generate cheating assistance

37. Discovery Engine Integration

Academic support threads should integrate with:

- Discovery Engine
- course linkage
- grade-level linkage
- institution linkage
- recommendation systems
- engagement systems

Students may search support infrastructure by:

- subject
- provider
- institution
- course
- academic level
- exam type
- pricing
- rating
- language
- availability
- response time
- verified status
- country

38. ONE Channel Relationship Infrastructure

Academic support threads may connect with:

- study circles
- study sessions
- communities
- courses
- cohorts
- mentorship
- advising
- MasterSessions
- institutional support programs

This creates:

- persistent academic relationships
- academic continuity
- engagement loops
- scalable support ecosystems

39. Trust & Safety Infrastructure

Academic support threads may support:

- moderation
- reporting
- audit logs
- communication restrictions
- blocking
- muting
- institutional oversight
- escalation systems

40. Institutional Infrastructure

Institution-linked support environments may support:

- approved providers
- intervention programs
- advisor-linked support
- institutional initiatives
- institution analytics
- institution oversight

41. Strategic Platform Principle

The AcademIQ Academic Support Infrastructure is NOT designed as a generic tutoring marketplace.

It is designed as:

A persistent academic support ecosystem operating through structured academic threads powered by ONE Channel.

Academic support threads become:

- the communication layer
- the academic memory layer
- the continuity layer
- the support layer
- the scheduling layer
- the analytics layer
- the governance layer
- the relationship layer

This infrastructure enables AcademIQ to deliver:

- scalable academic support
- intelligent recommendation systems
- institution-ready support infrastructure
- persistent learning continuity
- structured academic collaboration
- AI-assisted academic guidance
- long-term academic relationship infrastructure across the ecosystem.

Trust & Safety Infrastructure System

AcademIQ Trust & Safety Infrastructure System

DevSpec Document

1. Overview

The AcademIQ Trust & Safety Infrastructure System defines the platform-wide governance, moderation, protection, escalation, identity protection, reporting, communication restrictions, behavioral enforcement, and institutional safety architecture throughout the AcademIQ ecosystem.

This infrastructure is designed to ensure:

- academic safety
- student protection
- institutional trust
- communication accountability
- identity integrity
- platform governance
- abuse prevention
- moderation continuity
- scalable safety infrastructure
- protected academic collaboration
- secure educational networking

The Trust & Safety layer is foundational infrastructure and NOT merely a reporting feature.

This infrastructure integrates across:

- ONE Channel
- Threads
- Study Circles
- Study Sessions
- ASN
- Academic Networks
- Discovery Engine
- Moments
- Messaging
- Mentorship
- Tutoring
- Advising
- Institutional Ecosystems
- Emma
- Resource Sharing
- Collaboration Systems

2. Core Platform Safety Philosophy

AcademIQ is designed as:

- an academic platform
- a structured educational ecosystem
- a trust-centered collaboration environment
- a relationship-based academic infrastructure

NOT:

- an anonymous social platform
- an unrestricted messaging application
- an unmoderated tutoring marketplace
- an entertainment-first engagement system

The Trust & Safety Infrastructure exists to preserve:

- educational integrity
- institutional confidence
- student safety
- provider accountability
- identity transparency
- protected collaboration
- healthy academic engagement

3. Core Trust & Safety Objectives

The system is designed to:

- prevent abuse
- reduce harassment
- protect students
- enforce academic integrity
- support institutional moderation
- protect educational relationships
- support escalation workflows
- prevent impersonation
- reduce fraud
- maintain platform trust
- ensure communication accountability
- provide operational transparency

4. Identity Protection Infrastructure

The Trust & Safety system integrates directly with:

- verification infrastructure
- certification infrastructure
- institution linkage
- ASN trust architecture

Identity infrastructure may support:

- government ID verification
- email verification
- institution verification
- selfie/liveness verification
- credential verification
- institution affiliation verification
- role-based trust indicators

The system should clearly distinguish:

- verified users
- certified users
- institution-linked users
- peer users
- restricted users
- suspended users

5. Reporting Infrastructure

Users must be able to report:

- harassment
- spam
- impersonation
- academic misconduct
- inappropriate tutoring behavior
- abusive communication
- credential misuse
- policy violations
- inappropriate Moments
- unsafe conduct

- suspicious behavior

Reports may originate from:

- threads
- Moments
- profiles
- study circles
- sessions
- academic networks
- messages
- resources
- comments
- communities

6. Moderation Infrastructure

The system must support:

- manual moderation
- automated moderation
- institution moderation
- escalation workflows
- content restriction
- visibility enforcement
- communication restrictions
- moderation audit logs

Moderation actions may include:

- warning
- content removal
- visibility restriction
- temporary suspension
- permanent suspension
- feature limitation
- institution escalation
- ASN restriction
- thread restriction

7. Communication Restriction Infrastructure

The platform must support granular communication controls.

Examples:

- restrict direct messaging
- restrict thread posting
- restrict commenting
- restrict network invitations
- restrict study circle participation
- restrict session participation
- restrict Moments visibility
- restrict profile discoverability

Restrictions may be:

- temporary
- permanent
- institution-specific
- role-specific
- moderation-triggered

8. Blocking Infrastructure

Users must be able to block other users.

Blocking effects may include:

- prevent messaging
- prevent thread participation visibility
- prevent profile access
- prevent network requests
- prevent recommendations
- prevent Moments interactions
- prevent collaboration invitations

Blocking must remain:

- permission-aware
- privacy-aware
- institution-aware

9. Muting Infrastructure

Users may mute:

- users
- threads
- study circles
- communities
- Moments
- networks

Muting suppresses:

- notifications
- updates
- recommendations
- activity alerts

while preserving network relationships where applicable.

10. Restriction Infrastructure

Users and moderators may apply restrictions.

Restriction examples:

- limited messaging
- limited posting
- limited visibility
- limited participation
- limited network activity
- moderated posting approval

Restrictions support:

- progressive enforcement
- safety escalation
- institution governance
- abuse prevention

11. Academic Integrity Protection

Trust & Safety integrates directly with academic integrity systems.

The platform must prohibit:

- assignment completion services
- cheating assistance
- impersonation
- credential falsification
- academic fraud
- unauthorized grading activity
- institutional policy bypassing

The platform may monitor:

- suspicious tutoring activity
- abnormal engagement patterns
- repeated misconduct reports
- policy evasion attempts
- fraudulent credential claims

12. Institutional Governance Infrastructure

Institutions may configure:

- moderation rules
- visibility restrictions
- communication policies
- network permissions
- thread permissions
- Moments permissions
- escalation workflows
- advisor restrictions
- counseling visibility
- support access permissions

Institutions may also:

- assign moderators
- manage institution reports
- restrict external users
- enforce institution-specific conduct rules

13. Safety Escalation Infrastructure

The system must support escalation workflows.

Escalation examples:

- student safety concern
- mental health concern
- harassment
- predatory behavior
- impersonation
- institutional misconduct
- severe academic misconduct
- repeated abuse

Escalations may route to:

- platform moderation
- institution moderators
- support teams
- safety review teams
- emergency escalation workflows

14. Minor Protection Infrastructure

Additional protections may apply for minors.

Examples:

- restricted communication
- guardian visibility
- limited discoverability
- institution-only environments
- restricted external networking
- enhanced moderation review

Additional protections may vary by:

- jurisdiction
- institution policy
- age classification
- support category

15. Moments Moderation Infrastructure

Moments infrastructure must support:

- reporting
- moderation review
- visibility enforcement
- institution moderation
- creator restrictions
- educational integrity review

The platform may restrict:

- inappropriate content
- misleading educational claims
- harassment
- impersonation
- unsafe behavior
- non-educational misuse

16. Network Safety Infrastructure

Academic Networks must support:

- network blocking
- network removal
- request restrictions
- participation restrictions
- moderation review
- network visibility controls

Users must be able to:

- remove members
- block members
- mute members
- restrict participation
- disable discoverability

17. Session Safety Infrastructure

Live academic sessions may support:

- reporting
- moderation escalation
- participant removal
- emergency session termination
- recording review (where permitted)
- restricted access enforcement

Institutions may require:

- approved providers
- verified participants
- institution-only participation
- moderation oversight

18. Privacy & Permission Infrastructure

Trust & Safety must respect:

- privacy settings
- visibility permissions
- institutional permissions
- network permissions
- FERPA-conscious architecture
- role-based access

The platform must support:

- permission-aware recommendations
- visibility-aware discovery
- restricted-access environments
- protected educational spaces

19. Audit Log Infrastructure

The platform should maintain moderation audit logs.

Examples:

- moderation actions
- visibility changes
- restriction changes
- suspension history

- report resolutions
- escalation history
- institution moderation activity

Audit logs support:

- accountability
- transparency
- institutional reporting
- operational review
- abuse investigation

20. Emma Safety Infrastructure

Emma must remain:

- permission-aware
- privacy-aware
- institution-aware
- escalation-aware

Emma must NOT:

- bypass moderation restrictions
- expose protected information
- override institutional permissions
- reveal restricted content
- provide unsafe recommendations

Emma may escalate users when:

- mental health concerns arise
- institutional intervention is needed
- support exceeds AI limitations
- safety concerns are detected

21. Discovery Engine Safety Integration

Discovery Engine recommendations must respect:

- blocked users

- restricted users
- muted users
- visibility settings
- institution restrictions
- moderation restrictions
- protected-access environments

Restricted users should not receive unfair recommendation amplification.

22. Long-Term Safety Vision

The Trust & Safety Infrastructure is designed to evolve into a scalable educational governance system supporting:

- global educational ecosystems
- institution-ready moderation
- safe academic networking
- protected mentorship
- secure tutoring environments
- FERPA-conscious collaboration
- scalable educational trust systems

The infrastructure transforms AcademIQ into:

A structured, identity-aware, moderation-governed academic ecosystem built around:

- trust
- accountability
- educational integrity
- protected collaboration
- institutional confidence
- student safety
- long-term academic continuity.

Academic Network System

AcademIQ Academic Network System

DevSpec Document

1. Overview

The AcademIQ Academic Network System defines the platform-wide relationship, connection, engagement, continuity, collaboration, and academic-follow infrastructure across the entire AcademIQ ecosystem.

The Network System is NOT intended to function as a generic social media “followers” system.

Instead, the Academic Network System is designed to create:

- structured academic relationships
- persistent learning continuity
- academic engagement ecosystems
- collaborative educational communities
- subject-based learning networks
- institution-linked communities
- mentorship continuity
- academic support visibility
- educational discovery infrastructure
- role-based collaboration environments

The Network System functions as a foundational engagement layer throughout AcademIQ and integrates directly with:

- ASN
- ONE Channel
- Threads
- Study Circles
- Discovery Engine
- Messaging
- Resource Sharing
- Mentorship
- Sessions
- Academic Recommendations
- Cohort Infrastructure
- Institution Ecosystems

2. Core Objectives

The Academic Network System is designed to:

- facilitate learning continuity
- strengthen educational collaboration
- improve engagement retention
- support academic networking
- encourage community participation
- support academic discovery
- improve academic support accessibility
- enable persistent mentorship relationships
- support institution-linked engagement
- personalize educational experiences
- improve thread discoverability
- strengthen trust ecosystems

3. Network Philosophy

AcademIQ's "Network" infrastructure is academically centered and must NOT resemble traditional social media follower systems.

The Network System is intended to represent:

- educational relationships
- learning communities
- academic collaboration
- mentorship ecosystems
- educational continuity
- subject-based engagement
- professional academic networking

The system should prioritize:

- learning
- collaboration
- trust
- accountability
- educational engagement

over vanity metrics or social popularity.

4. Supported Network Roles

All user roles on AcademIQ may participate in the Network System.

Supported roles include:

4.1 Students

- General Students
- Institution Students
- Verified Students
- Peer Tutors
- Cohort Leaders
- Student Ambassadors

4.2 Instructors

- General Instructors
- Institution Instructors
- Certified Instructors
- Verified Instructors

4.3 Tutors

- General Tutors
- Verified Tutors
- Certified Tutors
- Peer Tutors

4.4 Advisors

- Academic Advisors
- Financial Aid Advisors
- Mentors
- Academic Coaches

4.5 Therapists & Counselors

- Licensed Therapists
- School Counselors
- Wellness Support Professionals

4.6 Institutions

Institution-linked ecosystems may participate in structured network environments where permitted.

5. Network Relationship Model

The Network System establishes directional academic relationships between users.

Examples:

- Student joins Instructor Network
- Student joins Tutor Network
- Peer Tutor joins Instructor Network
- Instructor joins Instructor Network
- Advisor joins Institution Instructor Network
- Student joins Study Circle Network
- Institution Student joins Cohort Network

6. Network Membership & Access Infrastructure

The AcademIQ Network System operates under a permission-granted membership architecture.

Users may NOT automatically join another user's Academic Network unless the network owner explicitly enables open joining.

By default:

network membership requires approval.

6.1 Network Join Request Workflow

The standard workflow is:

1. User submits a request to join a network.
2. Network owner receives the request.
3. Network owner may:
    - approve
    - decline
    - ignore
    - mute
    - restrict
    - or block the request/user
4. User becomes a network member only after approval is granted.

6.2 Network CTA States

Supported CTA/button states include:

- Request to Join Network
- Request Pending
- Joined Network
- Leave Network
- Request Declined
- Network Closed

6.3 Open Network Mode

Users may optionally enable:

Open Network Access

When enabled:

- approved role categories may join automatically
- no manual approval is required

Examples:

- public instructor networks
- public study communities
- institution-approved networks

Open joining remains subject to:

- moderation controls
- safety restrictions
- institution rules
- blocking systems
- role permissions

6.4 Network Requests Management

Users must have access to a dedicated:

Network Requests Management System

Users may manage:

- incoming requests
- approved members
- declined requests
- pending requests
- muted users
- restricted users
- blocked users
- removed members

7. Network Visibility & Permission Architecture

Network visibility settings determine discoverability and visibility scope only.

Visibility settings do NOT override:

- permissions
- moderation restrictions
- institution controls
- privacy rules
- blocking restrictions
- protected-access environments

All visibility remains:

permissions-granted-only.

Users must still possess appropriate permissions and approved access rights where applicable.

7.1 Public

Definition:

Visible to all AcademIQ users and publicly discoverable where enabled.

Examples:

- public instructor profiles
- public study circles
- public academic threads
- public academic networks

Public visibility does NOT automatically grant:

- network membership
- participation rights
- messaging rights
- thread posting rights
- restricted content access
- institution-restricted access

Additional permissions may still apply.

7.2 AcademIQ Members Only

Definition:

Visible only to authenticated AcademIQ users.

Content is hidden from:

- public visitors
- search engines
- unauthenticated users

Participation remains subject to:

- permissions
- moderation
- access rules
- role restrictions
- network approval workflows

7.3 Network Only

Definition:

Visible only to approved members within a user’s Academic Network.

Examples:

- network-only resources
- network-only study circles
- network-only announcements
- network-only discussions

Users outside the network cannot access visibility-restricted content unless approval is granted.

7.4 Institution Only

Definition:

Visible only to users associated with a specific institution ecosystem.

Examples:

- institution-only threads
- institution-only study circles
- institution-specific mentoring
- institution collaboration environments

Visibility remains restricted to:

- approved
- authorized
- institution-linked users only.

Additional institution permissions and restrictions may apply.

7.5 Private

Definition:

Visible only to the content owner and explicitly authorized users.

Private content is not discoverable through:

- Discovery Engine
- profile searches
- public indexing
- network recommendations

Examples:

- private mentorship notes
- draft resources
- restricted collaboration spaces

- internal planning threads

8. Network Controls

Users must have granular control over their network relationships and notifications.

8.1 Remove From Network

Users may remove members from their network.

Examples:

- instructor removes disruptive member
- student leaves instructor network
- tutor removes inappropriate participant

Removal effects:

- disconnect network relationship
- revoke network-only visibility
- stop network notifications
- remove access to restricted content

8.2 Leave Network

Users may voluntarily leave any network.

Example:
[Leave Network]

8.3 Mute Network

Users may mute a network while remaining connected.

Muted networks may suppress:

- notifications
- thread updates
- announcements
- study circle updates
- resource alerts

8.4 Notification Controls

Users must be able to configure notification categories individually.

Examples:

- Threads
- Study Circles
- Sessions
- Resources
- Messages
- Mentorship Activity
- Announcements
- Replies
- Recommendations

Notification settings may include:

- All Notifications
- Important Only
- Mentions Only
- Muted

9. Network Filtering Infrastructure

Network lists must be filterable.

This is critical for scalability.

9.1 Filter By Role

Users may filter networks by role.

Examples:

- Students
- Instructors
- Tutors
- Advisors
- Therapists
- Peer Tutors
- Mentors
- Institution Users

9.2 Filter By Institution

Examples:

- Institution Students
- Institution Instructors
- Institution Advisors

9.3 Filter By Relationship Type

Examples:

- Close Colleagues
- Mentors
- Study Partners
- Frequent Collaborators
- Favorite Instructors
- Academic Connections
- Cohort Members

9.4 Filter By Subject

Examples:

- Mathematics
- Computer Science
- Literature
- Biology
- Engineering

9.5 Filter By Activity

Examples:

- Most Active
- Recently Active
- Newest Connections
- Most Engaged
- Recently Joined

10. Academic Network Metrics

Profiles may display network metrics.

Examples:

- Network Members
- Active Members
- Study Circles Joined
- Thread Participation
- Collaborative Sessions
- Resources Shared

These metrics should support:

- trust
- engagement
- academic continuity

and should NOT prioritize vanity engagement.

11. Network Discovery Integration

The Discovery Engine must integrate with Network infrastructure.

Users may discover:

- instructors
- tutors
- study circles
- collaborators
- subject experts
- mentors
- peer tutors

through:

- network relationships
- mutual networks
- shared subjects
- shared institutions
- shared cohorts
- shared thread participation

12. ONE Channel Integration

Networks integrate directly with ONE Channel.

This allows:

- persistent collaboration
- academic continuity
- structured communication
- educational relationship history
- contextual engagement

13. Moderation & Safety

Network infrastructure must comply with:

- Trust & Safety systems
- moderation workflows
- academic integrity policies
- institutional restrictions
- safety governance

The platform may:

- restrict abusive users
- limit network activity
- suspend network privileges
- remove users from networks

14. Privacy & Consent

Users maintain ownership and control over:

- network visibility
- network participation
- notification settings
- discoverability
- relationship permissions

Users must be able to:

- block users
- restrict users
- mute users
- leave networks
- remove members
- disable discoverability

15. Long-Term Vision

The Academic Network System is intended to evolve into a global educational relationship infrastructure supporting:

- academic communities
- mentorship ecosystems
- institution-linked collaboration
- peer learning
- academic discovery
- educational continuity
- structured support ecosystems
- subject-based communities
- lifelong academic engagement

The system transforms AcademIQ from a traditional educational platform into a persistent academic relationship ecosystem built around:

- learning
- trust
- continuity
- collaboration
- identity
- engagement
- and educational growth.

Academic Support Matching System

AcademIQ Academic Support Matching System

DevSpec Document

1. Overview

The AcademIQ Academic Support Matching System defines the platform-wide infrastructure responsible for connecting students, instructors, tutors, advisors, mentors, peer tutors, study circles, threads, and academic support environments throughout the AcademIQ ecosystem.

This system functions as the foundational recommendation, discovery, routing, and engagement engine powering:

- ASN
- Discovery Engine
- ONE Channel
- Study Circles
- Study Sessions
- Threads
- Mentorship
- Academic Networks
- Session Recommendations
- Resource Recommendations
- Peer Collaboration
- AI Recommendations
- Personalized Academic Support

The Matching System is designed to facilitate learning continuity by intelligently connecting users with the most relevant academic support environments and academic relationships.

2. Core Objectives

The Academic Support Matching System is designed to:

- improve academic support discovery
- connect students to appropriate support providers
- reduce support friction
- improve engagement
- personalize academic experiences
- improve thread discovery
- improve study circle participation
- support mentorship continuity
- strengthen academic communities
- support peer collaboration
- improve learning outcomes
- increase platform retention

- support scalable academic networking
- facilitate educational continuity

3. Supported Roles

The Matching System supports all major AcademIQ user roles.

3.1 Students

- General Students
- Institution Students
- Verified Students
- Peer Tutors
- Cohort Leaders

3.2 Instructors

- General Instructors
- Institution Instructors
- Certified Instructors
- Verified Instructors

3.3 Tutors

- General Tutors
- Verified Tutors
- Certified Tutors
- Peer Tutors

3.4 Advisors

- Academic Advisors
- Financial Aid Advisors
- Academic Coaches
- Mentors

3.5 Therapists & Counselors

- Licensed Therapists
- School Counselors
- Wellness Professionals

4. Services Needed vs Services Offered Architecture

The platform must distinguish between:

Services Needed

and

Services Offered.

This distinction is foundational to the matching infrastructure.

4.1 Students

Students identify:

Services Needed.

4.2 Instructors, Tutors, Advisors & Support Providers

Support providers identify:

Services Offered.

5. Onboarding Collection Workflow

During onboarding, users must complete a structured academic support selection workflow.

The workflow must support:

- multi-select checkboxes
- searchable dropdowns
- custom entries
- categorized support types

6. Student Services Needed

Students may select one or more services needed.

Examples include:

- 1-on-1 Tutoring
- Study Sessions
- Study Circles
- Mentorship

- Homework Support
- Exam Preparation
- Thread-Based Support
- Collaborative Learning
- Accountability Support
- Group Learning
- Concept Clarification
- Resource Recommendations
- Academic Coaching

6.1 Other Option

Students must have access to:

Other

When selected:

- an input field must appear
- student may describe additional support needs

Example:
Other: “MCAT preparation accountability group”

7. Instructor/Tutor Services Offered

Support providers may select one or more services offered.

Examples include:

- 1-on-1 Tutoring
- Study Sessions
- Study Circles
- Mentorship
- Homework Support
- Exam Preparation
- Thread-Based Support
- Collaborative Learning
- Academic Coaching
- Group Instruction
- Resource Sharing
- Peer Support
- Concept Reinforcement

7.1 Other Option

Support providers must also have access to:

Other

When selected:

- custom service input field appears
- service may later be classified by system or moderation review

8. Academic Goals Infrastructure

The system must distinguish between:

support services

and

academic goals.

These are NOT interchangeable.

8.1 Examples of Academic Goals

- Improve GPA
- Pass Calculus
- Prepare for SAT
- Prepare for MCAT
- Improve Writing Skills
- Graduate Preparation
- Improve Study Habits
- Prepare for Final Exams
- Improve Time Management

Academic goals may be used for:

- recommendation systems
- support prioritization
- thread recommendations
- study circle matching
- AI personalization
- mentor pairing

9. Matching Factors

The Matching System must support multi-factor matching logic.

The platform may use:

- services needed
- services offered
- academic goals
- subjects
- courses
- institutions
- cohorts
- grade level
- major
- timezone
- language
- availability
- ratings
- verification status
- certifications
- thread participation
- study circle activity
- network relationships
- collaboration history
- engagement patterns
- response times
- student preferences

10. Discovery Engine Integration

The Discovery Engine must integrate directly with the Matching System.

The system may recommend:

- instructors
- tutors
- advisors
- mentors
- peer tutors
- study circles
- sessions
- threads
- resources
- academic communities

based on matching relevance.

11. Network Recommendation Infrastructure

The platform may recommend Academic Networks based on:

- shared courses
- shared services
- academic goals
- institutions
- thread participation
- study circle participation
- subject overlap
- peer collaboration
- engagement behavior

12. Thread Recommendation Infrastructure

The platform may recommend threads based on:

- services needed
- course enrollment
- thread activity
- academic goals
- subject relevance
- network participation
- institution participation
- saved resources
- collaboration history

13. Study Circle Matching Infrastructure

Study Circles may be recommended based on:

- support needs
- services offered
- academic goals
- subjects
- institutions
- collaboration preferences
- engagement activity
- timezone compatibility
- network overlap

14. Study Session Matching Infrastructure

Sessions may be recommended using:

- service compatibility
- instructor availability
- timezone alignment
- ratings
- subject expertise
- institution compatibility
- engagement patterns
- preferred session format

15. AI Recommendation Layer

The Matching System shall support future AI recommendation infrastructure including:

- personalized recommendations
- intelligent support routing
- academic pathway suggestions
- study partner recommendations
- mentorship suggestions
- thread prioritization
- session recommendations
- engagement optimization

This layer may integrate with:
Emma.

16. Other Option Classification Infrastructure

Custom “Other” entries must support classification workflows.

The system may:

- categorize custom entries
- map entries to existing services
- suggest standard categories
- flag entries for moderation review
- support future taxonomy expansion

17. Privacy & Permissions

Matching infrastructure must comply with:

- privacy controls
- institution restrictions
- moderation systems
- blocking systems
- network permissions
- FERPA-conscious architecture
- visibility settings

Matching recommendations must respect:

- blocked users
- muted users
- restricted environments
- institution-only ecosystems
- network-only visibility settings

18. Moderation & Trust Infrastructure

The platform may:

- restrict abusive users
- limit recommendation visibility
- suppress fraudulent services
- flag suspicious behavior
- remove misleading service claims

Verification and credential systems may influence:

- ranking
- visibility
- recommendation priority
- trust indicators

19. Long-Term Vision

The Academic Support Matching System is designed to evolve into a scalable educational relationship and recommendation infrastructure capable of supporting:

- global academic networking
- intelligent support discovery
- mentorship ecosystems
- collaborative learning
- institution-linked support
- peer learning
- personalized academic experiences
- lifelong learning continuity

The system transforms AcademIQ from a traditional educational platform into a structured academic relationship ecosystem centered around:

- learning
- collaboration
- trust
- continuity
- engagement
- personalization
- and educational growth.

Human Feedback & AI Reinforcement

AcademIQ Human Feedback & AI Reinforcement Infrastructure System

DevSpec Document

1. Executive Summary

The AcademIQ Human Feedback & AI Reinforcement Infrastructure System defines the complete post-session educational continuity, provider accountability, academic reinforcement, engagement, and support follow-up architecture throughout the AcademIQ ecosystem.

This infrastructure establishes a dual-layer academic support model consisting of:

1. mandatory human educational feedback
2. supplemental AI reinforcement infrastructure

This system applies across:

- tutoring
- study sessions
- study circles
- mentorship
- advising
- office hours
- institutional support programs
- academic recovery programs
- collaborative support environments

The infrastructure is designed to ensure:

- educational accountability
- structured continuity
- academic reinforcement
- persistent support memory
- scalable engagement
- institutional reporting
- learning continuity
- AI-assisted reinforcement
- human-led educational support

2. Human-Led Feedback Principle

All academic support sessions on AcademIQ must conclude with mandatory human provider feedback before session completion.

Human feedback must clearly display:

- the provider's full name
- provider role
- verification status (if applicable)
- institution affiliation (if applicable)
- submission timestamp

Examples

- Dr. Marcus Williams — Calculus Tutor
- Sarah Chen — Academic Advisor
- Michael Torres — Peer Tutor

AI-generated reinforcement must be clearly identified as originating from:

Emma

Examples

- Emma Reinforcement Summary
- Emma Study Recommendations
- Emma Continuity Insights

The system must maintain clear visual and structural separation between:

- human educational feedback

and

- Emma AI reinforcement

AI-generated reinforcement may supplement the support experience but may NOT replace required human educational feedback.

This principle ensures:

- educational accountability
- provider responsibility
- support quality
- institutional trust
- transparency
- ethical AI usage
- structured academic continuity
- human-led educational guidance

AcademIQ is designed as:

- human-led
- AI-assisted

NOT:

- AI-replaced education

3. Mandatory Session Completion Infrastructure

Academic support sessions may NOT be marked completed until mandatory human feedback is submitted.

Required human feedback components may include:

- session summary
- concepts covered
- objectives completed
- strengths observed
- areas needing improvement
- action plan
- follow-up recommendations
- continuation recommendations

Until feedback is submitted:

- session state remains “Pending Feedback”
- timeline reflects incomplete feedback
- completion analytics remain pending
- continuation workflows remain restricted

4. Session Lifecycle Infrastructure

Support session lifecycle states may include:

Draft
→ Scheduled
→ Active Session
→ Session Ended
→ Pending Human Feedback
→ Human Feedback Submitted
→ Emma Reinforcement Generated
→ Continuation Recommendations Activated
→ Session Completed

This infrastructure ensures:

- continuity
- accountability
- structured educational workflows
- operational visibility

5. Human Feedback Infrastructure

Human feedback should focus on:

- educational observations
- conceptual understanding
- student engagement
- learning progress
- participation quality
- study behaviors
- improvement opportunities
- mentorship guidance

Human feedback should remain:

- personalized
- educational
- actionable
- relationship-driven
- academically focused

All human feedback should clearly identify:

- provider name
- provider image/profile
- provider role
- provider verification status
- institution affiliation
- feedback submission time

Students should be able to:

- View Provider Profile
- Continue with Provider
- Save Provider
- Request Follow-Up Session

6. Tutor Feedback & Action Plan Infrastructure

Support providers should complete structured feedback workflows after each session.

Human Feedback May Include

Session Summary

- concepts covered
- objectives completed
- progress observed
- participation observations

Learning Assessment

- strengths demonstrated
- conceptual understanding
- pacing observations
- recurring challenges
- engagement quality

Areas for Improvement

- concept gaps
- practice weaknesses
- pacing concerns

- confidence issues
- participation concerns

Recommended Action Plan

- practice recommendations
- review priorities
- study strategies
- assignments
- additional exercises
- follow-up preparation
- reinforcement recommendations

Continuation Recommendations

- Schedule Follow-Up Session
- Join Study Circle
- Attend Study Session
- Practice Additional Problems
- Review Recommended Materials
- Attend Office Hours
- Request Continued Support
- Other

If “Other” is selected:

- custom recommendation field required

7. Mandatory Action Plan Principle

All human feedback submissions must include actionable academic guidance.

Feedback submissions should NOT consist solely of:

- generic comments
- attendance confirmation
- simple ratings
- incomplete summaries

Every feedback submission should provide:

- educational direction
- improvement recommendations
- structured next steps
- continuity guidance

This infrastructure transforms support sessions from:

- transactional interactions

into:

- structured academic continuity environments

8. Emma AI Reinforcement Infrastructure

Emma functions as an AI reinforcement and continuity layer.

Emma may generate:

- reinforcement summaries
- study recommendations
- review reminders
- practice suggestions
- continuation recommendations
- thread recommendations
- study circle recommendations
- milestone tracking
- engagement insights
- reinforcement planning

Emma should support:

- continuity
- organization
- reinforcement
- engagement
- workflow guidance

Emma should NOT:

- replace human educators

- replace provider accountability
- independently evaluate student performance
- function as the primary educational authority
- replace mandatory human feedback

All Emma-generated content should be clearly labeled as:

- Emma Feedback
- Emma Insights
- Emma Reinforcement
- Emma Recommendations

Emma-generated content should remain visually distinct from:

- human tutor feedback
- advisor feedback
- mentor feedback
- instructor observations

9. Emma Reinforcement Workflow

After human feedback submission, Emma may generate supplemental reinforcement infrastructure including:

Emma Reinforcement Summary

- concepts discussed
- major learning themes
- identified study areas
- reinforcement topics

Emma Personalized Study Recommendations

- practice exercises
- study materials
- videos
- study circles
- tutoring threads
- study sessions
- review plans

Emma Learning Continuity Suggestions

- recurring support recommendations
- follow-up reminders
- suggested timelines
- continuity scheduling
- milestone tracking

Emma Engagement Insights

- recurring support trends
- participation consistency
- reinforcement opportunities
- continuity recommendations

10. Human vs AI Role Separation Infrastructure

The human feedback layer and Emma reinforcement layer must remain structurally separated.

Human Layer Responsibilities

- mentorship
- educational observations
- conceptual analysis
- provider accountability
- learning assessment
- personalized guidance
- relationship-building
- action planning

Emma AI Responsibilities

- reinforcement
- summarization
- continuity support
- workflow intelligence
- engagement recommendations
- organizational assistance
- study recommendations
- reminder systems

This separation protects:

- educational integrity
- institutional trust
- provider accountability
- ethical AI usage
- human-led education principles

11. Thread Continuity Infrastructure

Feedback infrastructure should become persistent thread memory within ONE Channel.

Stored continuity data may include:

- prior feedback
- recurring recommendations
- improvement trends
- session history
- support continuity
- recurring learning gaps
- progress patterns
- reinforcement history

This enables:

- long-term academic continuity
- relationship persistence
- continuity intelligence
- recommendation quality
- support memory infrastructure

12. Feedback Timeline Infrastructure

Feedback events should appear inside:

- support threads
- thread timelines
- provider history
- student continuity history

Timeline Examples

- session completed
- feedback pending
- Dr. Marcus Williams submitted feedback
- Emma reinforcement generated
- follow-up recommendation created
- continuation session recommended
- study materials assigned

13. Adaptive Reinforcement Infrastructure

The system should dynamically adapt reinforcement recommendations based on:

- recurring learning gaps
- repeated subject difficulty
- engagement consistency
- support usage behavior
- provider observations
- continuation patterns
- regional trends
- curriculum requirements

Examples

- repeated derivative struggles may trigger additional Calculus recommendations
- recurring low engagement may trigger study planning recommendations
- repeated tutoring usage may recommend recurring support scheduling

This infrastructure enables:

- intelligent reinforcement
- adaptive support continuity
- personalization
- analytics-driven academic support

14. Feedback Completion SLA Infrastructure

Support providers should complete mandatory feedback within defined operational timeframes.

Examples

- immediately after session

- within 1 hour
- within 12 hours
- institution-defined requirements

If feedback remains incomplete:

- reminders should trigger automatically
- session remains incomplete
- continuation workflows remain pending
- analytics remain incomplete

This infrastructure supports:

- provider accountability
- operational consistency
- institutional reporting quality
- continuity integrity

Payment Hold Until Feedback Completion Infrastructure

For paid academic support services, provider payouts may remain pending until mandatory human feedback is completed.

If feedback remains incomplete, the system may:

- pause payout processing
- delay provider payment release
- maintain session in “Pending Feedback” state
- generate automated completion reminders

Reminder Examples

- “Please complete session feedback in order for payment processing to continue.”
- “Human feedback and action plan submission are required before payout release.”
- “Session remains pending until educational feedback is submitted.”

This infrastructure ensures:

- provider accountability
- educational continuity
- action plan completion

- support quality standards
- institution-ready operational governance

This infrastructure reinforces that AcademIQ compensates providers for:

- educational support
- continuity
- structured guidance
- academic accountability

NOT:

- transactional session attendance alone

Partial Completion State Infrastructure

Operational states may include:

- Session Complete
- Feedback Pending
- Action Plan Pending
- Emma Reinforcement Pending
- Payout Pending

This infrastructure supports:

- operational visibility
- provider transparency
- institutional reporting
- analytics accuracy
- workflow clarity

15. Student Continuation Infrastructure

After feedback completion, students may activate continuation workflows.

Continuation Actions May Include

- Schedule Follow-Up Session
- Continue with Same Provider
- Join Study Circle

- Attend Study Session
- Review Assigned Materials
- Practice More Problems
- Ask Emma for Reinforcement
- Request Different Provider
- Share Feedback with Approved Participant
- Other

If “Other” is selected:

- custom continuation field required

This infrastructure creates:

- engagement loops
- support continuity
- retention systems
- persistent academic relationships

16. Automated Reminder Infrastructure

The system may generate reminders related to:

- incomplete feedback
- continuation recommendations
- recurring support
- assigned practice
- study sessions
- follow-up scheduling
- milestone tracking

Reminder Examples

- “Dr. Marcus Williams submitted new recommendations.”
- “Emma generated new reinforcement suggestions.”
- “Your follow-up Calculus session is recommended this week.”
- “Practice materials remain incomplete.”

17. Feedback Visibility Governance

Human feedback and Emma reinforcement visibility should respect thread privacy settings.

Sensitive feedback environments should support:

- Private
- Institution
- Shared with Approved Participant

Public subject threads may support:

- limited public thread summaries
- anonymized engagement indicators
- public educational resources

Private educational observations should remain protected.

18. Shared Feedback Participation Infrastructure

If students choose to share feedback with approved participants, they must explicitly select:

- participant
- permission level
- visibility scope

Examples

- Parent / Guardian
- Advisor
- Mentor
- Counselor
- Instructor
- Institution Support Staff
- Other

Permission Examples

- View Only
- Comment
- Participate in Thread
- Receive Notifications

19. Analytics & Reporting Infrastructure

The infrastructure should support analytics related to:

- support effectiveness
- recurring learning gaps
- provider engagement
- continuation rates
- reinforcement completion
- recurring support behavior
- study consistency
- institutional outcomes
- engagement quality

Institutions may use aggregated analytics for:

- intervention programs
- support initiatives
- student success analysis
- retention programs
- academic recovery systems

20. Discovery Engine Integration

Feedback infrastructure may integrate with:

- Discovery Engine
- recommendation systems
- provider recommendations
- thread continuity systems
- study session recommendations
- study circle recommendations

This enables:

- smarter support pathways
- continuity recommendations
- reinforcement personalization
- engagement optimization

21. Strategic Platform Principle

The AcademIQ Human Feedback & AI Reinforcement Infrastructure is designed as:

A human-led academic continuity system supported by AI-assisted reinforcement infrastructure.

Human educational feedback remains:

- mandatory
- authoritative
- accountable
- relationship-driven

Emma functions as:

- reinforcement infrastructure
- continuity infrastructure
- engagement infrastructure
- workflow intelligence infrastructure

This architecture enables AcademIQ to deliver:

- structured academic continuity
- scalable educational reinforcement
- institution-ready support quality
- ethical AI integration
- persistent support ecosystems
- long-term student engagement
- human-centered educational infrastructure.

Assist Reinforcement Infrastructure System

Assist Reinforcement Infrastructure System

Master DevSpec Document

1. Executive Summary

The Assist Reinforcement Infrastructure System defines the complete AI-assisted educational reinforcement, continuity support, study reinforcement, preparation assistance, organizational guidance, and post-engagement academic support architecture throughout the AcademIQ ecosystem.

Assist functions as a supplemental academic reinforcement layer integrated across:

- ONE Channel
- Academic Support Threads
- Study Sessions
- Study Circles
- Tutoring Infrastructure
- Mentorship
- Advising
- Discovery Engine
- Library
- Moments
- Academic Networks
- Emma Infrastructure
- Human Feedback Infrastructure

Assist is NOT designed to function as:

- an autonomous educator
- a cheating system
- a replacement instructor
- a grading authority
- a fully independent tutoring system
- an academic integrity bypass tool

Instead, Assist functions as:

- a reinforcement layer
- a study support system
- a continuity assistant
- a preparation assistant
- a summarization system
- an organizational support system

- a learning reinforcement infrastructure

Assist exists to strengthen:

- educational continuity
- reinforcement
- study consistency
- organization
- engagement persistence
- long-term learning support

2. Core Assist Philosophy

Assist operates under:

Human-Led, AI-Assisted Reinforcement Infrastructure.

Human educators remain:

- authoritative
- accountable
- instructional leaders
- relationship drivers

Assist exists to:

- reinforce
- summarize
- organize
- guide study continuity
- support preparation
- improve educational accessibility

Assist must NEVER replace:

- instructors
- tutors
- advisors
- mentors
- counselors
- institutional authority

3. Core Objectives

Assist is designed to:

- reinforce learning
- improve study continuity
- improve organization
- improve preparation
- strengthen engagement
- reduce educational friction
- improve support follow-through
- improve reinforcement accessibility
- support continuity workflows
- improve educational retention

4. Core Assist Infrastructure Layers

Assist operates through multiple reinforcement layers.

4.1 Reinforcement Layer

Supports:

- reinforcement summaries
- study suggestions
- review guidance
- continuation prompts
- reinforcement planning

4.2 Preparation Layer

Supports:

- study preparation
- session preparation
- exam preparation
- assignment preparation
- organization assistance

4.3 Organizational Layer

Supports:

- reminders
- study planning
- continuity tracking
- task organization
- milestone reminders

4.4 Continuity Layer

Supports:

- follow-up recommendations
- recurring support reminders
- session continuity
- reinforcement continuity
- academic persistence

5. Assist vs Emma Separation

Assist and Emma serve different infrastructure purposes.

Emma

Focuses on:

- onboarding
- discovery
- navigation
- workflow guidance
- conversational support
- recommendations

Assist

Focuses on:

- reinforcement
- preparation
- organization
- summaries
- continuity support
- learning reinforcement

This separation is critical to maintain:

- architectural clarity
- workflow consistency
- scalable AI infrastructure

6. Assist Reinforcement Capabilities

Assist may generate:

- session summaries
- reinforcement summaries
- concept breakdowns
- review plans
- study recommendations
- follow-up reminders
- preparation reminders
- recurring study suggestions
- milestone tracking
- continuity suggestions

7. Assist Session Integration

Assist integrates directly with:

- tutoring sessions
- study sessions
- office hours
- mentorship sessions
- advising sessions
- study circles

After sessions, Assist may generate:

- reinforcement summaries
- study recommendations
- review priorities
- preparation guidance
- continuation reminders

8. Assist Summary Infrastructure

Assist may generate structured summaries including:

- concepts covered
- key takeaways
- reinforcement areas
- recommended review topics
- suggested next steps
- assigned materials
- study reminders

Summaries should remain:

- educational
- concise
- structured
- reinforcement-focused

9. Assist Study Reinforcement Infrastructure

Assist may provide reinforcement recommendations such as:

- additional practice
- concept review
- study resources
- review scheduling
- study session recommendations
- tutoring recommendations
- office hour reminders

Examples:

“You may benefit from reviewing derivatives before your next Calculus session.”

“Consider joining the upcoming Biology study circle this week.”

10. Assist Preparation Infrastructure

Assist may support preparation workflows including:

- exam preparation
- tutoring preparation
- study planning

- assignment preparation
- session readiness

Preparation support may include:

- review checklists
- recommended materials
- milestone reminders
- study sequencing
- continuity reminders

11. Assist Reminder Infrastructure

Assist may generate reminders related to:

- assignments
- tutoring sessions
- study sessions
- office hours
- practice completion
- reinforcement goals
- recurring support
- continuity recommendations

Examples:

“Your Chemistry tutoring session begins tomorrow at 4 PM.”

“You still have incomplete Calculus reinforcement materials.”

12. Assist Thread Integration

Assist operates directly inside:

- support threads
- tutoring threads
- study circle threads
- session threads
- mentorship threads
- office hour threads

This ensures reinforcement remains:

- contextual
- persistent
- searchable
- continuity-aware

13. Assist Discovery Engine Integration

Assist integrates directly with Discovery Engine infrastructure.

Assist may recommend:

- tutors
- study circles
- Moments
- resources
- sessions
- office hours
- support threads

based on:

- reinforcement needs
- recurring struggles
- engagement behavior
- continuity history
- subject relevance

14. Assist Library Integration

Assist may integrate with Library infrastructure.

Users may:

- save reinforcement summaries
- save study recommendations
- revisit reinforcement plans
- review prior summaries
- revisit study resources

This creates:

- searchable reinforcement memory

- continuity persistence
- long-term reinforcement tracking

15. Assist Continuity Infrastructure

Assist supports long-term continuity workflows.

Assist may:

- recommend recurring support
- suggest continuity sessions
- recommend milestone reviews
- track reinforcement consistency
- suggest follow-up support

Examples:

“You previously requested support for Organic Chemistry. Would you like continued reinforcement support?”

16. Assist Personalization Infrastructure

Assist recommendations may adapt using:

- support history
- tutoring history
- academic goals
- recurring struggles
- engagement patterns
- study consistency
- reinforcement completion
- institution participation
- subject preferences

17. Assist Notification Infrastructure

Assist notifications should remain:

- lightweight
- contextual
- reinforcement-focused
- non-spammy

Examples:

- study reminders
- reinforcement prompts
- continuity reminders
- review suggestions
- support recommendations

18. Assist Academic Integrity Restrictions

Assist must NOT:

- complete assignments
- generate cheating assistance
- bypass academic integrity systems
- impersonate instructors
- independently grade students
- provide exam answers

Assist should instead encourage:

- reinforcement
- tutoring
- study support
- collaboration
- office hours
- concept review

19. Assist Safety & Governance Infrastructure

Assist must remain:

- permission-aware
- institution-aware
- moderation-aware
- privacy-aware
- FERPA-conscious

Assist must respect:

- visibility settings

- moderation restrictions
- institution permissions
- blocked users
- restricted environments

20. Institutional Assist Infrastructure

Institutions may configure:

- reinforcement permissions
- notification policies
- institution-specific reminders
- support recommendations
- escalation rules
- continuity workflows

Institution-linked ecosystems may support:

- institution-approved reinforcement
- intervention support
- continuity monitoring
- academic recovery workflows

21. Analytics Infrastructure

Assist analytics may support:

- reinforcement engagement
- reminder completion
- continuity participation
- support follow-through
- tutoring continuation
- reinforcement effectiveness
- study consistency

Institutions may use aggregated analytics for:

- intervention planning
- student success initiatives
- academic recovery systems
- engagement optimization

22. Future Expansion Infrastructure

Future Assist capabilities may include:

- adaptive reinforcement planning
- AI-generated study pathways
- multilingual reinforcement
- predictive study support
- advanced milestone tracking
- collaborative reinforcement systems
- institution-specific reinforcement workflows
- advanced continuity intelligence

23. Long-Term Vision

Assist is designed to evolve into:

A scalable educational reinforcement infrastructure integrated throughout the AcademIQ ecosystem.

Assist transforms AI from:

- isolated study assistance

into:

- persistent educational continuity infrastructure

built around:

- reinforcement
- preparation
- continuity
- organization
- support persistence
- educational accessibility
- long-term academic engagement
- and human-led educational ecosystems.

Moments Infrastructure System

AcademIQ Moments Infrastructure System

FINAL DevSpec Document

1. Executive Summary

The AcademIQ Moments Infrastructure System defines the complete short-form academic content, discovery, engagement, educational conversion, creator interaction, and network amplification architecture throughout the AcademIQ ecosystem.

Moments are NOT designed to function as entertainment-first social media content.

Instead, Moments function as:

- academic discovery infrastructure
- educational micro-content infrastructure
- creator engagement infrastructure
- tutoring conversion infrastructure
- academic networking infrastructure
- course continuity infrastructure
- study support infrastructure
- institutional engagement infrastructure
- ONE Channel amplification infrastructure
- educational recommendation infrastructure

The primary purpose of Moments is to:

- increase academic engagement
- improve educational discovery
- connect students to support resources

- amplify educational creators
- drive participation in academic communities
- improve tutoring and study session discovery
- strengthen institutional communication
- create persistent academic continuity
- support educational networking
- integrate micro-learning into the AcademIQ ecosystem

Moments integrate directly with:

- Student Command Center
- Discovery Engine
- ONE Channel
- Study Circles
- Study Sessions
- Tutoring Infrastructure
- Institutional Communities
- Cohorts
- Library
- Emma AI Guidance System
- Creator Profiles
- Search Infrastructure

Moments are designed to create an academically structured ecosystem that prioritizes:

- educational value
- academic continuity
- relevance
- collaboration
- support
- conversion into learning activities

rather than:

- vanity metrics
- entertainment addiction
- passive scrolling
- meaningless engagement

2. Core System Philosophy

The Moments Infrastructure System operates on the following core principles:

2.1 Educational-First Architecture

Moments must prioritize:

- learning
- discovery
- academic engagement
- educational collaboration
- resource accessibility
- continuity
- academic support

Moments are NOT intended to replicate generic entertainment-focused short-form social media systems.

2.2 Persistent Academic Continuity

Unlike traditional social platforms where content is isolated and disposable, Moments integrate into the broader AcademIQ ecosystem.

Moments can be:

- linked to ONE Channel
- linked to study circles
- attached to courses
- attached to instructors
- attached to tutoring sessions
- searchable through Discovery Engine
- saved to Library
- recommended by Emma
- shared across cohorts
- referenced in academic discussions

2.3 Academic Conversion Infrastructure

Moments function as entry points into deeper educational engagement.

Examples:

- student watches a Moment
- student joins study session
- student joins study circle
- student books tutoring
- student opens ONE Channel
- student follows instructor network
- student saves educational resources

The Moment itself is NOT the final destination.

The system is designed to convert educational discovery into active academic participation.

3. Supported User Roles

The following user roles may create Moments depending on permissions and moderation policies.

3.1 Students

Students may create:

- educational Moments
- study tips
- peer support clips
- academic journey content
- collaboration requests

- study session recaps
- productivity content

Student Moments may include:

- verified student badges
- institution badges
- cohort badges
- peer tutor badges

3.2 Peer Tutors

Peer Tutors may create:

- concept explanations
- exam prep clips
- tutoring promotional Moments
- study circle Moments
- peer collaboration content

3.3 Tutors

Tutors may create:

- tutoring promotional content
- instructional clips
- academic guidance
- tutoring session highlights
- educational walkthroughs

3.4 Instructors

Instructors may create:

- mini lectures
- academic guidance
- office hour reminders
- course explanations
- assignment clarifications
- educational updates
- department announcements

3.5 Counselors & Advisors

Counselors may create:

- academic advising clips
- scholarship guidance
- FAFSA guidance
- transfer guidance
- career planning content
- wellness support guidance

3.6 Institutions

Institutions may create:

- campus announcements
- institutional updates
- department promotions
- orientation guidance
- event announcements
- academic support information

4. Moments Feed Infrastructure

4.1 Dedicated Moments Feed

AcademIQ shall include a dedicated Moments Feed.

The Moments Feed functions as the primary academic discovery environment for short-form educational content.

The Moments Feed shall exist independently from the Student Command Center while remaining fully integrated with the AcademIQ ecosystem.

4.2 Feed Accessibility

The Moments Feed shall be accessible from:

- Student Command Center
- Instructor Command Center
- Tutor Command Center
- Discovery Engine
- Hamburger Menu
- User Profiles
- ONE Channel
- Study Circles
- Emma Recommendations
- Search Results
- Notifications

4.3 Feed Objectives

The feed is designed to:

- surface academically relevant content
- improve educational discovery
- facilitate support matching
- increase study participation
- connect students to instructors and tutors
- promote educational communities
- improve collaboration
- strengthen academic continuity

5. Moments Content Structure

5.1 Supported Content Types

Moments may include:

- educational tips
- mini lectures
- quick concept explanations
- study strategies
- tutoring clips
- study session previews
- office hour reminders
- scholarship information
- career advice
- academic motivation
- cohort updates
- collaboration requests
- institutional announcements
- exam preparation content

5.2 Video Format

Recommended standards:

- vertical orientation

- mobile optimized
- caption support
- short-form structure
- accessibility support

Recommended duration:

- 30 seconds
- 45 seconds
- 60 seconds
- 90 seconds maximum recommended

5.3 Captions & Accessibility

Moments should support:

- captions
- transcript generation
- accessibility overlays
- screen reader compatibility
- multilingual support (future phase)

6. Moments Upload Workflow

6.1 Upload Flow

After selecting “Create Moment,” users proceed through guided upload flow.

Upload flow includes:

1. Upload Video
2. Add Title

3. Add Description
4. Add Tags
5. Select Subject/Course
6. Select Visibility
7. Select CTA
8. Configure CTA
9. Preview Moment
10. Publish

6.2 Metadata Requirements

Each Moment may include:

- title
- description
- creator identity
- creator role
- course association
- subject association
- institution association
- academic level
- hashtags/tags
- CTA configuration
- visibility type
- thread linkage
- timestamp

7. Visibility Infrastructure

7.1 Visibility Types

Creators must select visibility type before publishing.

Supported visibility options:

Public

Discoverable platform-wide.

Network Only

Visible only to connected users.

Institution Only

Visible only within institution ecosystem.

Cohort Only

Visible only to selected cohort.

Study Circle Only

Visible only to specific study circle.

Course Only

Visible only to enrolled course members.

Private Link

Accessible only through direct link.

7.2 Visibility Governance

Institutions may configure:

- creator permissions
- visibility restrictions
- moderation requirements
- approval workflows
- public posting limitations

8. CTA Infrastructure

8.1 CTA Philosophy

Moments shall support creator-configurable CTAs.

The creator chooses which CTA options appear after viewers watch the Moment.

CTAs function as educational conversion mechanisms.

8.2 CTA Selection Workflow

After upload, creators shall select CTA options through dropdown menu.

Creators may select:

- primary CTA
- secondary CTA

CTAs may vary based on:

- user role
- permissions
- institutional settings
- linked services

9. Supported CTA Types

9.1 Academic Engagement CTAs

- Join Study Session
- Join Study Circle
- Join my Network
- Open ONE Channel
- Continue Discussion
- Save to Library
- View Course Resources
- Ask Emma About Topic

9.2 Creator Growth CTAs

- Join My Network
- Follow Network
- View Profile
- View More Moments
- Subscribe to Updates

9.3 Tutoring & Service CTAs

- Book Tutoring Session
- Schedule Office Hours
- Book Advising Session
- Request Mentorship
- Contact Instructor

9.4 Institutional CTAs

- View Department
- Register for Event
- Join Institution Community

- Access Academic Support Services

10. Dynamic Role-Based CTA System

CTA options shall dynamically change based on creator role.

Example:

Instructor

Can attach:

- office hours
- study sessions
- course threads
- course resources

Tutor

Can attach:

- book tutoring
- study sessions
- study circles

Counselor

Can attach:

- advising appointments
- scholarship guidance
- career support sessions

Institution

Can attach:

- institution resources
- orientation events
- department communities

11. Moments Recommendation Engine

11.1 Personalized Feed Infrastructure

The Moments feed shall prioritize content using educational relevance.

Recommendations shall consider:

- enrolled courses
- academic level
- institution
- subject preferences
- cohorts
- tutoring history

- saved Moments
- study circles
- ONE Channel activity
- engagement history
- Discovery Engine settings
- linkage infrastructure settings

11.2 Educational Intent Ranking

The Moments recommendation engine must prioritize:

- educational relevance
- academic continuity
- learning outcomes
- support opportunities
- collaboration opportunities
- academic engagement

The system shall NOT prioritize:

- addictive engagement
- entertainment virality
- vanity metrics
- random trending behavior

12. Discovery Engine Integration

Moments integrate directly into Discovery Engine.

Users may search Moments using:

- course name
- subject
- instructor
- institution
- hashtags

- tutor name
- academic level
- topic
- cohort
- study circle
- creator role

13. ONE Channel Integration

13.1 Continue in Thread Infrastructure

Moments shall support “Continue in ONE Channel” functionality.

This enables:

- persistent discussion
- academic continuity
- collaborative engagement
- deeper educational interaction

13.2 Thread Linking

Moments may be linked directly to:

- ONE Channel threads
- study circles
- tutoring threads
- course discussions
- office hour channels

14. Emma Integration

Emma shall integrate directly with Moments.

Emma may:

- recommend Moments
- suggest study sessions
- suggest tutoring
- suggest related Moments
- suggest study circles
- recommend support resources
- answer Moment-related questions

Example:

“You watched several Calculus Moments. Would you like to join a Calculus II study circle?”

15. Library Integration

Users may:

- save Moments
- bookmark Moments
- organize Moments
- attach Moments to study resources
- revisit saved educational content

Saved Moments integrate directly into Library.

16. Creator Profiles

16.1 Creator Badges

Moments shall display creator identity markers.

Examples:

- verified badge
- institution badge
- instructor badge
- certified tutor badge
- peer tutor badge
- counselor badge
- ambassador badge

16.2 Profile Navigation

Users may navigate from Moments to:

- creator profile
- network profile
- tutoring profile
- instructor profile
- study circle
- office hours
- ONE Channel

17. Moderation & Governance

17.1 Moderation Infrastructure

Moments shall support:

- reporting
- moderation review
- institution governance
- flagging systems
- content restrictions
- visibility enforcement
- academic integrity policies

17.2 Institution Controls

Institutions may:

- restrict public Moments
- require approvals
- moderate institution Moments
- configure creator permissions
- manage visibility policies

18. Engagement Infrastructure

18.1 Supported Engagement Actions

Users may:

- like
- save
- share
- comment
- repost within network
- attach to thread
- send to study circle
- recommend to peers

18.2 Educational Sharing

Moments may be shared into:

- ONE Channel
- Study Circles
- Cohorts
- Messages
- Library
- Institution Communities

19. Notifications Infrastructure

Notifications may include:

- new Moments from followed creators
- recommended Moments
- study session invitations
- CTA interactions
- comments
- saves
- shares
- institutional Moments
- cohort Moments

20. Analytics Infrastructure

20.1 Creator Analytics

Creators may view:

- views
- saves
- shares
- engagement
- CTA clicks
- study session conversions
- tutoring conversions
- follower growth
- retention metrics

20.2 Institutional Analytics

Institutions may view:

- engagement by department
- student participation
- educational interaction trends
- support engagement metrics
- tutoring conversion data

21. Anti-Addiction & Educational Integrity Controls

AcademIQ Moments must maintain educational integrity.

The platform shall avoid:

- manipulative engagement systems
- excessive autoplay behavior
- entertainment-first ranking
- addictive algorithmic patterns
- vanity-focused ecosystems

The system must remain:

- academically focused
- productive
- supportive
- structured

- educationally relevant

22. Future Expansion Opportunities

Future infrastructure may include:

- live Moments
- collaborative Moments
- AI-assisted clip generation
- institution-sponsored Moments
- monetized educational creator programs
- verified educational creator marketplace
- AI-generated transcripts
- multilingual translation
- educational certification Moments
- assignment-linked Moments

23. Conclusion

The AcademIQ Moments Infrastructure System establishes a fully integrated educational short-form content ecosystem designed to:

- improve discovery
- strengthen academic engagement
- facilitate educational networking
- connect students to support resources
- amplify educational creators
- improve collaboration
- increase tutoring and study participation
- strengthen academic continuity
- integrate micro-learning into the AcademIQ ecosystem

Moments are not designed to function as entertainment infrastructure.

Moments function as academically contextual educational discovery infrastructure that connects students, instructors, tutors, counselors, institutions, and support systems into a persistent educational ecosystem.

AcademIQ Platform Stack

AcademIQ Platform Stack

1. Foundational Infrastructure Layer

This layer powers the core operational architecture of AcademIQ.

1.1 ONE Channel Infrastructure

Centralized thread-first communication and continuity architecture powering:

- communication
- collaboration
- tutoring
- advising
- study circles
- office hours
- academic continuity
- persistent engagement

1.2 Thread-Based Learning Infrastructure

Unified thread ecosystem powering:

- academic support
- collaboration
- mentorship
- sessions
- academic memory
- engagement continuity

1.3 Role-Based Thread Infrastructure

Role-adaptive thread workflows for:

- students
- instructors
- tutors
- advisors
- counselors
- institutions
- peer tutors
- mentors

Includes:

- permissions
- visibility
- metadata
- governance
- role-based workflows

2. Identity, Trust & Governance Layer

This layer powers trust, transparency, verification, moderation, and institutional accountability.

2.1 Verification Infrastructure

Validates:

- identity
- institutional affiliation
- legitimacy
- platform trust

Examples:

- verified student
- verified tutor
- verified instructor

2.2 Certification Infrastructure

Validates:

- professional qualifications
- licenses
- credentials
- institutional approvals

Examples:

- certified tutor
- licensed therapist

- certified instructor

2.3 Specialization Infrastructure

Powers:

- matching
- discovery
- recommendations
- expertise categorization

Examples:

- SAT Prep
- Calculus
- FAFSA
- MCAT Prep
- Organic Chemistry

2.4 ASN — Academic Support Network

Structured trust-based academic support ecosystem connecting:

- students
- instructors
- tutors
- advisors
- therapists
- peer tutors
- institutions

Built around:

- trust
- identity
- accountability
- verification
- academic continuity

2.5 Trust & Safety Infrastructure

Platform-wide:

- moderation
- reporting
- escalation
- blocking
- restrictions
- audit logs
- safety governance

3. Discovery, Relationship & Matching Layer

This layer powers educational networking, recommendations, discoverability, and relationship continuity.

3.1 Academic Network System

Structured academic relationship infrastructure supporting:

- mentorship
- collaboration
- networking
- cohort ecosystems
- educational continuity

3.2 Discovery Engine

Platform-wide educational discovery system powering:

- threads
- tutors
- study circles
- Moments
- instructors
- resources
- academic communities

3.3 Academic Support Matching System

Intelligent recommendation and matching engine connecting:

- students
- tutors

- advisors
- mentors
- study circles
- sessions
- threads

Based on:

- goals
- services needed
- services offered
- academic relevance
- engagement behavior

4. Academic Support & Continuity Layer

This layer powers long-term academic support ecosystems.

4.1 Academic Support Thread Infrastructure

Thread-based support system powering:

- tutoring
- study sessions
- mentorship
- office hours
- advising
- collaboration
- scheduling
- support continuity

4.2 Human Feedback & AI Reinforcement Infrastructure

Dual-layer reinforcement system consisting of:

- mandatory human educational feedback
- Emma AI reinforcement infrastructure

Supports:

- continuity
- accountability

- academic memory
- action plans
- reinforcement workflows

4.3 Academic Support Lifecycle Infrastructure

Operational support lifecycle management including:

- scheduling
- booking
- outlines
- follow-ups
- reminders
- continuation workflows
- engagement persistence

5. AI, Guidance & Reinforcement Layer

This layer powers conversational guidance, onboarding, reinforcement, and AI-assisted workflows.

5.1 Emma Infrastructure

AI guidance layer powering:

- onboarding
- discovery
- recommendations
- workflow guidance
- conversational support

Emma is:

- guidance-oriented
- thread-based
- contextual
- continuity-focused

5.2 Introductory Emma Thread System

Persistent onboarding and orientation thread automatically created for every user.

Functions as:

- onboarding memory
- support hub
- feature discovery environment
- conversational guidance layer

5.3 Assist Reinforcement Infrastructure

AI reinforcement layer supporting:

- summaries
- study recommendations
- preparation reminders
- continuity reinforcement
- learning support

Assist is:

- AI-assisted
- human-supportive
- reinforcement-oriented

NOT:

- academic replacement infrastructure

6. Engagement, Media & Educational Amplification Layer

This layer powers educational engagement, visibility, creator ecosystems, and academic media distribution.

6.1 Moments Infrastructure System

Short-form educational content ecosystem powering:

- academic discovery
- educational creator ecosystems
- tutoring conversion
- study session discovery
- micro-learning
- academic networking

Moments integrate with:

- ONE Channel
- Discovery Engine
- ASN
- Networks
- Study Circles
- Emma
- Library

6.2 Creator Infrastructure

Supports:

- educational creators
- instructors
- tutors
- peer tutors
- institutions
- counselors

Includes:

- creator profiles
- educational CTAs
- engagement analytics
- conversion infrastructure

6.3 Academic Conversion Infrastructure

Converts engagement into:

- tutoring
- study circles
- support threads
- office hours
- mentorship
- academic communities

7. Institutional Infrastructure Layer

This layer powers institution-ready governance and educational ecosystems.

7.1 Institution Ecosystem Infrastructure

Supports:

- institution-linked users
- institutional moderation
- institution visibility
- institution governance
- institutional communication

7.2 Cohort Infrastructure

Persistent cohort ecosystems supporting:

- student continuity
- collaboration
- peer engagement
- institutional networking

7.3 Institutional Analytics Infrastructure

Supports:

- engagement analytics
- intervention analytics
- support effectiveness
- student success initiatives
- retention programs

8. Educational Resource & Knowledge Layer

This layer powers educational storage, continuity, and resource accessibility.

8.1 Library Infrastructure

Persistent academic resource infrastructure supporting:

- saved threads
- saved Moments
- recordings
- study materials

- educational continuity

8.2 Resource Sharing Infrastructure

Supports:

- files
- notes
- recordings
- summaries
- assignments
- collaborative resources

8.3 Searchable Academic Memory Infrastructure

Persistent searchable ecosystem-wide academic memory layer supporting:

- continuity
- retrieval
- recommendations
- engagement history
- academic relationships

9. Future Expansion Infrastructure

Future ecosystem expansion may include:

- AI-assisted onboarding
- live Moments
- collaborative Moments
- educational creator monetization
- AI-generated clips
- multilingual infrastructure
- advanced mentorship ecosystems
- institution-sponsored creator ecosystems
- advanced adaptive recommendations
- academic pathway intelligence
- lifelong learning continuity systems

Final Platform Philosophy

AcademIQ is NOT:

- a tutoring marketplace
- a generic LMS
- a messaging platform
- a social media app
- a scheduling platform
- an isolated AI assistant

AcademIQ is:

A structured academic relationship, continuity, support, discovery, and engagement ecosystem built around:

- threads
- trust
- identity
- continuity
- collaboration
- educational networking
- AI-assisted guidance
- institutional integration
- long-term academic growth

AcademIQ Realtime Communication Infrastructure

AcademIQ Realtime Communication Infrastructure

LiveKit Educational Collaboration System

Developer Specification Document

1. Purpose

This document defines the architecture, permissions, integrations, workflows, communication systems, and educational collaboration capabilities for AcademIQ's in-house realtime communication infrastructure powered by LiveKit.

This system is NOT simply a video/chat feature.

It is a platform-wide educational collaboration infrastructure designed to power realtime interaction across the entire AcademIQ ecosystem.

The infrastructure must support:

- tutoring
- study circles
- study sessions
- ONE Channels
- academic collaboration
- instructor office hours
- counseling
- oral defenses
- integrity review sessions
- peer support
- cohort communication
- AI-powered interactions
- institutional collaboration

The infrastructure must be:

- scalable
- role-aware
- institution-aware
- permission-oriented
- AI-ready
- mobile-first
- educationally optimized
- audit-capable

2. Core Infrastructure Components

The system must support:

3. Platform-Wide Integration Areas

The realtime infrastructure must integrate directly into all major AcademIQ systems.

3.1 Tutoring Infrastructure

Capabilities:

- 1-on-1 tutoring
- group tutoring
- collaborative whiteboards
- live annotation
- assignment walkthroughs
- session recording
- tutor moderation tools
- attendance tracking
- tutoring analytics
- shared notes
- linked session files

3.2 Study Circles

Capabilities:

- persistent collaboration rooms
- group voice/video communication
- collaborative whiteboards
- study resource sharing
- session scheduling
- persistent circle chat
- peer collaboration
- moderator permissions
- shared session history

3.3 Study Sessions

Capabilities:

- temporary or scheduled rooms
- peer-led sessions
- instructor-led sessions
- hand raise system
- live collaboration
- attendance tracking
- structured agendas
- optional recordings

3.4 ONE Channels

Capabilities:

- persistent educational communication
- contextual conversations
- linked academic resources
- session continuity
- collaborative support
- integrated live sessions
- role-aware communication
- academic organization

3.5 Instructor Office Hours

Capabilities:

- waiting room queues
- scheduled sessions
- private student rooms
- institution-linked sessions
- session analytics
- attendance tracking

3.6 Counseling & Wellness

Capabilities:

- encrypted communication
- optional no-record mode
- confidentiality enforcement
- therapist controls
- wellness-safe communication

3.7 Integrity Review & Oral Defenses

Capabilities:

- mandatory recording support
- restricted participant access
- moderator controls
- audit logging
- identity verification
- evidence presentation
- session tracking

4. Live Chat Infrastructure (CRITICAL)

The AcademIQ live chat system is a foundational educational communication layer integrated across the entire platform.

This is NOT a generic messaging system.

The chat infrastructure must support:

- educational collaboration
- tutoring communication
- institution-linked messaging
- persistent ONE Channels
- realtime study collaboration
- AI-assisted educational communication
- structured academic support

4.1 Chat Types

4.2 Messaging Capabilities

Required:

- realtime messaging
- typing indicators
- read receipts
- reactions
- threaded replies
- pinned messages

- voice notes
- file sharing
- image sharing
- searchable history
- document previews
- persistent history
- session-linked messages
- role-aware permissions

Future Ready:

- AI-generated summaries
- AI suggested replies
- AI translation
- AI moderation
- contextual recommendations

4.3 Educational Context Awareness

Chats must support contextual academic organization.

Messages may be linked to:

- course
- institution
- assignment
- tutoring session
- study circle
- cohort
- assessment
- dispute case
- academic program

This enables:

- intelligent organization
- continuity
- AI contextual understanding
- improved searchability

4.4 Permission-Oriented Messaging

Messaging permissions must support:

- institution restrictions
- age-based restrictions
- instructor moderation
- tutoring-only access
- temporary room access
- blocked users
- external communication limitations
- file-sharing restrictions

4.5 Chat Moderation & Safety

Required:

- reporting system
- mute controls
- blocking system
- institution moderation visibility
- abuse detection
- harmful content flagging
- audit logging

4.6 Message Persistence & Continuity

Chats must support:

- persistent history
- cross-device synchronization
- reconnect handling
- archived conversations
- searchable records
- cloud synchronization

4.7 Live Session Chat Integration

Realtime sessions must support integrated live chat including:

- instructor announcements
- participant discussion

- private moderator chat
- collaborative file sharing
- timestamp-linked discussion

4.8 Emma & Assist Integration

The communication infrastructure must support future AI integrations including:

- Emma onboarding threads
- Assist educational support
- AI tutoring assistance
- AI summaries
- AI reminders
- AI recommendations
- intervention escalation

5. User Roles

6. Permission Architecture

Permissions must be dynamically assigned.

6.1 Core Session Permissions

6.2 Institution-Level Overrides

Institutions must be able to:

- disable recordings
- restrict external participants
- disable screen sharing
- disable file sharing
- require identity verification
- enforce moderation rules
- enforce retention policies
- restrict AI features

7. Educational Whiteboard System (CRITICAL)

The whiteboard is a CORE educational collaboration component.

Must support:

- realtime drawing
- stylus support
- math notation
- text insertion
- shapes
- diagram creation
- PDF annotation
- slide annotation
- image insertion
- multi-page boards
- export functionality
- persistent history
- collaborative editing

7.1 Educational Whiteboard Features

Required:

- graph plotting
- equation support
- chemistry diagrams
- annotation layers
- instructor lock mode
- participant follow mode
- live cursor collaboration

Future Ready:

- AI-generated diagrams
- AI tutoring overlays
- AI-assisted explanations

8. File Sharing Infrastructure

Must support:

- PDFs

- slides
- images
- Word documents
- spreadsheets
- code files
- audio/video files

Capabilities:

- drag-and-drop uploads
- previews
- permission-based downloads
- persistent file history
- session-linked attachments
- institution storage rules

9. Session Recording System

Must support:

- audio recording
- video recording
- screen recording
- whiteboard capture
- chat capture
- timestamped events

9.1 Recording Policies

10. Shared Notes System

Capabilities:

- collaborative notes
- export notes
- session-linked notes
- persistent note history
- ONE Channel integration

11. Hand Raise System

Must support:

- participant queues
- instructor prioritization
- moderation controls

12. Attendance & Engagement Tracking

Track:

- join/leave times
- participation duration
- speaking activity
- engagement metrics
- session participation

13. Session Analytics

Analytics may include:

- attendance
- engagement
- session duration
- tutoring effectiveness
- participation trends
- communication activity

14. AI Integration Readiness

Infrastructure must support future Emma & Assist integrations.

Future capabilities:

- realtime AI tutoring
- AI voice assistants
- AI summaries
- AI-generated notes
- AI-generated study guides
- AI moderation alerts
- AI translation

- AI accessibility support

15. Security & Compliance

Required:

- secure authentication
- encrypted communication
- role-based authorization
- institution isolation
- audit logs
- abuse reporting
- moderation tools
- device/session tracking

16. Mobile Requirements

Must fully support:

- iOS
- Android
- tablets

Requirements:

- adaptive layouts
- low bandwidth mode
- reconnect handling
- background audio
- mobile screen sharing
- push notifications

17. Scalability Requirements

System must support:

- thousands of concurrent users
- institution-scale deployments
- regional optimization
- distributed infrastructure

- failover systems
- dynamic room scaling

18. UI/UX Direction

Design must align with the AcademIQ ecosystem:

- modern academic aesthetic
- clean collaboration UI
- mobile-first design
- dark mode optimized
- educational workflows prioritized
- low-friction interaction

19. Future Expansion Readiness

Architecture must remain extensible for:

- virtual classrooms
- AI tutors
- webinars
- live courses
- livestreaming
- collaborative labs
- realtime translation
- VR/AR learning environments

20. Development Priority Order

Phase 1

- realtime chat
- audio/video
- permissions
- screen sharing
- session persistence
- mobile optimization

Phase 2

- whiteboards
- file sharing
- recordings
- attendance tracking
- analytics

Phase 3

- AI integrations
- breakout rooms
- collaborative notes
- transcription

Phase 4

- immersive educational systems
- AI voice tutoring
- advanced collaboration tooling

21. Final Development Directive

This infrastructure is NOT a standalone communication feature.

It is a foundational realtime educational collaboration layer powering the entire AcademIQ ecosystem.

All development decisions must prioritize:

- educational usability
- institutional scalability
- collaboration continuity
- permission-oriented architecture
- mobile accessibility
- AI readiness
- educational engagement
- long-term extensibility

The system must be designed to support millions of educational interactions across students, instructors, institutions, support staff, and future AI-powered educational systems.

AI Infra & Educational Intelligence Architecture

AcademIQ AI Infrastructure & Educational Intelligence Architecture

1. Introduction

This document defines the long-term AI infrastructure, orchestration strategy, and proprietary educational intelligence direction for AcademIQ.

The purpose of this document is to ensure that all AI systems within AcademIQ are built using a scalable, provider-agnostic, institutionally flexible, and future-proof architecture.

This document serves as a governing systems reference for:

- AI engineers
- backend engineers
- infrastructure architects
- platform engineers
- future machine learning teams
- institutional deployment teams

This document is NOT a feature specification.

It defines:

- how AI exists within AcademIQ
- how AI systems communicate
- how providers are orchestrated
- what infrastructure AcademIQ should own
- how proprietary educational intelligence evolves over time

SECTION 1 — AI INFRASTRUCTURE & ORCHESTRATION STRATEGY

2. Strategic AI Philosophy

AcademIQ should NOT attempt to become a frontier foundation model company.

Instead, AcademIQ should become:

an AI orchestration + educational intelligence platform.

The long-term competitive advantage of AcademIQ is NOT:

- raw LLM weights
- generalized intelligence

- competing with OpenAI, Google, or Anthropic

The long-term competitive advantage is:

- educational continuity
- support infrastructure
- institutional workflow intelligence
- ONE Channel continuity
- Real Talk infrastructure
- Emma orchestration
- reinforcement systems
- educational memory systems
- human + AI collaboration architecture
- educational routing systems

The model layer is infrastructure.

The workflow layer is the moat.

3. AI Infrastructure Evolution Strategy

Phase 1 — Pilot Stage

Primary AI Model:

- DeepSeek V4 Pro

Primary Objectives:

- validate infrastructure
- gather educational interaction data
- reduce operational burn
- refine UX workflows
- test institutional adoption
- validate thread continuity systems

DeepSeek Responsibilities:

- Emma pilot interactions
- onboarding guidance

- reinforcement generation
- ONE Channel continuity
- thread summaries
- moderation assistance
- support recommendations
- backend automation
- internal tooling

Goals:

- cost-efficient scaling
- rapid iteration
- infrastructure validation
- workflow experimentation

Phase 2 — Hybrid Infrastructure Transition

Additional Provider:

- Gemini 3.1 Pro

Gemini Responsibilities:

- multimodal reasoning
- PDF analysis
- image understanding
- screenshot analysis
- institutional document processing
- advanced reasoning tasks
- premium Emma workflows
- voice/video interactions

DeepSeek Responsibilities Continue:

- bulk inference
- thread continuity
- summaries
- reinforcement systems
- backend orchestration
- moderation support

- automation layers

Benefits:

- lower operational costs
- improved scalability
- multimodal expansion
- enterprise flexibility
- provider redundancy

Phase 3 — Intelligent AI Routing Architecture

AcademIQ evolves into a true AI orchestration platform.

Different models are dynamically selected depending on:

- task type
- cost
- latency
- modality
- institutional preference
- confidence thresholds
- compliance requirements

Example Routing Logic:

4. Unified AI Service Layer

Core Requirement

The frontend must NEVER communicate directly with AI providers.

AcademIQ must implement:

- ONE centralized AI orchestration layer
- provider abstraction architecture
- configurable routing engine
- model fallback support
- provider redundancy systems

This is a critical architectural requirement.

Unified AI Layer Responsibilities

Provider Abstraction

Responsibilities:

- isolate provider-specific APIs
- prevent infrastructure lock-in
- simplify provider switching
- normalize provider responses

Supported Providers:

- DeepSeek
- Gemini
- OpenAI
- GLM
- future frontier providers
- future internal models

AI Routing Engine

Responsibilities:

- task-based routing
- cost optimization
- latency optimization
- provider failover
- institution-specific routing
- multimodal delegation
- confidence-based escalation

Prompt Management System

Responsibilities:

- centralized prompt storage
- version control
- dynamic prompt injection

- institution-specific customization
- testing & experimentation

Context Injection Layer

Responsibilities:

- inject thread continuity
- inject educational history
- inject support context
- inject institutional metadata
- inject role permissions
- inject personalization layers

Memory & Continuity Layer

Responsibilities:

- ONE Channel continuity
- educational memory
- conversational continuity
- reinforcement history
- behavioral trend awareness
- contextual support persistence

AI Observability & Logging

Responsibilities:

- model monitoring
- latency monitoring
- failure tracking
- hallucination analysis
- routing visibility
- provider performance analytics
- institutional audit logs

Cost Governance System

Responsibilities:

- provider cost tracking
- inference optimization
- caching systems
- intelligent model selection
- institutional usage analytics
- API spend monitoring

5. Provider-Agnostic Architecture

AcademIQ must remain AI-provider agnostic.

The platform should never become:

- DeepSeek-dependent
- Gemini-dependent
- OpenAI-dependent

Benefits:

- future flexibility
- infrastructure resilience
- pricing leverage
- easier migration
- reduced vendor risk
- rapid experimentation

6. Failover & Redundancy Systems

The orchestration layer must support:

- provider failover
- degraded-mode inference
- retry systems
- latency-aware routing
- emergency provider switching

Example:
If Gemini experiences downtime:

- workflows automatically fallback

- continuity remains uninterrupted
- institution operations continue

7. Long-Term Infrastructure Direction

Potential Future Expansion:

- partial self-hosting
- sovereign AI infrastructure
- institution-hosted deployments
- regional inference nodes
- private institutional inference
- hybrid cloud architecture

Potential Institutional Demand:

- FERPA-sensitive deployments
- private inference
- regional data localization
- institution-owned infrastructure

SECTION 2 — EDUCATIONAL INTELLIGENCE & PROPRIETARY AI VISION

8. What AcademIQ Should Actually Own

AcademIQ's proprietary value is NOT:

- generalized intelligence
- frontier reasoning engines
- raw LLM weights

AcademIQ's proprietary value IS:

- educational workflows
- support continuity
- educational routing systems
- institutional intelligence
- reinforcement systems
- educational memory
- ONE Channel continuity

- Real Talk infrastructure
- Emma orchestration
- human + AI collaboration systems
- thread intelligence

9. Educational Intelligence Layer

This becomes the true product layer of AcademIQ.

The Educational Intelligence Layer sits ON TOP of foundation models.

Core Educational Intelligence Systems

Educational Memory Engine

Tracks:

- student continuity
- learning progression
- support history
- behavioral patterns
- academic context

Reinforcement Engine

Responsibilities:

- motivational reinforcement
- habit reinforcement
- academic encouragement
- engagement continuity
- adaptive messaging

Engagement Prediction Layer

Potential Responsibilities:

- disengagement detection
- support escalation prediction
- inactivity analysis
- participation forecasting

Burnout Detection System

Potential Responsibilities:

- burnout pattern detection
- support recommendation escalation
- emotional exhaustion indicators
- intervention recommendations

Educational Personalization Engine

Responsibilities:

- adaptive support
- personalized reinforcement
- personalized recommendations
- study optimization
- workflow personalization

Thread Intelligence Infrastructure

Responsibilities:

- thread continuity
- conversation understanding
- relationship persistence
- contextual support awareness
- support graph intelligence

Institutional Intelligence Systems

Responsibilities:

- institutional workflow analysis
- retention analytics
- support trends
- engagement metrics
- institutional insights

10. Long-Term Proprietary Model Strategy

AcademIQ should NOT attempt to build:

- a GPT competitor
- a generalized frontier model

Instead:
AcademIQ should eventually develop specialized educational intelligence systems.

Potential Future Internal Models

Examples:

- Academic Support Model
- Reinforcement Model
- Educational Classification Engine
- Student Success Prediction Layer
- Moderation Intelligence Layer
- Support Escalation Models
- Study Recommendation Engine
- Real Talk Wellness Routing Layer
- Institutional Workflow Models

11. Valuable Long-Term Data Infrastructure

Potential Long-Term Proprietary Assets:

- educational support patterns
- reinforcement effectiveness
- tutoring continuity
- engagement trends
- support escalation trends
- academic recovery pathways
- learning continuity maps
- anonymized educational behavior patterns

This becomes:
proprietary educational interaction infrastructure.

12. Ethical & Governance Requirements

AcademIQ must build responsibly.

Especially because the platform intersects with:

- education
- wellness
- student support
- institutional systems

Required Governance Systems

Consent Architecture

Requirements:

- explicit user consent
- institution-controlled permissions
- transparent data usage
- revocation capabilities

FERPA-Conscious Infrastructure

Requirements:

- educational data protection
- access controls
- audit logging
- institutional boundaries
- data isolation

Explainability Requirements

Requirements:

- explainable recommendations
- transparent AI behavior
- institution visibility
- traceable workflows

Auditability

Requirements:

- immutable logs
- workflow traceability
- inference history
- institutional oversight

Institutional Controls

Requirements:

- configurable permissions
- institution AI policies
- escalation controls
- AI enable/disable controls
- moderation configuration

13. Human + AI Collaboration Architecture

AcademIQ should never become:
fully AI-dependent.

The platform must maintain:

- human oversight
- instructor involvement
- counselor participation
- escalation systems
- support intervention capabilities

AI should support:
human educational ecosystems.

Not replace them.

14. Smartest Long-Term Positioning

AcademIQ should become:

an AI orchestration + educational intelligence company.

NOT:
another foundation model company.

15. Long-Term Strategic Recommendation

Short-Term

Use:

- DeepSeek
- Gemini

Medium-Term

Build:

- orchestration infrastructure
- continuity systems
- educational intelligence layers
- proprietary workflows
- memory systems

Long-Term

Develop:

- specialized educational models
- reinforcement systems
- classification engines
- moderation systems
- predictive educational systems

WHILE continuing to leverage frontier APIs where appropriate.

16. Why Hybrid Ownership Is the Optimal Strategy

Hybrid ownership allows AcademIQ to:

- benefit from frontier advancements
- remain flexible
- avoid provider lock-in

- reduce infrastructure rigidity
- optimize operational costs
- support institution-specific deployments
- rapidly adapt to AI industry evolution

17. Final Philosophy

Models will commoditize.

Educational infrastructure will not.

The long-term moat of AcademIQ is:

- continuity
- orchestration
- institutional intelligence
- educational workflows
- support ecosystems
- memory systems
- reinforcement systems
- human + AI integration

That is the long-term strategic direction of AI inside AcademIQ.

ONE Thread Infrastructure System

AcademIQ ONE Thread Infrastructure System

DevSpec Document

1. Introduction

This document defines the complete infrastructure architecture for the AcademIQ ONE Thread system.

ONE Thread is one of the foundational operating systems of the AcademIQ platform and serves as the primary continuity, communication, educational support, and contextual intelligence layer across the ecosystem.

This document establishes:

- persistent thread infrastructure
- educational continuity architecture
- conversational persistence
- role-aware participation systems
- AI continuity integration
- institutional governance
- moderation inheritance
- thread visibility systems
- searchable continuity
- reinforcement integration
- media persistence
- quick messaging logic
- educational support continuity

This document should be treated as:

- a master infrastructure document
- a platform-wide systems reference
- a dependency for multiple future infrastructure systems

Including:

- Emma Infrastructure
- Discovery Engine
- Messaging Infrastructure
- Reinforcement Infrastructure
- Academic Support Infrastructure

- Study Circle Infrastructure
- Institution Ecosystem Infrastructure

2. Core Philosophy of ONE Thread

Traditional communication systems are fragmented.

Users:

- lose educational continuity
- restart conversations repeatedly
- lose support history
- lose reinforcement progression
- disconnect from academic workflows
- lose contextual relationship memory

AcademIQ is designed differently.

ONE Thread is NOT:

- generic messaging
- isolated chat rooms
- temporary communication
- Slack-style channels

ONE Thread IS:

- persistent educational continuity
- relationship continuity
- support continuity
- academic workflow continuity
- AI-assisted continuity
- contextual educational infrastructure

The goal is to create:
a persistent educational relationship ecosystem.

3. What a ONE Thread Actually Is

A ONE Thread is:

a persistent contextual communication object that maintains:

- educational continuity
- relationship continuity
- support continuity
- academic progression
- conversational memory
- reinforcement continuity
- AI continuity
- media persistence
- institutional relevance

A ONE Thread may exist between:

- student ↔ student
- student ↔ tutor
- student ↔ counselor
- student ↔ instructor
- instructor ↔ instructor
- study circle participants
- cohorts
- institution groups
- Emma ↔ user

4. ONE Thread vs Quick Messaging

Quick Messaging

Quick Messaging exists INSIDE the ONE Thread infrastructure.

It is NOT a separate ecosystem.

Purpose:

- lightweight communication
- rapid interaction
- low-friction engagement
- immediate educational coordination

Examples:

- “Joining study session?”
- “Can you send notes?”
- “Are you available?”
- “Running late.”

Characteristics:

- lightweight
- conversational
- rapid-response focused
- optional persistence
- thread-linked continuity

Quick Messages may:

- remain lightweight

OR

- evolve into deeper persistent thread continuity

ONE Thread

Purpose:

- long-term educational continuity
- support continuity
- academic relationship continuity
- reinforcement continuity
- persistent contextual interaction

Characteristics:

- persistent
- searchable
- AI-aware
- institution-aware

- continuity-preserving
- reinforcement-enabled
- context-aware

5. Core Objectives

The ONE Thread system must support:

- persistent educational continuity
- conversational continuity
- role-aware participation
- AI continuity
- educational support
- institutional governance
- reinforcement integration
- searchable history
- moderation inheritance
- media persistence
- recommendation systems
- scalable infrastructure

6. Persistent Thread Object Architecture

Every ONE Thread should function as a structured infrastructure object.

Core Thread Object

Identity Layer

- thread_id
- thread_type
- institution_id (optional)
- cohort_id (optional)
- linked_support_objects
- thread_visibility_type

Participant Layer

- participant_ids
- participant_roles
- verification_states

- institution_affiliations
- trust_scores
- permission_levels

Context Layer

- academic_context
- support_context
- educational_tags
- course_associations
- continuity_metadata

AI Layer

- Emma_memory_context
- reinforcement_history
- AI_summary_objects
- contextual_embeddings
- recommendation_metadata

Communication Layer

- quick_messages
- persistent_messages
- media
- voice_notes
- reactions
- pinned_resources

Moderation Layer

- moderation_flags
- escalation_states
- report_history
- institution_rules
- trust_and_safety_status

Analytics Layer

- engagement_metrics
- continuity_scores
- participation_patterns
- support_effectiveness

7. Thread Types

The platform must support multiple thread classifications.

Academic Support Threads

Examples:

- student ↔ tutor
- student ↔ instructor
- student ↔ counselor

Peer Support Threads

Examples:

- accountability partnerships
- peer tutoring
- collaborative study groups

Study Circle Threads

Persistent continuity tied to:

- study circles
- recurring study sessions
- collaborative groups

Institutional Threads

Examples:

- department discussions
- cohort continuity
- institution advising
- academic recovery groups

Emma Threads

Persistent AI-guided continuity threads.

Examples:

- onboarding continuity
- reinforcement continuity
- support guidance
- academic progression

8. Thread Visibility Architecture

Every thread must support configurable visibility states.

This allows users and institutions to control:

- discoverability
- participation
- privacy
- moderation exposure
- academic boundaries

Public Threads

(Open to all)

Purpose:

- broad collaboration
- discoverability
- public academic engagement

Characteristics:

- searchable
- recommendation eligible
- joinable by eligible users
- AI-indexed
- moderation monitored

Examples:

- FAFSA Questions
- Calculus Help
- Career Advice
- Real Talk Public

Network-Only Threads

(Private within your network)

Purpose:

- trusted communication
- collaborative continuity
- safer interactions

Characteristics:

- only visible to accepted network members
- limited discovery exposure
- relationship-based participation

Examples:

- private study groups
- tutor communities
- accountability circles

Selected People Threads

(User-selected participants)

Purpose:

- highly controlled continuity
- private collaboration
- sensitive educational support

Characteristics:

- invitation-only
- manually selected participants
- hidden from discovery
- permission-controlled

Examples:

- tutoring groups
- counseling continuity
- project teams
- academic recovery groups

Institution-Only Threads

Restricted to:

- institution members
- departments
- programs
- cohorts
- courses

Examples:

- institution advising
- course discussions
- department support

Role-Restricted Threads

Restricted by:

- role
- permissions
- verification type

Examples:

- instructor-only
- counselor-only
- tutor collaboration

9. Thread Permission Infrastructure

Each thread should contain:

- visibility_type
- participant_permissions
- institution_scope
- network_scope
- role_restrictions
- invitation_rules
- moderation_level
- searchable_status
- recommendation_eligibility
- AI_visibility_permissions

10. Persistent Continuity Infrastructure

ONE Thread must preserve:

- educational continuity
- reinforcement continuity
- support continuity
- conversational continuity

Continuity should survive:

- semester changes
- device changes
- inactivity
- course transitions
- institutional progression

Continuity Features

Persistent History

- searchable interactions
- contextual retrieval
- reinforcement history

Relationship Persistence

- recurring educational continuity
- support relationship preservation

Educational Persistence

- course continuity
- academic progression continuity

11. Conversational Memory Architecture

ONE Thread must support:

- contextual conversational memory
- reinforcement memory
- educational continuity memory

Memory Types

Short-Term Memory

Recent interactions.

Long-Term Memory

Persistent educational context.

Reinforcement Memory

Tracks:

- motivational continuity
- intervention effectiveness
- engagement reinforcement

Institutional Memory

Institution-linked academic workflows.

12. Cross-Role Participation Logic

ONE Threads must support:

- multi-role participation
- role-aware permissions
- institutional restrictions

Supported Roles

- students
- instructors
- counselors
- tutors
- peer tutors
- institution admins
- Emma AI

Role Logic Examples

Student ↔ Counselor

- wellness protections enforced
- escalation enabled

Student ↔ Instructor

- academic integrity protections
- institution oversight

Student ↔ Student

- moderation protections
- trust scoring
- age protections

13. Institutional Thread Infrastructure

Institutions must support:

- institution-linked threads
- institution moderation authority
- institution governance controls
- institution analytics visibility

Institution Features

Institution-Branded Threads

Visual differentiation.

Institution Governance

Institution policies enforced.

Institution Moderation

Escalation authority.

Institutional Analytics

Engagement visibility.

14. AI Continuity Integration

ONE Thread is deeply integrated with:

- Emma
- reinforcement systems
- continuity engines
- Discovery Engine

AI Responsibilities

Summarization

Contextual thread summaries.

Continuity Awareness

Persistent support context.

Reinforcement

Motivational continuity.

Recommendation Systems

Suggest:

- tutors
- study circles
- resources
- support pathways

15. Emma Integration Layer

Emma acts as:

- continuity assistant
- educational guide
- support orchestrator

Emma must:

- understand thread context
- understand educational progression
- understand support continuity

Emma Capabilities

- contextual retrieval
- onboarding continuity
- educational guidance
- support routing
- reinforcement messaging
- academic recommendations

16. Reinforcement Integration

ONE Thread is deeply tied to:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement Features

Encouragement Tracking

Tracks support effectiveness.

Motivation Continuity

Maintains reinforcement consistency.

Academic Recovery Support

Supports struggling students.

Behavioral Reinforcement

Encourages productive habits.

17. Media & File Persistence

ONE Thread must support:

- file sharing
- voice notes
- screenshots
- recordings
- collaborative resources
- academic materials

File Infrastructure Requirements

Persistent Storage

Files tied to thread continuity.

AI Indexing

Searchable educational resources.

Moderation Scanning

Automated scanning for:

- harmful content
- abuse
- policy violations

18. Search & Retrieval Architecture

ONE Thread must support:

- semantic search
- contextual retrieval
- AI-assisted retrieval
- institution-aware filtering

Search Categories

- messages
- quick messages
- files
- summaries
- assignments

- resources
- reinforcement history

19. Notification Infrastructure

Notification systems must support:

- role-aware notifications
- reinforcement reminders
- institutional notices
- study session alerts
- AI-generated reminders

Notification Types

Real-Time

Immediate interaction alerts.

Reinforcement

Motivational reminders.

Academic

Deadlines and assignments.

Institutional

Official institution notices.

20. Thread Safety, Reporting & Moderation Infrastructure

Every thread must support:

- reporting
- moderation escalation
- safety enforcement
- institution review systems

This is especially critical because AcademIQ includes:

- minors
- wellness discussions
- peer interaction
- counseling support
- Real Talk infrastructure

Report Options

Users may report:

- harassment
- bullying
- abuse
- threats
- inappropriate behavior
- spam
- impersonation
- harmful content
- academic misconduct

Report Routing

Reports may route to:

- AI moderation
- human moderation
- institution moderation
- counselor escalation
- trust & safety teams

depending on:

- severity
- thread type
- institution policies
- participant age groups

Thread Restriction States

Threads may become:

- restricted
- read-only
- temporarily locked
- escalated
- institution-reviewed
- hidden pending review

Emergency Escalation

High-risk threads may trigger:

- urgent moderation review
- institution notification
- wellness escalation
- trust & safety intervention

Especially relevant for:

- Real Talk
- wellness discussions
- minors
- crisis-related interactions

21. Thread Lifecycle States

Every thread should support lifecycle states.

Example States

- active
- archived
- dormant
- restricted
- escalated
- institution-locked
- pending review
- deleted

22. Study Circle & Session Integration

Study circles automatically generate:
persistent ONE Threads.

These maintain:

- attendance continuity
- shared resources
- session history
- collaborative continuity
- reinforcement progression

23. Academic Support Integration

ONE Thread directly powers:

- tutoring continuity
- counseling continuity
- instructor continuity
- peer accountability
- academic recovery support

This becomes the primary educational continuity layer.

24. Thread Analytics Infrastructure

The system should track:

- engagement
- participation
- support effectiveness
- continuity effectiveness
- educational interaction patterns

Institutional Analytics

Institutions may view:

- aggregate engagement

- retention indicators
- support utilization
- continuity metrics

Subject to:

- permissions
- FERPA controls
- privacy governance

25. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of persistent threads
- AI indexing
- semantic embeddings
- scalable notifications
- distributed storage
- low-latency retrieval
- high concurrency

26. FERPA / Privacy / Governance Requirements

ONE Thread must maintain:

- educational privacy protections
- institutional governance
- consent systems
- auditability
- explainability

Required Governance Features

Data Governance

Clear ownership boundaries.

Audit Logs

Immutable interaction tracking.

Consent Controls

User-controlled permissions.

Institution Policies

Institution-level governance rules.

27. Future Expansion Architecture

Future capabilities may include:

- AI-generated study maps
- continuity intelligence graphs
- collaborative whiteboards
- multimodal continuity
- real-time translation
- advanced educational memory systems
- AI-generated meeting summaries

28. Final Engineering Recommendations

ONE Thread should be treated as:

- a platform operating system
- not a messaging feature

This infrastructure powers:

- educational continuity
- support ecosystems
- reinforcement systems
- AI orchestration
- institutional intelligence
- academic relationship persistence

Almost every major AcademIQ system depends on this architecture.

The dev team should prioritize:

- continuity preservation
- provider-agnostic AI integration
- moderation inheritance
- institutional governance
- thread persistence
- extensibility
- scalability
- safety systems
- role-aware infrastructure

Because ONE Thread is one of the foundational systems that defines how AcademIQ operates as an educational ecosystem.

Thread Lifecycle, Status Manag & Retention Arch

AcademIQ Thread Lifecycle, Status Management & Retention Architecture

Requirements & Developer Specification

1. Purpose

This document defines how threads are managed, organized, retained, archived, searched, and accessed throughout their lifecycle within AcademIQ.

The objective is to create a clean, scalable, and user-friendly lifecycle management system while preserving every thread as part of the student's permanent academic support history.

2. Core Principle

Threads are permanent academic records.

Threads serve as the official system of record for all AcademIQ services and interactions.

Thread statuses exist for organization and workflow management only.

Thread statuses never affect retention.

3. Permanent Thread Retention Policy

All AcademIQ threads shall be retained indefinitely.

Threads are never automatically deleted because of:

- Age
- Inactivity
- Service completion
- Storage limitations
- User inactivity

Threads remain part of the student's permanent academic support history.

4. Pre-Thread Workflow

Before a thread exists, users complete an outline.

Examples:

- Tutoring Outline
- Study Circle Outline
- Study Session Outline
- Advising Outline

The outline may be saved and edited before submission.

No thread is created until the outline is submitted.

Once submitted:

- Service record created
- Thread created
- Status assigned

5. Thread Status Types

The platform shall support the following thread statuses.

Upcoming

Service has been scheduled but has not yet started.

Examples:

- Tutoring appointment scheduled next week
- Study Session scheduled tomorrow
- Advising appointment scheduled next month

Characteristics:

- Visible in Upcoming
- Messaging enabled
- Resources may be shared
- Participants may prepare

Active

Service is currently active.

Examples:

- Ongoing tutoring engagement
- Active study circle
- Active advising relationship

Characteristics:

- Fully searchable
- Supports messaging
- Supports recordings
- Supports attendance
- Supports uploads

Completed

Primary service has concluded.

Examples:

- Tutoring session completed
- Study Session completed
- Advising session completed

Characteristics:

- Searchable
- Accessible
- Historical record preserved

Archived

Thread has been archived.

Examples:

- Historical tutoring engagements
- Completed semester study circles
- Older advising records

Characteristics:

- Searchable
- Viewable
- Downloadable
- Reportable
- Available in archive library

Reopened

Previously archived or completed thread becomes active again.

Examples:

- Student books another tutoring session
- Study circle resumes
- Advisor schedules follow-up session

Characteristics:

- Historical records preserved
- Thread returns to Active status

6. Thread Status Transitions

Tutoring

Upcoming

↓

Active

↓

Completed

↓

Archived

↓

Reopened

↓

Active

Study Session

Upcoming

↓

Active

↓

Completed

↓

Archived

Study Circle

Active

↓

Archived

↓

Reopened

↓

Active

Advising

Upcoming

↓

Active

↓

Completed

↓

Archived

↓

Reopened

↓

Active

7. Status Badges

Every thread shall display a visible status badge.

Examples:

- Upcoming
- Active
- Completed
- Archived
- Reopened

Status badges should appear in:

- Thread Library
- Search Results
- Thread Header
- Notifications
- Analytics

8. Thread Library Organization

The Thread Library shall support filtering by status.

Student View

- All Threads
- Upcoming
- Active
- Completed
- Archived
- Saved

Instructor View

- Upcoming
- Active

- Completed
- Archived

Institution View

- Active Services
- Completed Services
- Archived Services

9. Thread Searchability

All threads remain searchable regardless of status.

Search includes:

- Active Threads
- Completed Threads
- Archived Threads
- Saved Threads

Thread status shall never remove a thread from search.

10. Archive Behavior

Archived threads remain:

- Searchable
- Viewable
- Downloadable
- Bookmarkable
- Reportable

Archived threads continue serving as historical records.

Archived threads are not deleted.

11. Reopening Threads

Authorized users may reopen threads.

Student Example

Student books another tutoring session.

Result:

Archived thread becomes Active.

Instructor Example

Instructor restarts a Study Circle.

Result:

Archived thread becomes Active.

Advisor Example

Advisor schedules a follow-up meeting.

Result:

Archived thread becomes Active.

12. Pinned Threads

Users may pin important threads.

Examples:

- Primary tutor thread
- Scholarship advising thread
- Active study circle

Pinned threads appear above all status categories.

Pinning is independent of lifecycle status.

13. Saved Threads

Users may save threads for quick access.

Examples:

- Tutoring Threads
- Study Sessions
- Advising Threads
- Mentorship Threads

Saved threads appear under:

Library → Saved

Saving a thread does not change its status.

14. Thread Status Audit Log

Every status change shall be recorded.

Store:

- Status Change ID
- Previous Status
- New Status
- Trigger Source
- User
- Timestamp

Examples:

Upcoming → Active

Active → Completed

Completed → Archived

Archived → Reopened

15. Asset Retention Policy

Threads are permanent.

Assets may be managed separately.

Examples:

- Recordings
- Videos
- Audio
- PDFs
- Presentations
- Large attachments

Storage policies apply only to assets.

Storage policies never apply to thread retention.

16. Permanent Thread Components

The following shall remain permanently available.

Thread Metadata

- Thread Name
- Thread ID
- Subject
- Topic
- Service Type
- Institution
- Participants
- Status History

Outline Information

- Original Outline
- Goals
- Objectives

Communication

- Messages
- Announcements
- Notes

Academic Records

- Attendance
- Session Summaries
- Host Notes
- Emma Recommendations
- Feedback

Search Metadata

- Tags
- Dates
- Activity History

17. Emma Integration

Emma should consider thread status when generating recommendations.

Examples:

“You have an archived Calculus tutoring thread that may be useful as you prepare for Calculus II.”

“You previously completed a FAFSA advising thread. Consider scheduling a scholarship advising session.”

18. Analytics

Track:

- Upcoming Threads
- Active Threads
- Completed Threads
- Archived Threads
- Reopened Threads

Analytics should be available based on permissions.

Users:

- Students
- Instructors
- Advisors
- Institutions
- AcademIQ Administration

19. Notifications

Users should receive notifications when significant status changes occur.

Examples:

- Upcoming → Active

- Active → Completed
- Completed → Archived
- Archived → Reopened

20. Future Expansion

Future workflow statuses may include:

- Awaiting Action
- Awaiting Student Response
- Awaiting Instructor Review
- Awaiting Upload
- Pending Approval

These should be implemented as workflow states rather than lifecycle statuses.

21. Final Lifecycle Principle

Every service begins with an outline.

Every submitted outline creates a thread.

Every thread becomes part of the student’s permanent academic support history.

Threads are never automatically deleted.

Threads remain searchable throughout their lifecycle.

Thread statuses exist solely for organization, reporting, and user experience.

AcademIQ threads serve as the permanent system of record for all academic support services.

Service, Outline, Thread & Repository Archit

AcademIQ Service, Outline, Thread & Knowledge Repository Architecture

Developer Specification

1. Purpose

This document defines the architecture governing all academic support services within AcademIQ.

The objective is to create a unified ecosystem where every academic interaction is structured, searchable, recorded, retained, and accessible through a single source of truth.

Rather than creating isolated appointments, chats, or events, AcademIQ treats every service as part of a student's long-term academic support journey.

2. Core System Principle

Every service follows the same lifecycle:

Outline → Thread → Service Activity → Recording → Transcript → Searchable Knowledge Repository

No service may exist without:

- A completed outline
- A linked thread
- Service records
- Searchable history

The thread serves as the permanent source of truth.

3. Services Covered

The following services must follow this architecture:

- Tutoring
- Study Circles
- Study Sessions
- Academic Advising
- Financial Advising
- Career Advising
- Mentorship
- Institution Support Services
- Sponsored Study Sessions
- Future AcademIQ Support Services

4. Outline Requirement

Every service begins with an outline.

The outline establishes:

- Purpose
- Expectations
- Goals
- Preparation requirements
- Desired outcomes
- Context for participants

The system shall not allow service creation, publication, registration, or booking until the outline is completed.

5. Tutoring Outline

Required before booking tutoring.

Required Fields

- Subject
- Course
- Topic(s) needing support
- Current challenges
- Current grade (optional)
- Desired outcome
- Assignment details
- Files and attachments
- Urgency level
- Additional notes

Example

Subject:
Calculus

Topic:
Derivatives

Goal:
Prepare for Midterm Exam

Expected Outcome:
Improve understanding of derivatives and related problem-solving techniques.

6. Study Circle Outline

Required before creating a Study Circle.

Required Fields

- Circle name
- Subject
- Purpose
- Topics covered
- Meeting frequency
- Membership expectations
- Study goals
- Attachments

Example

Name:
Calculus Midterm Study Circle

Purpose:
Prepare for Exam 1

Topics:
Limits, Derivatives, Optimization

7. Study Session Outline

Required before creating a Study Session.

Applicable to:

- Students
- Tutors
- Instructors
- Advisors
- Institutions
- AcademIQ Staff

Required Fields

- Session title
- Host
- Subject
- Topics covered
- Learning objectives
- Expected outcomes
- Capacity
- Visibility settings
- Materials needed
- Date and time

8. Thread Naming Standard

Thread names must be automatically generated by the system using information provided in the outline.

Users should not manually create thread names.

Examples

Tutoring

- Calculus - Derivatives - Tutoring
- Biology - Cellular Respiration - Tutoring
- Chemistry - Stoichiometry - Tutoring

Study Sessions

- Calculus - Midterm Review - Study Session
- Biology - Genetics Review - Study Session

Study Circles

- Calculus - Exam 1 Prep - Study Circle
- Organic Chemistry - MCAT Prep - Study Circle

Advising

- Academic Advising - Course Selection
- Financial Advising - FAFSA Planning
- Career Advising - Medical School Planning

The purpose of descriptive naming is to improve search, organization, analytics, and knowledge retrieval.

9. Thread Architecture

Every service must have a thread.

Every outline must be attached to a thread.

Every service activity must be recorded within the thread.

Every thread serves as the permanent record for that service.

10. Thread Contents

Each thread stores:

- Original outline
- Messages
- Announcements
- Shared resources
- Uploaded files
- Session notes
- Session summaries
- Attendance records
- Session recordings
- Transcripts
- Emma recommendations
- Feedback
- Follow-up actions
- Status history

11. Tutoring Thread Structure

A tutoring request generates a tutoring thread.

Example:

Calculus - Derivatives - Tutoring

The thread contains:

- Outline
- Tutor communication
- Student communication

- Shared resources
- Session recordings
- Session notes
- Attendance records
- Recommendations
- Feedback

12. Study Circle Thread Structure

Each Study Circle receives a dedicated thread.

Example:

Calculus - Exam 1 Prep - Study Circle

The thread contains:

- Discussions
- Resources
- Announcements
- Polls
- Notes
- Recordings
- Summaries
- Recommendations

13. Study Session Thread Structure

Each Study Session receives a dedicated thread.

Example:

Calculus - Midterm Review - Study Session

The thread contains:

- Session outline
- Registration records
- Session chat
- Questions
- Session recording
- Transcript
- Session summary
- Follow-up materials

14. Session Recording Requirement

All live sessions must be recorded.

This applies to:

- Tutoring Sessions
- Study Sessions
- Academic Advising
- Financial Advising
- Career Advising
- Mentorship Sessions
- Sponsored Sessions

Recording begins automatically when the session starts.

Recording ends automatically when the session ends.

Recordings are automatically attached to the associated thread.

15. Recording Metadata

Store:

- Recording ID
- Thread ID
- Service ID
- Session ID
- Recording URL
- Recording Duration
- Host
- Date
- Transcript Status
- Creation Timestamp

16. AI Transcription Requirement

Every recording must generate a transcript.

Benefits include:

- Accessibility
- Searchability

- Knowledge retrieval
- Session review
- Emma recommendations

17. Session Summaries

Every recorded session should generate:

AI Summary

Including:

- Topics discussed
- Questions asked
- Key takeaways
- Resources shared
- Suggested next steps

Host Notes

Hosts may add:

- Recommendations
- Observations
- Action items

Both are stored inside the thread.

18. Attendance Tracking

All sessions should track attendance.

Store:

- Participant
- Join Time
- Leave Time
- Attendance Duration
- Participation Metrics

Attendance records become part of the thread history.

19. Thread Library

AcademIQ shall provide a centralized Thread Library.

Users can access:

- Active Threads
- Saved Threads
- Archived Threads
- Institution Threads
- Global Threads
- Sponsored Threads

20. Bookmarking

Users may bookmark:

- Threads
- Recordings
- Resources
- Files
- Summaries
- Messages

Bookmarked items appear in:

Library → Saved

21. Save to Library

Users may save:

Entire Threads

Including:

- Outline
- Messages
- Files
- Recordings
- Notes
- Summaries
- Recommendations

Individual Assets

Including:

- PDFs
- Resources
- Recordings
- Transcripts
- Messages
- Summaries

22. Global Search

AcademIQ should support platform-wide search.

Searchable content includes:

Threads

- Title
- Description
- Participants
- Tags

Messages

- Messages
- Announcements
- Discussions

Outlines

- Tutoring outlines
- Study Circle outlines
- Study Session outlines

Files

- PDFs
- Notes
- Presentations

- Study Guides

Recordings

- Titles
- Metadata
- Transcript Content

Summaries

- AI Summaries
- Host Notes

23. Searchable Metadata

Every thread should store:

- Thread ID
- Subject
- Topic
- Service Type
- Course
- Institution
- Creator
- Host
- Semester/Term
- Tags
- Status
- Creation Date
- Last Activity Date

All metadata should be indexed for search.

24. Transcript Search

Students should be able to search keywords inside transcripts.

Example:

Search:
“Chain Rule”

Results:

- Calculus Tutoring Recording
- Calculus Study Session Recording
- Study Circle Discussion
- Session Summary
- Related Resources

Results should support timestamp navigation.

25. Emma Integration

Emma should utilize:

- Outlines
- Attendance
- Thread activity
- Recordings
- Transcripts
- Summaries
- Engagement data

Example:

“Based on your tutoring outline and recent activity, I recommend joining the upcoming Calculus Optimization Study Session.”

26. Visibility Architecture

Institution Mode

Display:

- Institution Name
- Institution Logo
- Institution Personnel
- Institution Services

Example:

SDSU - Calculus - Derivatives - Tutoring

Global Mode

Display:

- AcademIQ Branding
- Global Tutors
- Public Services
- Global Study Circles

Example:

Global - Calculus - Derivatives - Tutoring

Sponsored Sessions

Sponsored sessions must be visible in both Institution and Global environments without requiring users to switch modes.

27. Permissions

Thread access should be role-based.

Possible participants:

- Student
- Instructor
- Tutor
- Academic Advisor
- Financial Advisor
- Career Advisor
- Institution Administrator
- Parent (where permitted)
- AcademIQ Staff

Permissions should control:

- Viewing
- Messaging
- Uploading
- Downloading
- Recording Access
- Editing Rights

28. Analytics

Track:

- Attendance
- Engagement
- Resource Usage
- Recording Views
- Thread Activity
- Learning Outcomes

Analytics should be available according to permissions.

29. Knowledge Repository Architecture

Threads are not temporary conversations.

Threads become permanent academic knowledge repositories.

Each thread accumulates:

- Outlines
- Messages
- Files
- Resources
- Notes
- Recordings
- Transcripts
- Summaries
- Recommendations
- Attendance
- Feedback

Over time, every thread becomes a searchable academic knowledge asset.

30. Future Expansion

Any future service added to AcademIQ must follow:

Outline → Thread → Activity → Recording → Transcript → Searchable Repository

No exceptions.

31. Final System Principle

Every service begins with an outline.

Every service lives inside a thread.

Every session is recorded.

Every recording is transcribed.

Every thread is searchable.

Every thread can be bookmarked.

Every thread can be saved to a personal library.

Every thread becomes a long-term academic knowledge repository.

The thread serves as the permanent source of truth for the student’s academic support experience.

🟧 Emma Conversational Intelligence Infras System

AcademIQ Emma Conversational Intelligence Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for Emma, the conversational intelligence and educational continuity system of AcademIQ.

Emma is not a simple chatbot.

Emma functions as:

- an educational continuity assistant
- an AI orchestration layer
- a contextual guidance system
- a support routing engine
- a reinforcement intelligence system
- a continuity-aware conversational assistant
- an educational workflow assistant

Emma serves as one of the foundational intelligence layers of AcademIQ and operates across:

- onboarding
- ONE Threads
- academic support
- reinforcement systems
- study systems
- scheduling systems
- institutional workflows
- educational personalization

This document defines:

- conversational intelligence architecture
- contextual memory systems
- educational continuity systems
- AI orchestration logic
- reinforcement infrastructure
- support escalation pathways
- role-aware guidance systems
- institutional governance boundaries

- moderation boundaries
- ethical AI controls

This document should be treated as:

- a master AI infrastructure document
- a conversational intelligence systems reference
- a dependency for multiple AcademIQ systems

Including:

- ONE Thread Infrastructure
- Discovery Engine Infrastructure
- Human Feedback & AI Reinforcement Infrastructure
- Institution Ecosystem Infrastructure
- Scheduling Infrastructure
- Study Circle Infrastructure
- Onboarding Infrastructure

2. Core Philosophy of Emma

Traditional AI assistants are often:

- transactional
- short-term
- context-limited
- productivity-focused
- generic

Emma is fundamentally different.

Emma is designed to provide:

- educational continuity
- academic support continuity
- contextual educational guidance
- reinforcement continuity
- personalized educational assistance
- support orchestration
- intelligent educational routing

Emma should feel:

- supportive
- context-aware
- educationally intelligent
- continuity-aware
- human-centered
- institutionally safe

The goal is NOT:

- replacing instructors
- replacing counselors
- replacing human educational systems

The goal IS:

- strengthening educational ecosystems
- improving continuity
- improving support accessibility
- improving educational organization
- supporting human-led educational environments

3. What Emma Actually Is

Emma is:

an AI-powered educational continuity and orchestration system that intelligently assists users throughout their educational journey.

Emma continuously evaluates:

- educational context
- support history
- thread continuity
- academic progression
- scheduling data
- onboarding preferences
- engagement patterns
- reinforcement effectiveness

- institutional context

Emma then assists users through:

- conversational guidance
- recommendations
- support routing
- continuity awareness
- educational personalization
- reinforcement systems

4. Core Objectives

Emma must support:

- conversational continuity
- educational personalization
- reinforcement continuity
- intelligent support routing
- role-aware guidance
- onboarding continuity
- academic support assistance
- scheduling assistance
- continuity-aware recommendations
- institution-safe AI assistance
- scalable conversational infrastructure

5. Emma System Architecture

Emma should operate as a layered conversational intelligence system.

Layer 1 — Conversational Interface Layer

Handles:

- user interaction
- conversational responses
- voice interactions
- multimodal communication
- thread participation

Layer 2 — Context & Memory Layer

Maintains:

- conversational memory
- educational continuity
- reinforcement history
- support context
- onboarding history
- thread awareness

Layer 3 — Educational Intelligence Layer

Processes:

- educational recommendations
- support pathways
- continuity intelligence
- engagement awareness
- academic assistance
- reinforcement logic

Layer 4 — AI Orchestration Layer

Routes requests to:

- DeepSeek
- Gemini
- future AI providers
- internal educational intelligence systems

Layer 5 — Governance & Safety Layer

Enforces:

- institution rules
- moderation boundaries
- FERPA protections
- safety systems
- ethical AI controls

6. Emma Conversational Infrastructure

Emma should support:

- persistent continuity
- thread-aware conversations
- educational context awareness
- conversational memory
- role-aware communication
- institutional-safe responses

Supported Conversation Types

Academic Guidance

Examples:

- study planning
- assignment organization
- course navigation
- educational recommendations

Support Guidance

Examples:

- tutoring recommendations
- counselor recommendations
- study circle recommendations
- accountability support

Reinforcement Conversations

Examples:

- motivational support
- productivity reinforcement
- engagement encouragement
- academic recovery support

Institutional Guidance

Examples:

- onboarding assistance
- institution navigation
- program discovery
- scheduling support

7. Conversational Memory Architecture

Emma must maintain:

- continuity-aware memory
- educational memory
- reinforcement memory
- support continuity memory

Memory Types

Short-Term Conversational Memory

Tracks recent interactions.

Long-Term Educational Memory

Tracks:

- academic goals
- educational interests
- support pathways
- continuity patterns

Reinforcement Memory

Tracks:

- motivational history
- reinforcement effectiveness
- engagement interventions

Institutional Memory

Tracks:

- institution-specific workflows
- institution policies
- onboarding continuity

8. Thread-Aware Intelligence Infrastructure

Emma must deeply integrate with:
ONE Thread Infrastructure.

Emma should understand:

- thread context
- thread relationships
- educational continuity
- participant roles
- support progression
- collaboration history

Emma Capabilities Inside Threads

Examples:

- summarize threads
- recommend resources
- suggest study circles
- provide reinforcement
- suggest next actions
- recommend support escalation
- organize continuity

9. Onboarding Intelligence Infrastructure

Emma should play a major role during onboarding.

Emma Onboarding Responsibilities

Examples:

- guide onboarding flow
- explain platform features
- recommend threads
- recommend communities
- recommend study circles
- personalize educational pathways
- initialize support systems

Emma Introductory Thread

Every user may receive:

- a persistent Emma onboarding thread
- onboarding continuity
- educational setup assistance
- guided discovery support

This becomes the user’s first continuity layer inside AcademIQ.

10. Educational Guidance Infrastructure

Emma should assist users with:

- organization
- continuity
- productivity
- educational planning
- support discovery
- academic navigation

Educational Guidance Examples

Study Assistance

Examples:

- study recommendations
- session planning
- focus guidance
- reinforcement reminders

Academic Planning

Examples:

- scheduling assistance
- workload organization
- assignment prioritization
- deadline awareness

Support Discovery

Examples:

- tutoring recommendations
- counselor suggestions
- peer support recommendations
- study circle discovery

11. Reinforcement Infrastructure

Emma must deeply integrate with:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement Responsibilities

Examples:

- motivational support
- habit reinforcement
- engagement encouragement
- academic recovery support
- productivity reinforcement

Reinforcement Awareness

Emma should evaluate:

- engagement trends
- continuity gaps
- inactivity

- support participation
- study behavior

12. Support Escalation Infrastructure

Emma should intelligently route users toward human support systems when appropriate.

Escalation Pathways

Examples:

- tutor recommendations
- counselor recommendations
- instructor guidance
- peer support recommendations
- institutional support routing

Escalation Signals

Potential signals:

- repeated academic struggles
- disengagement patterns
- burnout indicators
- support requests
- educational frustration

13. Emotional & Wellness Routing Infrastructure

Emma may support:

- wellness-aware educational guidance
- emotional support routing
- Real Talk continuity assistance

However:
Emma is NOT a therapist.

Wellness Boundaries

Emma should:

- encourage human support
- route users toward counselors
- escalate risk situations
- avoid clinical advice
- avoid diagnostic behavior

Crisis Escalation

High-risk interactions may trigger:

- emergency escalation workflows
- trust & safety review
- institution escalation
- counselor escalation

Subject to:

- institution policies
- age protections
- privacy rules

14. Role-Aware Guidance Infrastructure

Emma must adapt dynamically based on user role.

Student Guidance

Examples:

- study support
- tutoring recommendations
- organization assistance
- reinforcement support

Instructor Guidance

Examples:

- course organization

- student support insights
- scheduling support
- educational continuity insights

Counselor Guidance

Examples:

- support continuity awareness
- escalation visibility
- educational support insights

Tutor Guidance

Examples:

- session continuity
- reinforcement insights
- study support organization

15. Discovery Engine Integration

Emma should deeply integrate with:
Discovery Engine Infrastructure.

Emma Recommendation Capabilities

Emma may recommend:

- threads
- communities
- cohorts
- tutors
- counselors
- study circles
- educational resources
- opportunities

Based on:

- educational context
- onboarding data
- continuity history
- reinforcement systems
- engagement behavior

16. Scheduling & Productivity Integration

Emma should integrate with:
Scheduling Infrastructure.

Scheduling Capabilities

Examples:

- reminder generation
- study planning
- office hours reminders
- tutoring scheduling
- availability awareness
- session recommendations

17. Study Circle & Session Integration

Emma should assist with:

- study circle continuity
- recurring session organization
- collaborative support
- accountability reinforcement

Study Circle Assistance Examples

Examples:

- recommending circles
- summarizing sessions
- generating reminders
- surfacing resources
- encouraging participation

18. AI Orchestration Infrastructure

Emma should NEVER directly depend on one AI provider.

Emma must operate through:
Unified AI Service Layer.

AI Routing Examples

DeepSeek

Used for:

- summaries
- reinforcement
- lightweight conversations
- continuity tasks

Gemini

Used for:

- multimodal reasoning
- PDFs
- screenshots
- advanced reasoning
- voice/video workflows

Future Provider Support

Emma must support:

- provider abstraction
- routing flexibility
- fallback systems
- future AI providers
- internal educational intelligence models

19. Moderation & Safety Infrastructure

Emma must integrate deeply with:

- Trust & Safety Infrastructure
- institution governance
- moderation systems

Safety Systems

Emma should support:

- moderation-aware responses
- restricted content handling
- escalation triggers
- abuse detection awareness
- institution-specific moderation policies

Unsafe Content Handling

Emma should:

- avoid harmful guidance
- avoid unsafe escalation
- redirect users appropriately
- escalate severe situations

20. Institutional Governance Infrastructure

Institutions must maintain governance control over Emma.

Institution Controls

Institutions may configure:

- AI visibility
- AI participation permissions
- moderation rules
- escalation pathways
- institution AI policies
- support routing permissions

21. Analytics Infrastructure

Emma should track:

- engagement trends
- reinforcement effectiveness
- support routing outcomes
- educational continuity metrics
- productivity interactions
- onboarding effectiveness

Institutional Analytics

Institutions may view:

- aggregate engagement
- support utilization
- continuity trends
- educational participation

Subject to:

- FERPA protections
- permissions
- privacy governance

22. FERPA / Privacy / Governance Requirements

Emma must maintain:

- educational privacy protections
- consent systems
- auditability
- explainability
- institution governance

Required Governance Features

Consent Controls

Users control:

- AI participation

- memory permissions
- personalization settings

Audit Logs

The system should maintain:

- AI interaction history
- escalation visibility
- moderation traceability

Explainability

Emma should support:

- understandable recommendations
- transparent support routing
- institutional visibility

23. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of conversations
- persistent continuity memory
- semantic retrieval
- AI orchestration
- scalable embeddings
- distributed memory systems
- low-latency interactions
- multimodal workflows

24. Future Expansion Architecture

Potential future capabilities:

- advanced educational memory systems
- predictive educational assistance
- burnout-aware reinforcement
- adaptive educational pathways

- multilingual Emma support
- multimodal educational continuity
- institutional AI copilots
- collaborative AI support systems

25. Final Engineering Recommendations

Emma should be treated as:

- a continuity intelligence system
- an educational orchestration layer
- a support infrastructure system
- not a generic chatbot

The goal is to optimize:

- educational continuity
- support accessibility
- reinforcement continuity
- productivity
- institutional support systems
- educational personalization

The dev team should prioritize:

- conversational continuity
- memory infrastructure
- provider-agnostic orchestration
- reinforcement integration
- moderation-aware AI behavior
- institution governance
- educational context awareness
- scalable conversational systems

Because Emma becomes one of the primary intelligence systems responsible for orchestrating continuity across the entire AcademIQ ecosystem.

Academic Memory Search

Emma Thread Discovery & Academic Memory Search (AMS)

Purpose

Emma shall function not only as a recommendation engine but also as the primary navigation and retrieval layer for AcademIQ's knowledge repository.

Students should not be required to manually locate threads, recordings, transcripts, resources, or summaries when Emma can retrieve them through natural language requests.

Emma serves as the student's academic concierge, helping users quickly locate information across their entire AcademIQ history.

Academic Memory Search (AMS)

AcademIQ shall include an Academic Memory Search (AMS) system accessible through Emma.

AMS enables Emma to search across all authorized academic content and return relevant results using natural language.

Users should not be required to know:

- Exact thread names
- Session dates
- Tutor names
- Resource locations
- Recording titles

Emma should identify and retrieve the requested content automatically.

Searchable Sources

Emma shall be capable of searching:

Threads

- Thread titles
- Thread descriptions
- Thread tags
- Thread metadata

Outlines

- Tutoring outlines
- Study Circle outlines
- Study Session outlines
- Advising outlines

Messages

- Student messages
- Instructor messages
- Tutor messages
- Advisor communications
- Announcements

Files

- PDFs
- Study guides
- Presentations
- Notes
- Uploaded resources

Recordings

- Recording titles
- Recording metadata
- Recording summaries

Transcripts

- Transcript content
- Timestamped transcript segments

Summaries

- AI-generated summaries
- Instructor notes
- Tutor notes
- Advisor notes

Saved Content

- Saved threads
- Saved resources
- Saved recordings
- Saved files
- Saved summaries

Bookmarks

- Thread bookmarks
- Resource bookmarks
- Recording bookmarks

Example Student Requests

Find Thread

Student:

“Find my derivatives tutoring thread.”

Emma:

“I found the following thread:

• Calculus - Derivatives - Tutoring

Would you like me to open it?”

Find Multiple Related Threads

Student:

“Find everything related to derivatives.”

Emma returns:

• Calculus - Derivatives - Tutoring

• Calculus - Midterm Review - Study Session

• Calculus - Exam 1 Prep - Study Circle

• Derivatives Study Guide

• Derivatives Recording Library

Find By Course

Student:

“Show me all my Calculus support.”

Emma retrieves:

- Tutoring threads
- Study sessions
- Study circles
- Resources
- Saved content

Find By Instructor

Student:

“Show me all sessions with Professor Smith.”

Emma returns:

- Threads
- Recordings
- Resources
- Notes
- Summaries

Find By Date

Student:

“Show me the tutoring session before my Calculus midterm.”

Emma identifies:

- Relevant thread
- Session date
- Recording
- Notes
- Resources

Find Information Inside Recordings

Student:

“Find where my tutor explained the Chain Rule.”

Emma searches transcripts and returns:

Thread:
Calculus - Derivatives - Tutoring

Recording:
Session 3

Timestamp:
34:17

Summary:
Chain Rule explanation and examples.

Find Saved Content

Student:

“Find the derivatives study guide I saved.”

Emma searches:

- Saved files
- Saved resources
- Saved threads
- Saved recordings

Permission-Based Search

Emma may only retrieve information the user has permission to access.

Examples:

Student may access:

- Their own tutoring threads
- Their own advising sessions
- Their own saved content

Instructor may access:

- Authorized course threads
- Student support records where permitted

Institution administrators may access:

- Institution-level data according to permissions

Emma must always respect the platform permission model.

Search Ranking Logic

Results should prioritize:

1. Exact matches
2. Recent activity
3. Saved items
4. Bookmarked items
5. Frequently accessed content
6. High-engagement resources

This ensures the most relevant information appears first.

Emma as Academic Navigator

Emma should provide direct navigation actions.

Examples:

- Open Thread
- Open Recording
- Open Transcript
- Open Resource
- Open Summary
- Open Saved Item

The goal is to reduce manual navigation throughout the platform.

Future Expansion

Future versions of AMS may support:

- Cross-course knowledge retrieval
- Learning pattern analysis
- Personalized study recommendations
- Academic history timelines
- Predictive resource suggestions

Final Principle

Emma should function as the primary discovery, retrieval, and navigation layer of the AcademIQ ecosystem.

Students should be able to locate academic information through natural language rather than manually searching folders, libraries, threads, recordings, or resources.

Academic Memory Search transforms AcademIQ from a collection of stored information into an intelligent academic knowledge system.

🟥 Discovery Engine Infrastructure System

AcademIQ Discovery Engine Infrastructure System

DevSpec Document

1. Introduction

This document defines the complete infrastructure architecture for the AcademIQ Discovery Engine.

The Discovery Engine is one of the foundational intelligence systems of the AcademIQ ecosystem.

Its purpose is to intelligently connect users with:

- people
- threads
- study circles
- tutors
- counselors
- instructors
- institutions
- communities
- cohorts
- opportunities
- support systems
- educational resources
- sessions
- academic pathways

The Discovery Engine is not simply a search system.

It is:

- a contextual educational intelligence layer
- a recommendation infrastructure system
- a relationship orchestration engine
- a continuity-aware matching system
- a role-aware discovery architecture

This infrastructure powers:

- onboarding recommendations
- thread recommendations

- support matching
- academic networking
- Emma recommendations
- study circle suggestions
- institutional discovery
- educational personalization

This document should be treated as:

- a master infrastructure document
- an AI-assisted recommendation architecture reference
- a dependency for multiple AcademIQ systems

Including:

- ONE Thread Infrastructure
- Emma Infrastructure
- Academic Support Network (ASN)
- Human Feedback & AI Reinforcement Infrastructure
- Study Circle Infrastructure
- Institution Ecosystem Infrastructure
- Onboarding Infrastructure

2. Core Philosophy of the Discovery Engine

Traditional platforms rely heavily on:

- follower systems
- popularity metrics
- generic social feeds
- engagement addiction loops
- entertainment-driven algorithms

AcademIQ is fundamentally different.

The Discovery Engine exists to optimize:

- educational relevance
- support continuity
- productive collaboration

- academic growth
- healthy educational relationships
- institutional outcomes
- personalized educational ecosystems

The goal is NOT:

- infinite scrolling
- viral engagement
- attention maximization

The goal IS:

- meaningful educational connection
- support accessibility
- continuity-driven discovery
- productive academic interaction
- intelligent educational recommendations

3. What the Discovery Engine Actually Is

The Discovery Engine is:

an intelligent recommendation and retrieval infrastructure system that dynamically connects users to the most relevant educational people, threads, resources, and support opportunities.

The Discovery Engine continuously evaluates:

- educational context
- onboarding data
- role type
- academic needs
- continuity history
- engagement patterns
- institutional relationships
- reinforcement patterns
- support pathways
- thread activity

It then generates:

- personalized recommendations
- educational matches
- contextual discovery opportunities
- support routing suggestions

4. Core Objectives

The Discovery Engine must support:

- educational relevance
- intelligent matching
- role-aware recommendations
- institutional-safe discovery
- onboarding-based personalization
- AI-assisted retrieval
- continuity-aware recommendations
- academic support routing
- personalized educational ecosystems
- scalable recommendation infrastructure

5. Discovery Categories

The system should support discovery across multiple infrastructure layers.

Thread Discovery

Users may discover:

- public threads
- academic support threads
- institution threads
- study circle threads
- Real Talk discussions
- subject-specific threads
- cohort discussions

People Discovery

Users may discover:

- tutors
- counselors
- instructors
- peer tutors
- accountability partners
- study partners
- institution peers

Study Circle Discovery

Users may discover:

- active study circles
- recurring sessions
- collaborative groups
- institution-linked circles
- subject-based circles

Session Discovery

Users may discover:

- tutoring sessions
- office hours
- workshops
- study sessions
- webinars
- live events

Resource Discovery

Users may discover:

- study guides
- notes
- shared resources
- academic materials
- recordings
- recommended learning content

Community Discovery

Users may discover:

- academic communities
- support communities
- institution communities
- subject-based communities
- wellness communities
- Real Talk communities
- career-focused communities
- collaborative learning communities

Cohort Discovery

Users may discover:

- academic cohorts
- institution cohorts
- incoming student cohorts
- program-based cohorts
- graduation cohorts
- study cohorts
- support cohorts
- cohort-based collaboration groups

Institutional Discovery

Users may discover:

- departments
- programs
- institution support systems
- institution events
- institution communities

6. Discovery Engine Architecture

The Discovery Engine should operate as a layered intelligence system.

Layer 1 — Data Collection Layer

Collects:

- onboarding data
- role data
- academic interests
- institution affiliations
- engagement behavior
- support interactions
- thread participation
- reinforcement history
- educational goals

Layer 2 — Educational Context Layer

Builds educational understanding using:

- subjects
- educational level
- academic pathways
- support needs
- study behavior
- continuity history
- institutional context

Layer 3 — Recommendation Intelligence Layer

Processes:

- recommendation ranking
- educational relevance scoring
- support matching
- continuity weighting
- trust scoring
- moderation eligibility

Layer 4 — Delivery Layer

Surfaces recommendations across:

- onboarding
- home feed
- Emma

- threads
- study circles
- profiles
- search systems
- notifications

7. Recommendation Infrastructure

Recommendations must be:

- contextual
- educationally relevant
- continuity-aware
- institution-safe
- role-aware
- reinforcement-aware

Recommendation Categories

Suggested Threads

Examples:

- Calculus Support
- FAFSA Questions
- Study Accountability
- Real Talk

Suggested People

Examples:

- peer tutors
- instructors
- counselors
- accountability partners

Suggested Study Circles

Examples:

- Biology Study Circle
- SAT Prep Circle
- Institution Cohort Circle

Suggested Resources

Examples:

- study materials
- tutoring resources
- recordings
- AI-generated summaries

Suggested Opportunities

Examples:

- internships
- mentorship
- scholarships
- institution programs

8. Educational Relevance Scoring

The Discovery Engine should prioritize educational relevance over popularity.

Relevance Signals

The system may evaluate:

- academic subjects
- educational level
- onboarding preferences
- support needs
- institution affiliation
- thread engagement
- continuity relationships
- reinforcement effectiveness
- educational goals
- study behavior

Weighting Factors

Potential weighting:

- educational match quality
- support relevance
- trust scoring
- institutional alignment
- continuity probability
- moderation status
- verification status

9. Role-Aware Discovery Infrastructure

Discovery must dynamically adapt based on user role.

Student Discovery

Students may discover:

- tutors
- peer tutors
- instructors
- study circles
- academic support threads
- support resources

Instructor Discovery

Instructors may discover:

- students needing support
- peer instructors
- study groups
- institution communities

Counselor Discovery

Counselors may discover:

- support escalation pathways

- academic recovery systems
- wellness-related support opportunities

Tutor Discovery

Tutors may discover:

- tutoring opportunities
- support requests
- relevant educational communities

10. Institution-Safe Discovery Architecture

Institutions must maintain control over:

- visibility
- discovery exposure
- moderation restrictions
- institution boundaries
- approved interactions

Institution Controls

Institutions may configure:

- allowed discovery scopes
- role restrictions
- institution-only recommendations
- moderation filtering
- age restrictions
- visibility permissions

11. Thread Discovery Infrastructure

Thread discovery is one of the core systems of AcademIQ.

Discovery Sources

Threads may be recommended based on:

- onboarding data
- educational interests
- institution affiliation
- continuity patterns
- support needs
- existing network relationships
- reinforcement systems
- Emma recommendations

Visibility Enforcement

Discovery must respect:

- public threads
- network-only threads
- selected people threads
- institution-only threads
- role-restricted threads

Private or restricted threads must NEVER appear in unauthorized discovery systems.

12. Search Infrastructure

The Discovery Engine must support:

- semantic search
- contextual search
- AI-assisted retrieval
- educational filtering
- institution-aware search

Search Categories

Users may search:

- threads
- people
- study circles
- sessions
- institutions
- resources
- courses

- academic support

Search Filters

Examples:

- subject
- semester
- educational level
- institution
- role
- verification type
- language
- support type
- availability

13. AI-Assisted Retrieval Infrastructure

AI systems should assist users in discovering:

- relevant threads
- educational pathways
- study opportunities
- support systems
- institutional resources

Emma Discovery Assistance

Emma may recommend:

- study circles
- support threads
- tutors
- counselors
- resources
- recovery pathways
- reinforcement opportunities

Based on:

- educational context
- engagement patterns
- support history
- academic progression

14. Onboarding-Based Personalization

The Discovery Engine should begin personalization immediately during onboarding.

Onboarding Signals

Examples:

- role type
- support needs
- subjects
- academic goals
- institution
- preferred support style
- study preferences
- tutoring interests

Immediate Recommendation Examples

After onboarding:

- suggested threads
- suggested study circles
- suggested people
- Emma onboarding thread
- institution communities

15. Continuity-Aware Discovery

The Discovery Engine should prioritize continuity.

This means recommendations should consider:

- existing support relationships
- recurring collaboration
- reinforcement effectiveness
- prior educational engagement

- continuity probability

The goal is not random discovery.

The goal is:
productive long-term educational continuity.

16. Reinforcement Integration

The Discovery Engine should integrate with:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement-Aware Discovery

Examples:

- recommending accountability groups
- surfacing productive study habits
- encouraging healthy educational participation
- reinforcing support systems

17. Trust, Safety & Moderation Integration

The Discovery Engine must integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- institution governance

Moderation-Aware Discovery

Discovery systems should evaluate:

- moderation history
- trust scoring
- thread safety
- verification status
- abuse reports
- institution restrictions

Unsafe or restricted content should be:

- hidden
- deprioritized
- removed from recommendations
- escalated when necessary

18. Verification & Credential Integration

The Discovery Engine should prioritize verified educational relationships.

Examples:

- verified instructors
- verified tutors
- institution-verified students
- certified counselors

Verification should influence:

- ranking
- recommendation visibility
- trust weighting
- support recommendations

19. Academic Support Matching Infrastructure

The Discovery Engine should support intelligent educational matching.

Matching Examples

Student ↔ Tutor

Based on:

- subjects
- educational level
- availability
- support style

Student ↔ Counselor

Based on:

- academic support needs
- wellness pathways
- institution rules

Study Partner Matching

Based on:

- shared goals
- courses
- schedules
- study preferences

20. Network-Aware Discovery

Recommendations should evaluate:

- existing network relationships
- trusted educational relationships
- recurring collaboration
- institution affiliation

This creates:

- safer discovery
- stronger continuity
- higher trust interactions

21. Analytics Infrastructure

The Discovery Engine should track:

- recommendation effectiveness
- support outcomes
- educational engagement
- continuity metrics
- participation quality

- academic support interactions

Institutional Analytics

Institutions may view:

- support engagement
- educational participation
- continuity trends
- thread activity
- study circle participation

Subject to:

- permissions
- FERPA controls
- privacy governance

22. AI Routing & Orchestration Integration

The Discovery Engine should integrate with:

- Emma orchestration
- AI routing systems
- recommendation intelligence
- educational memory systems

AI may dynamically:

- rerank recommendations
- adjust support pathways
- prioritize interventions
- optimize educational matching

23. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of users

- real-time recommendations
- semantic embeddings
- AI-assisted retrieval
- low-latency ranking
- distributed indexing
- scalable search systems
- high concurrency

24. FERPA / Privacy / Governance Requirements

The Discovery Engine must maintain:

- educational privacy protections
- institution boundaries
- consent systems
- explainability
- auditability

Required Governance Features

Consent Controls

Users control:

- visibility
- discovery participation
- recommendation eligibility

Institution Governance

Institutions control:

- discovery boundaries
- moderation restrictions
- recommendation permissions

Auditability

Recommendation systems must maintain:

- audit logs
- moderation visibility
- institutional traceability

25. Future Expansion Architecture

Potential future capabilities:

- AI-generated educational pathways
- predictive support recommendations
- burnout-aware recommendations
- adaptive educational ecosystems
- learning graph infrastructure
- continuity intelligence graphs
- multilingual discovery systems
- institution-specific AI ranking systems

26. Final Engineering Recommendations

The Discovery Engine should be treated as:

- a core educational intelligence system
- not a generic recommendation feed

The goal is to optimize:

- educational relevance
- continuity
- support accessibility
- healthy collaboration
- institutional outcomes
- academic growth

The dev team should prioritize:

- educational relevance scoring
- continuity-aware recommendations
- moderation-aware discovery
- institution-safe visibility
- scalable indexing

- semantic retrieval
- provider-agnostic AI integration
- trust-aware ranking systems

Because the Discovery Engine becomes one of the primary systems responsible for connecting the entire AcademIQ ecosystem together.

Messaging & Quick Messaging Infras System

AcademIQ Messaging & Quick Messaging Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Messaging & Quick Messaging system.

Messaging within AcademIQ is not designed to function as a traditional social messaging platform.

Instead, messaging exists to support:

- educational continuity
- academic coordination
- support communication
- study collaboration
- institutional workflows
- reinforcement continuity
- productive educational interaction

This document establishes:

- messaging architecture
- quick messaging infrastructure
- thread-linked communication
- real-time communication systems
- unread state systems
- online and availability systems
- typing indicators
- moderation-aware messaging
- institution-safe communication
- attachment infrastructure
- escalation systems
- communication lifecycle infrastructure

This document should be treated as:

- a master messaging infrastructure document
- a dependency for ONE Thread Infrastructure
- a real-time communication systems reference

Including dependencies with:

- ONE Thread Infrastructure
- Emma Infrastructure
- Discovery Engine Infrastructure
- Trust & Safety Infrastructure
- Institution Ecosystem Infrastructure
- Scheduling Infrastructure

2. Core Philosophy of Messaging

Traditional messaging systems often optimize for:

- endless engagement
- entertainment
- social addiction
- high-volume conversation

AcademIQ messaging is fundamentally different.

Messaging should optimize for:

- productive educational interaction
- support accessibility
- educational continuity
- healthy communication
- academic collaboration
- contextual communication

The goal is NOT:

- replacing educational workflows
- becoming a generic social messenger
- encouraging non-productive engagement

The goal IS:

- supporting educational ecosystems
- enabling efficient communication

- strengthening continuity
- improving academic collaboration

3. Messaging Infrastructure Overview

Messaging within AcademIQ consists of:

Persistent ONE Thread Communication

Long-term continuity communication.

Examples:

- tutoring continuity
- counseling continuity
- instructor communication
- recurring study collaboration

Quick Messaging Infrastructure

Lightweight communication within thread ecosystems.

Examples:

- scheduling coordination
- quick academic questions
- study session coordination
- availability checks

Real-Time Communication Infrastructure

Supports:

- live messaging
- typing indicators
- read states
- online presence
- quick interactions

4. Quick Messaging Infrastructure

Quick Messaging exists INSIDE the ONE Thread ecosystem.

It is NOT a separate communication platform.

Purpose of Quick Messaging

Quick Messaging is designed for:

- rapid coordination
- lightweight communication
- low-friction educational interaction
- quick academic collaboration

Example Quick Messages

Examples:

- “Joining study session?”
- “Can you send notes?”
- “Are you available?”
- “Office hours moved.”
- “Need help with question 3.”

Quick Message Characteristics

Quick Messages should support:

- lightweight delivery
- rapid response
- conversational continuity
- optional persistence
- thread linkage
- moderation scanning

Escalation Into Persistent Continuity

Quick Messages may evolve into:

- persistent educational threads
- tutoring continuity
- study support continuity

- recurring collaboration

This ensures communication continuity remains unified.

5. Persistent Messaging Infrastructure

Persistent messaging powers:

- long-term support continuity
- tutoring relationships
- instructor communication
- counseling continuity
- recurring study collaboration

Persistent communication should maintain:

- searchable history
- contextual continuity
- reinforcement continuity
- educational progression awareness

6. Messaging Object Architecture

Every messaging object should contain:

Identity Layer

- message_id
- thread_id
- sender_id
- participant_ids
- institution_id
- message_type

Content Layer

- message_content
- attachments
- voice_notes
- reactions

- AI-generated summaries

Context Layer

- educational_context
- subject_tags
- support_context
- continuity_metadata

Status Layer

- delivered_state
- read_state
- edited_state
- deleted_state
- moderation_state

Moderation Layer

- moderation_flags
- report_status
- escalation_status
- trust_scoring

7. Real-Time Communication Infrastructure

Messaging must support:

- low-latency delivery
- real-time synchronization
- live interaction systems
- scalable messaging concurrency

Real-Time Features

Typing Indicators

Examples:

- “Emma is typing…”
- “Instructor is typing…”
- “Study group member is typing…”

Online Presence

Examples:

- online
- available
- busy
- in tutoring session
- in study circle
- offline

Read Receipts

Supports:

- delivered state
- seen state
- unread state
- muted state

Presence Synchronization

Presence should dynamically update based on:

- session activity
- study circle participation
- tutoring activity
- scheduling status
- manual status selection

8. Availability & Busy Infrastructure

Availability systems should support:

- educational scheduling awareness
- session conflict awareness

- support accessibility

Example Availability States

Available

Open for communication.

Busy

Examples:

- in tutoring session
- in study circle
- in class
- in counseling session

Away

Temporarily inactive.

Invisible

Hidden presence mode.

9. Messaging Permissions Infrastructure

Messaging permissions must be:

- role-aware
- institution-aware
- moderation-aware
- age-aware

Permission Categories

Direct Messaging Permissions

Who may initiate communication.

Thread Participation Permissions

Who may message within threads.

Institution Messaging Restrictions

Institution-level communication boundaries.

Media Sharing Permissions

Controls attachment sharing.

10. Thread Messaging Integration

Messaging is deeply integrated with:
ONE Thread Infrastructure.

Messaging should preserve:

- continuity
- educational context
- reinforcement history
- support relationships

Messaging Inside Threads

Examples:

- quick messaging
- persistent communication
- collaborative discussion
- tutoring continuity
- reinforcement interaction

11. Attachment & File Infrastructure

Messaging must support:

- document sharing
- screenshots
- recordings
- study resources
- voice notes
- collaborative educational files

Attachment Features

Persistent Storage

Files tied to thread continuity.

AI Indexing

Educational files become searchable.

Institution Restrictions

Institutions may restrict:

- file types
- sharing permissions
- external content

12. Moderation & Safety Infrastructure

Messaging must integrate deeply with:

- Trust & Safety Infrastructure
- institution governance
- moderation systems

Messaging Moderation Systems

Examples:

- abuse detection
- harassment detection
- bullying detection
- inappropriate content detection
- spam detection
- impersonation detection

Message Reporting Infrastructure

Users must be able to:

- flag messages
- report abuse
- report harassment
- report threats
- report spam
- report inappropriate behavior

Report Routing

Reports may route to:

- AI moderation
- human moderation
- institution moderation
- trust & safety review
- emergency escalation systems

13. Thread Restriction & Enforcement Infrastructure

Threads or messaging access may become:

- restricted
- read-only
- muted
- temporarily suspended
- institution-reviewed
- escalated

Enforcement Examples

Examples:

- communication restrictions
- temporary suspensions
- institution intervention
- trust & safety escalation

14. Emergency Escalation Infrastructure

High-risk interactions may trigger:

- emergency moderation review
- institution escalation
- counselor escalation
- trust & safety intervention

Especially relevant for:

- minors
- wellness discussions
- Real Talk
- crisis-related interactions

15. Emma Messaging Integration

Emma should participate intelligently within messaging ecosystems.

Emma Messaging Capabilities

Examples:

- reinforcement reminders
- study reminders
- educational organization
- thread summaries
- support recommendations
- continuity assistance

Emma Messaging Boundaries

Emma should:

- avoid unsafe guidance
- respect institution policies
- avoid replacing human support systems
- escalate appropriately when necessary

16. Scheduling Integration

Messaging should integrate with:
Scheduling Infrastructure.

Scheduling Features

Examples:

- session reminders
- tutoring reminders
- office hours alerts
- study session coordination
- availability synchronization

17. Discovery Engine Integration

Messaging should integrate with:
Discovery Engine Infrastructure.

Discovery-Aware Messaging

Examples:

- thread invitations
- study circle invitations
- support recommendations
- accountability invitations
- onboarding recommendations

18. Institution Governance Infrastructure

Institutions must maintain governance control over messaging.

Institution Controls

Institutions may configure:

- communication permissions
- role restrictions
- moderation rules
- attachment restrictions
- escalation workflows
- visibility controls

19. Analytics Infrastructure

Messaging systems should track:

- communication engagement
- support participation
- continuity effectiveness
- response behavior
- reinforcement effectiveness

Institutional Analytics

Institutions may view:

- aggregate communication trends
- support participation
- educational collaboration metrics
- moderation trends

Subject to:

- FERPA protections
- permissions
- privacy governance

20. FERPA / Privacy / Governance Requirements

Messaging systems must maintain:

- educational privacy protections
- consent systems
- auditability
- explainability
- institution governance

Required Governance Features

Consent Controls

Users control:

- messaging permissions

- visibility settings
- notification preferences

Audit Logs

The system should maintain:

- communication history
- moderation visibility
- escalation traceability

Institution Governance

Institutions may enforce:

- communication boundaries
- moderation policies
- role restrictions

21. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of concurrent messages
- low-latency synchronization
- real-time communication
- distributed messaging infrastructure
- scalable notifications
- high concurrency
- AI-assisted moderation
- semantic indexing

22. Future Expansion Architecture

Potential future capabilities:

- multilingual messaging
- real-time translation
- AI-generated summaries
- collaborative whiteboards

- voice messaging intelligence
- multimodal educational communication
- AI meeting summaries
- institution communication copilots

23. Final Engineering Recommendations

Messaging should be treated as:

- an educational continuity infrastructure system
- not a generic social messaging platform

The goal is to optimize:

- productive communication
- educational continuity
- support accessibility
- academic collaboration
- reinforcement continuity
- institution-safe interaction

The dev team should prioritize:

- thread-linked continuity
- moderation-aware communication
- scalable real-time infrastructure
- institution governance
- reinforcement integration
- AI-assisted messaging systems
- educational context preservation
- safety systems

Because messaging becomes one of the foundational communication systems powering the AcademIQ educational ecosystem.

🟨 Verification & Credential Infra System

AcademIQ Verification & Credential Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Verification & Credential system.

The Verification & Credential Infrastructure exists to establish:

- trust
- authenticity
- institutional credibility
- role legitimacy
- educational transparency
- safer educational interactions

This system powers:

- verification workflows
- institutional verification
- credential validation
- trust scoring
- role authentication
- badge systems
- moderation escalation
- educational identity assurance

The Verification & Credential Infrastructure is foundational to:

- Academic Support Network (ASN)
- ONE Thread Infrastructure
- Discovery Engine Infrastructure
- Institution Ecosystem Infrastructure
- Messaging Infrastructure
- Study Circle Infrastructure
- onboarding systems

This document defines:

- verification workflows
- credential architecture
- badge infrastructure
- trust systems
- moderation escalation
- institutional verification systems
- reverification infrastructure
- verification analytics
- governance controls

2. Core Philosophy of Verification

AcademIQ is an educational ecosystem.

Trust and authenticity are critical because the platform includes:

- students
- minors
- instructors
- counselors
- tutors
- institutions
- wellness-related support
- academic collaboration

Verification is NOT intended to:

- create elitism
- reduce accessibility
- discourage participation

Verification IS intended to:

- improve safety
- improve trust
- improve educational legitimacy
- improve institutional confidence
- improve support quality
- improve moderation visibility

The goal is to create:

- trusted educational relationships
- transparent educational identity systems
- institution-safe participation

3. Verification Infrastructure Overview

The Verification & Credential system should support:

- user verification
- institution verification
- credential verification
- educational verification
- certification validation
- trust scoring
- role legitimacy
- reverification workflows
- moderation escalation

4. Supported Verification Types

The platform should support multiple verification classifications.

Student Verification

Examples:

- verified student
- institution-verified student
- verified peer tutor

Verification methods may include:

- school email verification
- institution roster validation
- student ID review
- institution-issued confirmation
- Transcripts

Instructor Verification

Examples:

- verified instructor
- institution instructor
- certified instructor

Verification methods may include:

- institutional employment validation
- credential review
- institution-issued verification

Tutor Verification

Examples:

- verified tutor
- certified tutor
- institution tutor

Verification methods may include:

- credential uploads
- institution verification
- certification review

Counselor Verification

Examples:

- verified counselor
- licensed counselor
- academic counselor
- therapist verification

Verification methods may include:

- license validation
- institution verification
- manual credential review

Institution Verification

Examples:

- verified institution
- accredited institution

Verification methods may include:

- official documentation
- accreditation review
- institutional agreements

5. Verification Workflow Infrastructure

Verification workflows should support:

- automated verification
- manual review workflows
- institution-assisted verification
- moderation escalation
- reverification systems

Verification Workflow Stages

Stage 1 — Submission

User submits:

- documents
- credentials
- institution information
- identification information

Stage 2 — Validation

The system validates:

- document authenticity
- institution matching
- email verification
- credential structure

Stage 3 — Review

Verification may route to:

- automated systems
- manual moderation review
- institution review
- trust & safety review

Stage 4 — Approval or Denial

Users receive:

- approved status
- denied status
- pending review status
- escalation requests

Stage 5 — Reverification

The system may require:

- periodic reverification
- expiration renewal
- updated documentation

6. Verification States Infrastructure

Every user should maintain verification states.

Example Verification States

- unverified

- pending verification
- verified
- institution-verified
- certified
- partially verified
- expired verification
- suspended verification
- rejected verification
- under review

7. Badge Infrastructure

Verification should integrate with:
visual badge systems.

Badge Types

Examples:

- verified student badge
- institution student badge
- verified instructor badge
- certified tutor badge
- licensed counselor badge
- verified institution badge

Badge Display Infrastructure

Badges should display consistently across:

- profiles
- threads
- messaging
- study circles
- communities
- search results
- Discovery Engine recommendations

Badge Metadata

Each badge may contain:

- badge type
- issuing authority
- expiration status
- institution affiliation
- verification timestamp

8. Credential Upload Infrastructure

The platform must support:

- secure document uploads
- credential validation
- institutional document review
- certification management

Supported Credential Types

Examples:

- teaching certifications
- counselor licenses
- tutoring certifications
- institutional credentials
- academic achievements
- educational licenses

Credential Security Requirements

The system should support:

- encrypted storage
- access restrictions
- moderation review
- institution visibility controls

9. Trust Scoring Infrastructure

The Verification system should integrate with:
Trust & Safety Infrastructure.

Trust Scoring Signals

Potential signals:

- verification status
- institution affiliation
- moderation history
- report history
- educational participation
- support continuity
- credential legitimacy

Trust Score Usage

Trust scoring may influence:

- Discovery Engine visibility
- recommendation ranking
- messaging permissions
- support eligibility
- moderation escalation

10. Institution Verification Infrastructure

Institutions should maintain authority over:

- institution-issued verification
- student legitimacy
- instructor legitimacy
- institutional roles
- cohort participation

Institution Controls

Institutions may:

- verify users
- revoke verification
- issue badges
- review credentials
- restrict participation
- manage institution trust systems

11. Reverification & Expiration Infrastructure

Verification systems must support:

- expiration tracking
- reverification reminders
- institutional updates
- certification renewal

Reverification Examples

Examples:

- expired counseling license
- updated institution enrollment
- instructor contract expiration
- outdated tutor credentials

12. Moderation & Escalation Infrastructure

Verification systems should integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- institution governance

Verification Escalation Examples

Examples:

- fraudulent credentials
- impersonation attempts
- institution disputes
- fake certifications
- suspicious verification behavior

Escalation Routing

Escalations may route to:

- moderation review
- institution review
- trust & safety teams
- manual verification specialists

13. Discovery Engine Integration

Verification status should influence:

- recommendation visibility
- search ranking
- trust weighting
- educational credibility

Discovery Examples

Examples:

- verified instructors prioritized
- institution tutors surfaced higher
- certified counselors prioritized in recommendations

14. ONE Thread Integration

Verification systems should integrate deeply with:
ONE Thread Infrastructure.

Thread Verification Features

Examples:

- verified badges in threads
- trust visibility
- role legitimacy indicators
- institution-linked verification

15. Messaging Integration

Verification status may influence:

- messaging permissions
- communication restrictions
- support routing
- moderation visibility

Messaging Examples

Examples:

- verified counselors allowed wellness communication
- institution instructors allowed institution threads
- restricted messaging for suspicious accounts

16. Community & Cohort Verification Infrastructure

Communities and cohorts may support:

- verification requirements
- institution restrictions
- role-specific participation

Examples

Examples:

- verified medical students only
- institution-only cohorts
- verified tutor communities
- counselor-only collaboration spaces

17. Analytics Infrastructure

The system should track:

- verification approval rates
- trust metrics
- moderation escalations
- reverification trends
- institution verification participation

Institutional Analytics

Institutions may view:

- verified student counts
- instructor verification rates
- support credential metrics
- moderation trends

Subject to:

- permissions
- FERPA protections
- privacy governance

18. FERPA / Privacy / Governance Requirements

Verification systems must maintain:

- educational privacy protections
- secure document handling
- consent systems
- auditability
- institution governance

Required Governance Features

Consent Controls

Users control:

- credential visibility
- verification visibility
- profile display permissions

Audit Logs

The system should maintain:

- verification history
- credential review history
- moderation actions
- institution verification actions

Access Restrictions

Sensitive verification data should support:

- restricted visibility
- encrypted storage
- institution-specific access

19. AI-Assisted Verification Infrastructure

AI systems may assist with:

- document validation
- fraud detection
- credential classification
- moderation assistance
- suspicious behavior detection

However:
final verification authority should support human oversight.

20. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of users
- secure credential storage
- scalable verification review
- distributed moderation systems
- AI-assisted validation
- institution-linked verification
- high concurrency

21. Future Expansion Architecture

Potential future capabilities:

- blockchain credential verification
- institution-issued digital credentials
- cross-institution verification portability
- AI-powered fraud detection
- global educational verification systems
- regional compliance systems
- sovereign institution verification

22. Final Engineering Recommendations

The Verification & Credential system should be treated as:

- a trust infrastructure system
- a legitimacy infrastructure layer
- an institution safety system
- not just a badge feature

The goal is to optimize:

- trust
- authenticity
- educational legitimacy
- safer collaboration
- institutional confidence
- moderation visibility

The dev team should prioritize:

- secure credential handling
- moderation-aware verification
- institution-linked trust systems
- scalable review infrastructure
- provider-agnostic AI validation
- role-aware verification
- reverification systems
- badge consistency

Because the Verification & Credential Infrastructure becomes one of the foundational trust systems powering the AcademIQ ecosystem.

Onboarding Infra System

AcademIQ Onboarding Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Onboarding system.

The Onboarding Infrastructure is one of the foundational operating systems of AcademIQ because it initializes:

- educational identity
- support pathways
- thread continuity
- Discovery Engine personalization
- network relationships
- institutional linkage
- Emma continuity
- educational preferences
- role permissions
- verification workflows

Onboarding is not simply account creation.

It is:

- educational ecosystem initialization
- continuity initialization
- personalization initialization
- support system initialization
- institutional onboarding
- relationship infrastructure initialization

This document defines:

- role selection systems
- onboarding flow architecture
- dynamic onboarding logic
- Emma onboarding infrastructure
- institutional onboarding
- support matching systems
- profile initialization
- cohort linkage

- discovery initialization
- verification onboarding
- privacy initialization
- educational personalization

This document should be treated as:

- a master onboarding infrastructure document
- a dependency for multiple AcademIQ systems
- an educational initialization systems reference

Including:

- ONE Thread Infrastructure
- Discovery Engine Infrastructure
- Emma Infrastructure
- Verification & Credential Infrastructure
- Academic Support Network (ASN)
- Institution Ecosystem Infrastructure
- Human Feedback & AI Reinforcement Infrastructure

2. Core Philosophy of Onboarding

Traditional onboarding systems are often:

- static
- generic
- transactional
- profile-focused only

AcademIQ onboarding is fundamentally different.

The onboarding process should:

- initialize educational continuity
- initialize support systems
- personalize educational experiences
- establish trust systems
- connect users into educational ecosystems
- reduce friction to support access

- initialize long-term continuity infrastructure

The goal is NOT:

- rapidly forcing account creation
- maximizing short-term engagement
- overwhelming users with complexity

The goal IS:

- intelligent educational initialization
- meaningful continuity setup
- support accessibility
- trust initialization
- educational personalization

3. What the Onboarding System Actually Is

The Onboarding Infrastructure is:

an adaptive educational initialization system that dynamically configures a user's educational ecosystem inside AcademIQ.

The system initializes:

- user identity
- educational pathways
- support needs
- thread recommendations
- Discovery Engine personalization
- Emma continuity
- verification workflows
- institutional relationships
- scheduling preferences
- communication preferences

4. Core Objectives

The onboarding system must support:

- dynamic onboarding flows
- role-aware onboarding
- institution-aware onboarding
- support personalization
- continuity initialization
- thread initialization
- Discovery Engine initialization
- Emma onboarding continuity
- verification onboarding
- educational personalization
- scalable onboarding infrastructure

5. Role Selection Infrastructure

The onboarding system must support multiple user roles.

Primary Role Types

Student

Examples:

- general student
- institution student
- peer tutor student

Instructor

Examples:

- general instructor
- institution instructor
- certified instructor

Tutor

Examples:

- peer tutor
- verified tutor
- institution tutor

Counselor

Examples:

- academic counselor
- career counselor
- financial counselor
- therapist
- academic coach
- wellness support

6. Dynamic Onboarding Infrastructure

Onboarding flows should dynamically adapt based on:

- role selection
- institution affiliation
- educational goals
- support needs
- verification status
- age group
- onboarding responses

Dynamic Flow Examples

Student Onboarding

Focuses on:

- support needs
- educational goals
- study preferences
- academic continuity
- cohort matching

Instructor Onboarding

Focuses on:

- credentials

- teaching subjects
- institution linkage
- support offerings
- verification workflows

Counselor Onboarding

Focuses on:

- specialization
- licensing verification
- support pathways
- institution governance

7. Educational Personalization Infrastructure

The onboarding system should initialize:

- educational interests
- academic pathways
- support preferences
- study preferences
- educational goals

Personalization Signals

Examples:

- subjects
- educational level
- learning goals
- support needs
- study style
- communication preferences
- tutoring interests
- scheduling preferences

8. Support Needs Infrastructure

Users should identify support needs during onboarding.

Support Categories

Examples:

- tutoring
- study accountability
- counseling
- career support
- financial aid guidance
- wellness support
- peer support
- academic coaching
- test preparation
- other

Custom Support Input

Users should support:

- “Other” option
- custom input field
- flexible support descriptions

9. Emma Onboarding Infrastructure

Emma should play a central role during onboarding.

Emma Responsibilities

Examples:

- guide onboarding
- explain platform systems
- personalize onboarding
- initialize support systems
- recommend threads
- recommend communities
- recommend study circles
- initialize continuity

Emma Introductory Thread

Every user may receive:

- a persistent Emma onboarding thread
- educational setup guidance
- onboarding continuity
- support assistance
- Discovery Engine recommendations

This becomes the user’s first persistent continuity layer.

10. Verification Onboarding Infrastructure

Verification systems should initialize during onboarding.

Verification Examples

Student Verification

Examples:

- school email verification
- institution verification
- student ID upload

Instructor Verification

Examples:

- credential uploads
- institution validation
- certification review

Counselor Verification

Examples:

- licensing validation
- institution verification
- manual review workflows

Verification Status Initialization

Users may begin as:

- unverified
- pending verification
- institution-verified
- partially verified

11. Institution Onboarding Infrastructure

The onboarding system must support:

- institution-linked onboarding
- institution roster matching
- institution governance initialization
- institution-specific onboarding flows

Institution Initialization Examples

Examples:

- institution branding
- department linkage
- cohort assignment
- institution permissions
- institution-specific recommendations

12. Cohort Initialization Infrastructure

Users may be automatically linked to:

- academic cohorts
- institution cohorts
- incoming student cohorts
- study cohorts
- support cohorts

Cohort Infrastructure Goals

The system should:

- improve continuity
- improve peer connection
- improve onboarding support
- improve educational collaboration

13. Community Initialization Infrastructure

The onboarding system should recommend:

- academic communities
- support communities
- subject communities
- institution communities
- wellness communities
- Real Talk communities

14. Discovery Engine Initialization

Onboarding should initialize:
Discovery Engine personalization.

Initial Recommendation Examples

After onboarding users may receive:

- suggested threads
- suggested communities
- suggested study circles
- suggested tutors
- suggested support systems
- suggested accountability groups

15. ONE Thread Initialization

The onboarding system should initialize:

- educational continuity
- support continuity
- Emma continuity

- thread participation pathways

Initial Thread Examples

Examples:

- Emma onboarding thread
- institution onboarding thread
- cohort thread
- study accountability thread
- Real Talk onboarding thread

16. Profile Initialization Infrastructure

The onboarding system should initialize:

- profile visibility
- badges
- role permissions
- educational preferences
- support preferences
- scheduling preferences

Profile Elements Initialized

Examples:

- verification state
- support interests
- educational interests
- institution affiliation
- availability settings
- onboarding badges

17. Privacy & Permission Initialization

Users should configure:

- profile visibility
- messaging permissions

- thread visibility
- discovery participation
- AI participation permissions
- notification preferences

Visibility Examples

Examples:

- public profile
- network-only visibility
- institution-only visibility
- selected people visibility

18. Messaging Initialization Infrastructure

Onboarding should initialize:

- messaging permissions
- communication preferences
- online visibility
- availability systems
- notification preferences

19. Scheduling Initialization Infrastructure

Users should initialize:

- availability
- preferred study times
- scheduling preferences
- tutoring availability
- office hours visibility

Scheduling Examples

Examples:

- timezone setup
- recurring availability

- busy state defaults
- reminder preferences

20. Reinforcement Initialization Infrastructure

The onboarding system should initialize:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement Initialization Examples

Examples:

- motivational preferences
- study reinforcement preferences
- productivity reminders
- accountability systems

21. Accessibility Infrastructure

The onboarding system should support:

- accessibility preferences
- language preferences
- communication accessibility
- educational accommodations

Accessibility Examples

Examples:

- screen reader support
- caption preferences
- font scaling
- language preferences
- accessibility accommodations

22. Moderation & Safety Initialization

Safety systems should initialize during onboarding.

Safety Initialization Examples

Examples:

- age protections
- moderation boundaries
- institution moderation rules
- trust systems
- reporting education

User Education

Users should receive:

- community guidelines
- trust & safety guidance
- reporting instructions
- privacy explanations

23. AI Governance & Consent Infrastructure

Users should explicitly configure:

- AI participation permissions
- memory permissions
- personalization permissions
- recommendation permissions

Consent Categories

Examples:

- Emma participation
- personalization
- reinforcement systems
- Discovery Engine recommendations
- continuity memory

24. Analytics Infrastructure

The onboarding system should track:

- onboarding completion
- onboarding drop-off
- personalization effectiveness
- support matching effectiveness
- thread participation
- Discovery Engine engagement

Institutional Analytics

Institutions may view:

- onboarding completion rates
- cohort engagement
- support participation
- onboarding effectiveness

Subject to:

- FERPA protections
- permissions
- privacy governance

25. FERPA / Privacy / Governance Requirements

The onboarding system must maintain:

- educational privacy protections
- consent systems
- auditability
- institution governance
- explainability

Required Governance Features

Consent Controls

Users control:

- onboarding personalization

- AI participation
- discovery participation
- messaging permissions

Audit Logs

The system should maintain:

- onboarding history
- verification actions
- consent actions
- moderation actions

Institution Governance

Institutions may configure:

- onboarding requirements
- verification requirements
- moderation rules
- visibility permissions

26. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of onboarding flows
- dynamic onboarding logic
- scalable personalization
- AI-assisted onboarding
- institution-linked onboarding
- distributed onboarding infrastructure
- real-time recommendations

27. Future Expansion Architecture

Potential future capabilities:

- AI-adaptive onboarding
- predictive support initialization

- multilingual onboarding
- institution-specific onboarding copilots
- advanced continuity initialization
- educational pathway mapping
- onboarding intelligence scoring

28. Final Engineering Recommendations

The onboarding system should be treated as:

- an educational initialization system
- a continuity infrastructure system
- a personalization infrastructure layer
- not just account creation

The goal is to optimize:

- educational continuity
- support accessibility
- trust initialization
- onboarding personalization
- educational relationship building
- institutional integration

The dev team should prioritize:

- dynamic onboarding architecture
- continuity initialization
- Discovery Engine personalization
- Emma onboarding integration
- institution-safe onboarding
- scalable onboarding flows
- consent infrastructure
- moderation-aware onboarding

Because onboarding becomes the foundational initialization layer that shapes how every user enters and experiences the AcademIQ ecosystem.

Study Circle & Study Session Infra System

AcademIQ Study Circle & Study Session Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Study Circle & Study Session system.

Study Circles and Study Sessions are foundational collaborative learning systems inside the AcademIQ ecosystem.

These systems are designed to support:

- collaborative learning
- educational continuity
- accountability
- peer support
- tutoring workflows
- instructor-led sessions
- institutional learning communities
- recurring educational collaboration

This infrastructure powers:

- study circles
- study sessions
- collaborative learning groups
- recurring educational communities
- tutoring collaboration
- accountability systems
- session continuity
- AI-assisted collaboration
- educational reinforcement

This document defines:

- study circle architecture
- session lifecycle systems
- scheduling infrastructure
- participant permissions
- recurring session systems
- attendance infrastructure

- moderation systems
- AI summaries
- collaboration tools
- institutional integration
- analytics infrastructure
- recording infrastructure
- educational continuity systems

This document should be treated as:

- a master collaborative learning infrastructure document
- a dependency for multiple AcademIQ systems
- a continuity-driven educational collaboration systems reference

Including:

- ONE Thread Infrastructure
- Messaging Infrastructure
- Discovery Engine Infrastructure
- Emma Infrastructure
- Scheduling Infrastructure
- Institution Ecosystem Infrastructure
- Human Feedback & AI Reinforcement Infrastructure
- Verification & Credential Infrastructure

2. Core Philosophy of Study Circles

Traditional group collaboration systems are often:

- temporary
- unstructured
- disconnected
- socially driven
- continuity-limited

AcademIQ Study Circles are fundamentally different.

Study Circles should optimize:

- educational continuity

- accountability
- productive collaboration
- recurring educational engagement
- support accessibility
- reinforcement continuity
- collaborative academic growth

The goal is NOT:

- casual social groups
- entertainment-focused communication
- temporary collaboration only

The goal IS:

- persistent educational collaboration
- structured study continuity
- support ecosystems
- collaborative accountability
- academic success support

3. What Study Circles Actually Are

A Study Circle is:

a persistent collaborative educational continuity system designed around recurring academic interaction.

Study Circles maintain:

- recurring participation
- educational continuity
- shared resources
- accountability systems
- session history
- reinforcement continuity
- collaborative support
- AI-assisted organization

A Study Session is:

a scheduled collaborative learning interaction occurring within or outside a Study Circle.

4. Core Objectives

The Study Circle system must support:

- recurring educational collaboration
- persistent continuity
- scheduling infrastructure
- attendance tracking
- collaborative communication
- AI-assisted organization
- reinforcement systems
- institutional collaboration
- role-aware permissions
- moderation-aware collaboration
- scalable educational sessions

5. Study Circle Infrastructure Overview

The system should support:

- public study circles
- network-only circles
- institution study circles
- tutoring circles
- instructor-led circles
- peer accountability circles
- subject-specific circles
- cohort-linked circles

Study Session Types

Examples:

- peer study sessions
- tutoring sessions
- instructor office hours
- collaborative review sessions
- accountability sessions
- institution workshops
- exam preparation sessions

6. Study Circle Object Architecture

Each Study Circle should function as a structured infrastructure object.

Identity Layer

- study_circle_id
- institution_id
- linked_thread_id
- creator_id
- visibility_type
- educational_tags

Participant Layer

- participant_ids
- participant_roles
- permission_levels
- verification_states
- attendance_history

Scheduling Layer

- recurring_schedule
- timezone
- upcoming_sessions
- availability_rules
- reminder_settings

Collaboration Layer

- shared_resources
- recordings
- whiteboards
- collaborative_notes
- AI-generated summaries

Moderation Layer

- moderation_flags
- report_history
- escalation_states
- institution_rules

Analytics Layer

- engagement_metrics
- attendance_patterns
- reinforcement_scores
- participation_quality

7. Visibility Infrastructure

Study Circles should support multiple visibility states.

Public Study Circles

Open discovery and participation.

Examples:

- SAT Prep Circle
- Calculus Study Circle
- Real Talk Study Accountability

Network-Only Study Circles

Visible only to accepted network members.

Examples:

- accountability groups
- peer collaboration circles
- private tutoring groups

Selected Participants Circles

Invitation-only circles.

Examples:

- project groups
- tutoring continuity groups
- academic recovery groups

Institution-Only Study Circles

Restricted to:

- institution students
- departments
- programs
- cohorts

8. Study Session Infrastructure

Study Sessions are scheduled educational interactions.

Session Features

The system should support:

- recurring sessions
- one-time sessions
- collaborative sessions
- AI-assisted sessions
- instructor-led sessions
- tutoring sessions
- hybrid participation

Session Capabilities

Examples:

- live collaboration
- video communication
- audio communication
- chat communication
- whiteboards
- file sharing
- collaborative notes

- screen sharing

9. Live Collaboration Infrastructure

The platform should support:

- live video
- live audio
- live messaging
- collaborative whiteboards
- real-time collaboration
- AI-assisted educational tools

Collaboration Tools

Examples:

- whiteboards
- shared notes
- collaborative study boards
- file sharing
- assignment collaboration
- live annotations

10. Attendance Infrastructure

The system should support:

- attendance tracking
- recurring attendance history
- participation analytics
- accountability tracking

Attendance Features

Examples:

- attendance logs
- recurring participation metrics
- late join tracking

- session completion tracking
- institution attendance integration

11. Recurring Session Infrastructure

Study Circles should support recurring educational continuity.

Recurring Session Features

Examples:

- daily sessions
- weekly sessions
- monthly sessions
- recurring tutoring
- recurring accountability groups

Recurring Continuity Goals

The system should improve:

- accountability
- consistency
- educational continuity
- reinforcement effectiveness

12. ONE Thread Integration

Every Study Circle should generate:

- persistent ONE Threads
- continuity-aware communication
- collaborative continuity
- session continuity

Thread Features

Examples:

- session discussions

- shared resources
- AI summaries
- accountability continuity
- reinforcement messaging

13. Messaging Integration

Study Circles should integrate deeply with:
Messaging Infrastructure.

Messaging Features

Examples:

- quick messaging
- session coordination
- reminders
- collaborative discussion
- live session chat

14. Scheduling Integration

Study Circles should integrate deeply with:
Scheduling Infrastructure.

Scheduling Features

Examples:

- recurring scheduling
- timezone handling
- conflict detection
- availability matching
- reminder systems
- office hours integration

15. Emma Integration

Emma should assist with:

- study organization
- continuity reminders

- accountability reinforcement
- session summaries
- study recommendations
- participation encouragement

Emma Capabilities

Examples:

- recommending circles
- generating summaries
- surfacing resources
- reinforcing participation
- identifying continuity gaps

16. Discovery Engine Integration

Study Circles should integrate deeply with:
Discovery Engine Infrastructure.

Discovery Examples

Users may discover:

- active study circles
- recurring sessions
- accountability groups
- institution circles
- tutoring circles

Recommendations may evaluate:

- educational relevance
- continuity probability
- support needs
- scheduling compatibility

17. Reinforcement Infrastructure

Study Circles should integrate with:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement Examples

Examples:

- attendance encouragement
- accountability reminders
- productivity reinforcement
- collaborative participation encouragement

18. Moderation & Safety Infrastructure

Study Circles must integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- institution governance

Moderation Features

Examples:

- reporting
- participant moderation
- abuse detection
- harassment detection
- institution review

Report Options

Users may report:

- harassment
- inappropriate behavior
- spam
- bullying
- academic misconduct
- unsafe interactions

19. Role-Aware Permission Infrastructure

Permissions must support:

- role-aware collaboration
- institution restrictions
- verification requirements
- moderation-aware permissions

Permission Examples

Examples:

- instructor moderators
- tutor session controls
- counselor-only circles
- verified participant requirements

20. Recording & Resource Infrastructure

The system should support:

- session recordings
- collaborative resources
- AI-generated summaries
- searchable educational content

Resource Features

Examples:

- downloadable resources
- AI indexing
- collaborative notes
- session archives
- educational playback

21. Institutional Integration Infrastructure

Institutions should support:

- institution study circles
- department circles
- cohort sessions
- institution workshops
- institution moderation

Institution Controls

Institutions may configure:

- participation permissions
- moderation rules
- attendance policies
- recording permissions
- visibility restrictions

22. Analytics Infrastructure

The system should track:

- participation
- attendance
- continuity effectiveness
- collaboration quality
- reinforcement effectiveness
- session engagement

Institutional Analytics

Institutions may view:

- attendance trends
- collaboration participation
- educational engagement
- continuity metrics
- study support effectiveness

Subject to:

- permissions

- FERPA protections
- privacy governance

23. FERPA / Privacy / Governance Requirements

The system must maintain:

- educational privacy protections
- consent systems
- recording permissions
- auditability
- institution governance

Required Governance Features

Consent Controls

Users control:

- recording participation
- visibility settings
- communication permissions

Audit Logs

The system should maintain:

- session history
- attendance history
- moderation actions
- institution actions

Institution Governance

Institutions may enforce:

- participation restrictions
- moderation policies
- recording policies
- attendance policies

24. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of sessions
- large-scale collaboration
- low-latency communication
- distributed session infrastructure
- scalable video/audio systems
- AI-assisted summaries
- real-time synchronization

25. Future Expansion Architecture

Potential future capabilities:

- AI-generated study plans
- adaptive study circles
- multilingual collaboration
- advanced collaborative whiteboards
- AI-powered educational copilots
- predictive accountability systems
- immersive collaboration environments

26. Final Engineering Recommendations

Study Circles and Study Sessions should be treated as:

- collaborative educational infrastructure systems
- continuity-driven learning ecosystems
- support reinforcement systems
- not generic group chats or video rooms

The goal is to optimize:

- educational collaboration
- accountability
- continuity
- support accessibility
- productive learning environments

- institutional collaboration

The dev team should prioritize:

- persistent continuity
- scalable collaboration systems
- scheduling integration
- moderation-aware collaboration
- AI-assisted organization
- institution governance
- reinforcement integration
- educational context preservation

Because Study Circles become one of the foundational collaborative learning systems powering the AcademIQ ecosystem.

Scheduling & Calendar Infra System

AcademIQ Scheduling & Calendar Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Scheduling & Calendar system.

The Scheduling & Calendar Infrastructure is one of the foundational operational systems of the AcademIQ ecosystem.

This infrastructure powers:

- tutoring scheduling
- study session scheduling
- office hours
- study circle scheduling
- recurring educational sessions
- institutional scheduling
- availability systems
- reminder systems
- continuity-aware scheduling
- AI-assisted planning
- academic workflow coordination

This document defines:

- scheduling architecture
- calendar systems
- recurring scheduling infrastructure
- timezone handling
- availability systems
- busy status systems
- conflict detection
- AI-assisted scheduling
- institutional scheduling controls
- reminder infrastructure
- office hours infrastructure
- session coordination systems
- analytics and governance systems

This document should be treated as:

- a master scheduling infrastructure document
- a dependency for multiple AcademIQ systems
- a continuity-aware educational scheduling systems reference

Including:

- ONE Thread Infrastructure
- Messaging Infrastructure
- Study Circle Infrastructure
- Emma Infrastructure
- Discovery Engine Infrastructure
- Institution Ecosystem Infrastructure
- Human Feedback & AI Reinforcement Infrastructure

2. Core Philosophy of Scheduling

Traditional scheduling systems are often:

- isolated
- static
- productivity-only focused
- disconnected from continuity

AcademIQ scheduling is fundamentally different.

Scheduling should optimize:

- educational continuity
- support accessibility
- collaborative learning
- recurring engagement
- accountability
- institutional coordination
- reinforcement continuity

The goal is NOT:

- simple calendar management
- generic appointment booking
- disconnected scheduling workflows

The goal IS:

- educational continuity coordination
- intelligent support organization
- recurring academic collaboration
- adaptive educational planning

3. What the Scheduling System Actually Is

The Scheduling Infrastructure is:

an intelligent educational coordination system that manages recurring educational interaction, support accessibility, and continuity-aware scheduling across the AcademIQ ecosystem.

The system coordinates:

- tutoring sessions
- office hours
- study circles
- accountability sessions
- institution events
- collaborative study sessions
- counselor appointments
- educational reminders
- recurring support interactions

4. Core Objectives

The Scheduling system must support:

- recurring scheduling
- timezone-aware coordination
- educational continuity
- availability matching
- conflict detection
- AI-assisted scheduling
- institutional scheduling
- reminder systems
- accountability scheduling
- scalable calendar infrastructure

5. Scheduling Infrastructure Overview

Study Sessions are one of the core operational scheduling systems inside AcademIQ.

Study Sessions should support:

- peer collaboration
- recurring accountability
- tutoring continuity
- instructor-led learning
- collaborative review sessions
- exam preparation
- institution workshops
- academic recovery support
- AI-assisted educational organization

Study Sessions may function as:

- one-time sessions
- recurring sessions
- public educational sessions
- network-only sessions
- institution-only sessions
- invitation-only sessions

The Scheduling Infrastructure must treat Study Sessions as persistent educational continuity systems rather than temporary meetings.

The platform should support:

- tutoring schedules
- instructor office hours
- study sessions
- study circles
- recurring accountability sessions
- institution events
- counseling appointments
- workshops and webinars

Study Session Scheduling Infrastructure

Study Session scheduling should support:

- instant session creation
- recurring scheduling
- timezone-aware participation
- session reminders
- attendance tracking
- collaborative session continuity
- AI-assisted organization
- office hours integration
- accountability scheduling

Study Session Types

Peer Study Sessions

Student-led collaborative sessions.

Tutoring Sessions

Structured tutoring support sessions.

Instructor-Led Study Sessions

Instructor-guided academic review sessions.

Accountability Sessions

Recurring productivity and study accountability sessions.

Exam Preparation Sessions

Focused exam review and preparation sessions.

Institution Study Sessions

Institution-sponsored workshops and collaborative sessions.

Scheduling Categories

One-Time Sessions

Examples:

- exam review sessions

- tutoring appointments
- institution events

Recurring Sessions

Examples:

- weekly study circles
- recurring tutoring
- office hours
- accountability meetings

Institution Scheduling

Examples:

- department events
- orientation sessions
- cohort meetings
- institutional workshops

6. Calendar Infrastructure Architecture

Each scheduling object should function as a structured infrastructure object.

Identity Layer

- schedule_id
- institution_id
- linked_thread_id
- creator_id
- event_type
- visibility_type

Participant Layer

- participant_ids
- participant_roles
- availability_states
- attendance_permissions

Time Layer

- start_time
- end_time
- timezone
- recurrence_rules
- reminder_rules

Context Layer

- educational_context
- course_associations
- support_context
- continuity_metadata

Moderation Layer

- moderation_flags
- institution_rules
- escalation_states

Analytics Layer

- attendance_metrics
- participation_patterns
- engagement_scores
- continuity_metrics

7. Availability Infrastructure

The system should support:

- dynamic availability
- recurring availability
- real-time status awareness
- educational scheduling awareness

Availability States

Available

Open for scheduling and communication.

Busy

Examples:

- in tutoring session
- in study circle
- in class
- in counseling session

Away

Temporarily inactive.

Invisible

Presence hidden.

8. Busy Status Infrastructure

Busy states should dynamically update based on:

- active sessions
- tutoring participation
- study circle participation
- scheduling conflicts
- manual user settings

Busy Status Examples

Examples:

- In Study Session
- In Tutoring Session
- In Office Hours
- In Counseling Session
- Focus Mode

9. Recurring Scheduling Infrastructure

Recurring educational continuity is one of the foundational goals of the system.

Recurring Features

Examples:

- weekly tutoring
- recurring office hours
- recurring study circles
- recurring accountability meetings
- monthly workshops

Recurrence Infrastructure

The system should support:

- daily recurrence
- weekly recurrence
- monthly recurrence
- custom recurrence patterns
- recurring reminders

10. Timezone Infrastructure

The system must support:

- global scheduling
- timezone conversion
- international collaboration
- institution-aware scheduling

Timezone Features

Examples:

- automatic timezone conversion
- local time rendering
- travel-aware scheduling
- daylight savings support

11. Conflict Detection Infrastructure

The scheduling system should intelligently detect:

- overlapping sessions
- conflicting appointments
- unavailable participants
- institutional conflicts

Conflict Examples

Examples:

- tutoring overlaps
- study session conflicts
- office hours overlap
- institution scheduling conflicts

Conflict Resolution Assistance

The system may:

- recommend alternative times
- suggest open slots
- reroute scheduling recommendations
- surface availability insights

12. Reminder Infrastructure

The system should support:

- real-time reminders
- recurring reminders
- reinforcement reminders
- educational notifications

Reminder Types

Session Reminders

Examples:

- tutoring reminders
- office hours reminders
- study circle reminders

Reinforcement Reminders

Examples:

- accountability reminders
- productivity encouragement
- participation reminders

Institutional Reminders

Examples:

- workshops
- orientation sessions
- deadlines

13. ONE Thread Integration

Scheduling should integrate deeply with:
ONE Thread Infrastructure.

Thread Scheduling Features

Examples:

- scheduling discussions
- session coordination
- attendance continuity
- reminder continuity
- collaborative planning

14. Messaging Integration

Scheduling should integrate deeply with:
Messaging Infrastructure.

Messaging Features

Examples:

- schedule confirmations
- quick scheduling coordination
- attendance reminders
- live session notifications

15. Study Circle & Study Session Integration

Scheduling should integrate deeply with:
Study Circle Infrastructure.

Study Circle & Study Session Scheduling Features

Examples:

- recurring sessions
- attendance coordination
- accountability schedules
- collaborative session planning
- instant study session creation
- session reminders
- recurring study continuity
- session-based ONE Threads

16. Office Hours Infrastructure

The system should support:

- instructor office hours
- tutor office hours
- counselor availability
- recurring support windows

Office Hours Features

Examples:

- recurring availability
- student booking
- institution scheduling
- timezone-aware support

17. Emma Scheduling Integration

Emma should assist with:

- schedule organization
- study planning
- session reminders
- accountability reinforcement
- conflict resolution
- productivity support

Emma Scheduling Capabilities

Examples:

- recommending schedules
- surfacing availability
- generating reminders
- identifying continuity gaps
- encouraging participation

18. Discovery Engine Integration

Scheduling should integrate deeply with:
Discovery Engine Infrastructure.

Discovery-Aware Scheduling

Examples:

- surfacing open study sessions
- recommending tutoring availability
- suggesting office hours
- recommending accountability sessions

19. Institutional Scheduling Infrastructure

Institutions should support:

- institution calendars
- department scheduling
- cohort scheduling
- institution events
- academic support scheduling

Institution Controls

Institutions may configure:

- scheduling permissions
- office hours requirements
- attendance policies
- scheduling restrictions
- institution visibility rules

20. Attendance Infrastructure

The scheduling system should support:

- attendance tracking
- participation logging
- recurring attendance analytics
- accountability tracking

Attendance Features

Examples:

- check-ins
- late join tracking
- session completion tracking
- recurring participation metrics

21. Moderation & Safety Infrastructure

Scheduling systems must integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- institution governance

Safety Features

Examples:

- session reporting
- participant restrictions
- institution escalation
- moderation review

22. Analytics Infrastructure

The system should track:

- scheduling engagement
- attendance trends
- support participation
- continuity effectiveness
- scheduling conflicts
- reinforcement effectiveness

Institutional Analytics

Institutions may view:

- office hours participation
- tutoring participation
- attendance trends
- collaboration metrics
- educational engagement

Subject to:

- permissions

- FERPA protections
- privacy governance

23. FERPA / Privacy / Governance Requirements

The scheduling system must maintain:

- educational privacy protections
- consent systems
- auditability
- institution governance
- scheduling visibility controls

Required Governance Features

Consent Controls

Users control:

- visibility settings
- scheduling permissions
- reminder preferences
- attendance visibility

Audit Logs

The system should maintain:

- scheduling history
- attendance history
- moderation actions
- institution actions

Institution Governance

Institutions may enforce:

- scheduling policies
- office hours requirements
- participation restrictions

- attendance policies

24. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of schedules
- real-time synchronization
- scalable calendar infrastructure
- distributed reminder systems
- timezone-aware coordination
- AI-assisted planning
- high concurrency

25. Future Expansion Architecture

Potential future capabilities:

- AI-generated study schedules
- predictive scheduling recommendations
- adaptive productivity planning
- multilingual scheduling systems
- institution AI scheduling copilots
- burnout-aware scheduling recommendations
- smart accountability scheduling

26. Final Engineering Recommendations

The Scheduling & Calendar system should be treated as:

- an educational coordination infrastructure system
- a continuity-aware planning system
- a support accessibility infrastructure layer
- not just a generic calendar feature

The goal is to optimize:

- educational continuity
- accountability
- support accessibility

- productive collaboration
- recurring educational engagement
- institutional coordination

The dev team should prioritize:

- recurring continuity systems
- timezone-aware scheduling
- AI-assisted coordination
- messaging integration
- reinforcement integration
- institution governance
- scalable synchronization
- educational context preservation

Because Scheduling & Calendar Infrastructure becomes one of the foundational operational systems coordinating continuity across the entire AcademIQ ecosystem.

Resource Sharing & File Infra System

AcademIQ Resource Sharing & File Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Resource Sharing & File system.

The Resource Sharing Infrastructure is one of the foundational educational collaboration systems inside the AcademIQ ecosystem.

This infrastructure powers:

- educational file sharing
- collaborative resources
- study materials
- academic document organization
- institutional resource distribution
- AI-assisted resource indexing
- continuity-aware file systems
- collaborative educational workflows
- session resources
- thread-linked educational content

This document defines:

- file infrastructure architecture
- resource sharing systems
- collaborative document systems
- file permissions
- moderation scanning
- AI indexing infrastructure

- searchable educational resources
- institutional restrictions
- storage architecture
- continuity-linked resources
- analytics and governance systems

This document should be treated as:

- a master educational resource infrastructure document
- a dependency for multiple AcademIQ systems
- a continuity-aware educational collaboration systems reference

Including:

- ONE Thread Infrastructure
- Messaging Infrastructure
- Study Circle Infrastructure
- Discovery Engine Infrastructure
- Emma Infrastructure
- Institution Ecosystem Infrastructure
- Verification & Credential Infrastructure
- Scheduling Infrastructure

2. Core Philosophy of Resource Sharing

Traditional file sharing systems are often:

- disconnected
- storage-focused only
- collaboration-limited
- difficult to organize
- continuity-unaware

AcademIQ Resource Sharing is fundamentally different.

The system should optimize:

- educational continuity
- collaborative learning
- contextual organization
- searchable educational resources
- institutional collaboration
- support accessibility
- continuity-aware resource management

The goal is NOT:

- becoming generic cloud storage
- disconnected file hosting
- unstructured document dumping

The goal IS:

- educational resource continuity
- collaborative educational infrastructure
- intelligent educational organization
- searchable academic knowledge systems

3. What the Resource Infrastructure Actually Is

The Resource Sharing Infrastructure is:

an educational collaboration and continuity-aware file infrastructure system that intelligently organizes, secures, indexes, and distributes educational resources across the AcademIQ ecosystem.

The system supports:

- thread-linked resources
- collaborative educational files
- study materials
- tutoring resources
- institution resources
- session resources
- shared educational libraries
- AI-assisted resource organization

4. Core Objectives

The Resource Sharing system must support:

- educational collaboration
- continuity-aware storage
- AI-assisted indexing
- searchable educational resources
- institutional governance
- moderation-aware resource sharing
- scalable file infrastructure
- collaborative document systems
- secure educational file management
- educational workflow continuity

5. Resource Infrastructure Overview

The platform should support:

- educational document sharing
- study resources
- tutoring materials
- collaborative notes
- session recordings
- assignment resources
- institution libraries
- AI-generated summaries
- collaborative whiteboards
- educational media

Resource Categories

Study Resources

Examples:

- notes
- study guides
- flashcards
- review packets

Tutoring Resources

Examples:

- tutoring worksheets
- lesson materials
- tutoring recordings
- collaborative examples

Institutional Resources

Examples:

- syllabi
- department resources
- institutional guides
- onboarding materials

Session Resources

Examples:

- study session recordings
- collaborative notes
- whiteboards
- session summaries

Collaborative Resources

Examples:

- shared documents
- collaborative boards
- group notes
- project files

6. Resource Object Architecture

Each resource object should function as a structured infrastructure object.

Identity Layer

- resource_id
- institution_id
- linked_thread_id
- creator_id
- resource_type
- visibility_type

Content Layer

- file_name
- file_format
- file_size
- resource_description
- embedded_media
- collaborative_elements

Context Layer

- educational_tags
- subject_tags
- course_associations
- support_context
- continuity_metadata

Permission Layer

- participant_permissions
- institution_permissions
- visibility_controls
- download_permissions
- editing_permissions

Moderation Layer

- moderation_flags
- report_history
- institution_rules
- escalation_states

Analytics Layer

- engagement_metrics
- download_metrics
- collaboration_metrics
- educational_usage_patterns

7. File Sharing Infrastructure

The platform should support:

- direct file uploads
- thread-linked resources
- collaborative sharing
- institutional resource distribution
- educational media sharing

Supported File Types

Examples:

- PDFs
- presentations
- spreadsheets

- study guides
- images
- audio recordings
- video recordings
- collaborative notes
- whiteboards

Upload Infrastructure

The system should support:

- drag-and-drop uploads
- mobile uploads
- bulk uploads
- institution uploads
- session-linked uploads

8. Visibility & Permission Infrastructure

Resources should support multiple visibility states.

Public Resources

Open educational resources.

Examples:

- public study guides
- community learning resources
- public tutoring materials

Network-Only Resources

Restricted to accepted network relationships.

Examples:

- accountability group resources
- private tutoring notes
- collaborative project files

Selected Participant Resources

Invitation-only visibility.

Examples:

- tutoring session documents
- counseling resources
- private collaboration files

Institution-Only Resources

Restricted to:

- institutions
- departments
- programs
- cohorts
- courses

9. Collaborative Document Infrastructure

The system should support:

- collaborative editing
- shared educational notes
- collaborative whiteboards
- real-time collaboration
- educational teamwork

Collaboration Features

Examples:

- live editing
- annotations
- comments
- collaborative study boards
- version history
- educational markup

10. ONE Thread Integration

Resources should integrate deeply with:

ONE Thread Infrastructure.

Thread Resource Features

Examples:

- thread-linked study guides
- collaborative continuity
- AI-generated summaries
- persistent educational resources
- continuity-aware organization

11. Messaging Integration

Resource sharing should integrate deeply with:

Messaging Infrastructure.

Messaging Features

Examples:

- quick file sharing
- collaborative discussion
- session resources
- attachment continuity

12. Study Circle & Study Session Integration

Resources should integrate deeply with:

Study Circle Infrastructure.

Study Circle Features

Examples:

- collaborative study notes
- session recordings
- accountability documents
- recurring session resources
- collaborative whiteboards

13. Discovery Engine Integration

Resources should integrate deeply with:

Discovery Engine Infrastructure.

Discovery Features

Users may discover:

- study guides
- collaborative resources
- tutoring materials
- institution resources
- educational libraries

Recommendations may evaluate:

- educational relevance
- continuity relationships
- support needs
- institution permissions

14. Emma Integration

Emma should assist with:

- resource recommendations
- educational organization
- summarization
- continuity-aware retrieval
- collaborative organization

Emma Resource Capabilities

Examples:

- recommending resources
- generating summaries
- surfacing study materials
- organizing collaborative resources
- identifying educational gaps

15. AI Indexing & Search Infrastructure

The system should support:

- semantic indexing
- OCR support
- AI-assisted retrieval
- searchable educational resources
- contextual search

AI Indexing Examples

Examples:

- PDF indexing
- screenshot understanding
- whiteboard extraction
- transcript generation
- AI-generated summaries

Search Categories

Users may search:

- study guides
- notes
- recordings
- collaborative files
- institution documents
- tutoring resources

16. Academic Integrity Infrastructure

The system should support:

- academic integrity protections
- plagiarism-aware resource sharing
- institution governance
- assignment protection systems

Integrity Features

Examples:

- restricted assignment sharing

- institution policy enforcement
- AI-assisted integrity scanning
- educational misuse detection

17. Moderation & Safety Infrastructure

Resource systems must integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- institution governance

Moderation Features

Examples:

- harmful content scanning
- inappropriate content detection
- spam detection
- malware scanning
- institution moderation review

Reporting Infrastructure

Users may report:

- harmful content
- inappropriate resources
- academic misconduct
- spam
- malicious uploads

18. Recording Infrastructure

The system should support:

- session recordings
- tutoring recordings
- collaborative playback
- searchable recordings
- AI-generated transcripts

Recording Features

Examples:

- timestamped playback
- transcript generation
- AI summaries
- educational indexing
- institution recording controls

19. Institutional Resource Infrastructure

Institutions should support:

- institutional libraries
- department resources
- program resources
- onboarding materials
- educational repositories

Institution Controls

Institutions may configure:

- upload permissions
- sharing restrictions
- moderation rules
- retention policies
- visibility permissions

20. Analytics Infrastructure

The system should track:

- resource engagement
- download behavior
- collaboration participation
- educational usage trends
- continuity effectiveness

Institutional Analytics

Institutions may view:

- resource utilization
- collaboration metrics
- educational engagement
- study material usage
- support participation

Subject to:

- permissions
- FERPA protections
- privacy governance

21. FERPA / Privacy / Governance Requirements

The resource system must maintain:

- educational privacy protections
- consent systems
- auditability
- institution governance
- secure file handling

Required Governance Features

Consent Controls

Users control:

- resource visibility
- download permissions
- collaboration permissions

Audit Logs

The system should maintain:

- upload history
- download history
- collaboration history
- moderation actions
- institution actions

Institution Governance

Institutions may enforce:

- resource restrictions
- retention policies
- sharing boundaries
- moderation rules

22. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of files
- distributed storage systems
- AI-assisted indexing
- scalable collaboration systems
- real-time synchronization
- secure educational storage
- high concurrency

23. Future Expansion Architecture

Potential future capabilities:

- AI-generated study packets
- adaptive educational libraries
- multilingual resource indexing
- collaborative AI whiteboards
- intelligent educational repositories
- institution AI resource copilots
- predictive educational resource recommendations

24. Final Engineering Recommendations

The Resource Sharing & File system should be treated as:

- an educational collaboration infrastructure system
- a continuity-aware educational repository
- a collaborative academic workflow layer
- not generic cloud storage

The goal is to optimize:

- educational continuity
- collaborative learning
- support accessibility
- educational organization
- institutional collaboration
- searchable educational knowledge

The dev team should prioritize:

- continuity-aware resource organization
- AI-assisted indexing
- moderation-aware sharing
- institution governance
- scalable storage systems
- collaborative educational workflows

- academic integrity protections
- secure educational file handling

Because the Resource Sharing Infrastructure becomes one of the foundational educational collaboration systems powering the AcademIQ ecosystem.

Institution Ecosystem Infra System

AcademIQ Institution Ecosystem Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Institution Ecosystem.

The Institution Ecosystem Infrastructure is one of the foundational operating systems of the AcademIQ platform.

This infrastructure powers:

- institution onboarding
- departments
- programs
- cohorts
- Community
- institution governance
- institution moderation
- institution analytics
- institution communication
- institution scheduling
- institution study circles
- institution support systems
- institution-linked Discovery Engine behavior
- institution continuity systems
- LMS integrations
- institution identity systems

This document defines:

- institution architecture
- institutional hierarchy systems
- institution governance infrastructure
- institution permission systems
- department infrastructure
- program infrastructure
- cohort infrastructure
- institution moderation systems
- institution communication systems
- institution analytics systems
- institution onboarding systems
- LMS integration architecture

- FERPA and governance infrastructure

This document should be treated as:

- a master institutional operating system document
- a dependency for nearly all AcademIQ systems
- the institutional backbone of the AcademIQ ecosystem

Including dependencies with:

- ONE Thread Infrastructure
- Discovery Engine Infrastructure
- Emma Infrastructure
- Messaging Infrastructure
- Scheduling Infrastructure
- Study Circle Infrastructure
- Verification & Credential Infrastructure
- Resource Sharing Infrastructure
- Human Feedback & AI Reinforcement Infrastructure

2. Core Philosophy of the Institution Ecosystem

Traditional educational platforms are often:

- fragmented
- siloed
- disconnected
- workflow-limited
- continuity-unaware

AcademIQ institutions are fundamentally different.

The Institution Ecosystem should optimize:

- educational continuity
- institutional collaboration
- support accessibility
- academic coordination
- educational personalization
- cross-role communication

- continuity-aware educational systems
- institution-wide educational intelligence

The goal is NOT:

- functioning as a generic LMS only
- replacing institutional leadership
- creating isolated institutional silos

The goal IS:

- strengthening institutional ecosystems
- improving support coordination
- improving continuity
- improving educational engagement
- improving collaboration between students, instructors, counselors, tutors, and administrators

3. What the Institution Ecosystem Actually Is

The Institution Ecosystem is:

an institution-aware educational infrastructure system that coordinates governance, collaboration, communication, continuity, and educational intelligence across institutional environments.

The system coordinates:

- departments
- programs
- cohorts
- institution users
- institution threads
- institution study circles
- institution scheduling
- institution communication
- institution analytics
- institutional support pathways

4. Core Objectives

The Institution Ecosystem must support:

- institution hierarchy systems
- institutional governance
- institution-aware continuity
- institutional collaboration
- department coordination
- cohort infrastructure
- institution analytics
- institution moderation
- institution communication
- institution onboarding
- scalable institution infrastructure

5. Institution Architecture Overview

Each institution should function as a structured infrastructure object.

Institution Object Layers

Identity Layer

- institution_id
- institution_name
- institution_type
- accreditation_status
- branding_assets
- institution_domains

Governance Layer

- institution_admins
- department_admins
- moderation_permissions
- governance_rules
- escalation_rules

Academic Layer

- departments
- programs

- courses
- cohorts
- educational pathways

Communication Layer

- institution threads
- announcements
- institution messaging
- emergency communication

Analytics Layer

- engagement_metrics
- retention_metrics
- continuity_metrics
- support_metrics
- educational participation

Integration Layer

- LMS integrations
- SIS integrations
- authentication systems
- institution APIs

6. Institution Hierarchy Infrastructure

Institutions should support structured hierarchy systems.

Hierarchy Levels

Institution Level

Top-level institution governance.

Department Level

Examples:

- Mathematics Department
- Engineering Department
- Psychology Department

Program Level

Examples:

- Nursing Program
- Computer Science Program
- MBA Program

Cohort Level

Examples:

- Incoming Freshman Cohort
- Nursing Cohort 2028
- Transfer Student Cohort

Course Level

Examples:

- Calculus I
- Organic Chemistry
- Statistics

7. Department Infrastructure

Departments should function as operational educational ecosystems.

Department Features

Examples:

- department threads
- department study circles
- department scheduling
- department announcements

- department moderation
- department analytics

Department Permissions

Departments may configure:

- visibility rules
- moderation rules
- communication permissions
- participation restrictions

8. Program Infrastructure

Programs should support:

- program continuity
- academic pathways
- support coordination
- program communities
- educational collaboration

Program Features

Examples:

- program onboarding
- program-specific threads
- program communities
- support pathways
- program scheduling

9. Cohort Infrastructure

Cohorts are one of the foundational continuity systems inside AcademIQ.

Cohort Features

Examples:

- cohort onboarding
- cohort threads
- cohort study circles
- cohort scheduling
- cohort announcements
- cohort collaboration
- cohort continuity tracking

Cohort Goals

The system should improve:

- peer continuity
- accountability
- support accessibility
- academic collaboration
- retention

10. Institution Onboarding Infrastructure

Institutions should support:

- institution onboarding
- student onboarding
- instructor onboarding
- cohort assignment
- institution verification
- institution continuity initialization

Institution Onboarding Features

Examples:

- institution branding
- role assignment
- department assignment
- cohort assignment
- onboarding threads
- institution recommendations

11. Institution Verification Infrastructure

Institutions should maintain authority over:

- institution verification
- student legitimacy
- instructor legitimacy
- counselor legitimacy
- institutional participation

Verification Examples

Examples:

- institution-issued badges
- verified students
- verified instructors
- verified counselors
- accredited institutions

12. Institution Permission Infrastructure

Institution permissions must support:

- role-aware permissions
- department restrictions
- moderation controls
- visibility controls
- communication permissions
- scheduling permissions

Permission Examples

Examples:

- institution-only threads
- department-specific resources
- instructor communication restrictions
- cohort visibility rules

13. Institution Communication Infrastructure

Institutions should support:

- institution-wide communication
- department communication
- cohort communication
- emergency alerts
- academic announcements
- continuity-aware messaging

Communication Systems

Examples:

- institution announcements
- department threads
- cohort discussions
- academic support threads
- office hours communication

14. ONE Thread Integration

Institutions should integrate deeply with:
ONE Thread Infrastructure.

Institution Thread Examples

Examples:

- institution onboarding threads
- cohort continuity threads
- department support threads
- program collaboration threads
- academic support threads

Visibility Enforcement

Institution threads may support:

- institution-only visibility
- department-only visibility

- cohort-only visibility
- role-specific participation

15. Messaging Integration

Institutions should integrate deeply with:
Messaging Infrastructure.

Institution Messaging Features

Examples:

- institution communication
- department coordination
- office hours messaging
- cohort reminders
- emergency communication

16. Scheduling Integration

Institutions should integrate deeply with:
Scheduling Infrastructure.

Institution Scheduling Features

Examples:

- office hours scheduling
- institution events
- cohort scheduling
- tutoring coordination
- academic support scheduling

17. Study Circle & Study Session Integration

Institutions should support:

- institution study circles
- department study circles
- cohort sessions
- instructor-led study sessions
- tutoring collaboration

Institution Collaboration Examples

Examples:

- engineering study circles
- cohort accountability sessions
- exam review workshops
- department tutoring sessions

18. Discovery Engine Integration

Institutions should integrate deeply with:
Discovery Engine Infrastructure.

Institution Discovery Features

Users may discover:

- institution communities
- cohort study circles
- department resources
- institution events
- support pathways

Recommendations may evaluate:

- institution affiliation
- department relevance
- educational continuity
- support needs

19. Emma Institutional Integration

Emma should support:

- institution onboarding
- institution continuity
- institutional support routing
- academic guidance
- institution navigation

Emma Institution Capabilities

Examples:

- recommending institution resources
- surfacing office hours
- recommending support systems
- onboarding continuity guidance
- cohort support recommendations

20. LMS & SIS Integration Infrastructure

Institutions should support:

- LMS integrations
- SIS integrations
- roster synchronization
- assignment synchronization
- authentication systems
- academic continuity integration

LMS Integration Examples

Examples:

- Canvas
- Blackboard
- Moodle
- Google Classroom
- institution proprietary systems

SIS Integration Examples

Examples:

- roster syncing
- enrollment syncing
- cohort assignment
- academic standing synchronization

21. Institutional Moderation Infrastructure

Institutions must integrate deeply with:

- Trust & Safety Infrastructure
- moderation systems
- governance systems

Institution Moderation Features

Examples:

- institution moderation teams
- escalation workflows
- department moderation
- thread moderation
- communication enforcement

Moderation Actions

Examples:

- participation restrictions
- communication suspension
- institution escalation
- moderation review
- trust & safety escalation

22. Institutional Analytics Infrastructure

Institutions should support:

- engagement analytics
- continuity analytics
- retention analytics
- support analytics
- collaboration analytics
- educational participation metrics

Institutional Dashboard Examples

Examples:

- cohort engagement
- office hours participation
- tutoring utilization
- study circle engagement
- support participation
- continuity trends

23. FERPA / Privacy / Governance Requirements

The Institution Ecosystem must maintain:

- FERPA protections
- institution governance
- educational privacy protections
- consent systems
- auditability
- moderation traceability

Required Governance Features

Consent Controls

Users control:

- visibility settings
- communication permissions
- discovery participation
- AI participation

Audit Logs

The system should maintain:

- institution actions
- moderation actions
- onboarding history
- participation history

- governance actions

Institution Governance

Institutions may enforce:

- moderation policies
- communication boundaries
- onboarding rules
- participation restrictions
- scheduling policies

24. Scalability & Infrastructure Requirements

Infrastructure must support:

- thousands of institutions
- millions of users
- institution-level isolation
- distributed institution infrastructure
- scalable analytics
- real-time synchronization
- AI-assisted orchestration
- high concurrency

25. Future Expansion Architecture

Potential future capabilities:

- institution AI copilots
- predictive retention systems
- adaptive institutional support systems
- multilingual institutional ecosystems
- sovereign institution deployments
- institution-hosted AI infrastructure
- regional educational intelligence nodes

26. Final Engineering Recommendations

The Institution Ecosystem should be treated as:

- an institutional operating system
- a continuity-aware educational infrastructure layer
- a collaborative institutional intelligence system
- not just an LMS extension

The goal is to optimize:

- educational continuity
- institutional collaboration
- support accessibility
- academic coordination
- student success
- educational engagement
- continuity-aware institutional intelligence

The dev team should prioritize:

- scalable institution architecture
- continuity-aware collaboration
- institution governance systems
- moderation-aware infrastructure
- LMS/SIS integrations
- institutional analytics
- support coordination
- educational context preservation

Because the Institution Ecosystem becomes the institutional backbone coordinating continuity across the entire AcademIQ platform.

AI Boundary Governance Infra System

AcademIQ AI Boundary Governance Infrastructure System

DevSpec Document

1. Introduction

This document defines the governance architecture, operational boundaries, ethical controls, and institutional AI safety infrastructure for the AcademIQ ecosystem.

This document is critically important because AI systems inside AcademIQ interact with:

- students
- minors
- instructors
- counselors
- institutions
- wellness-related discussions
- academic support systems
- educational continuity systems
- reinforcement systems
- institutional workflows

AcademIQ AI systems include:

- Emma
- Discovery Engine intelligence
- reinforcement systems
- AI summaries
- AI recommendations
- educational routing systems
- AI moderation assistance
- institutional AI workflows
- continuity-aware AI systems

This document defines:

- AI behavioral boundaries
- educational AI governance
- human oversight systems
- institution AI controls
- moderation-aware AI behavior
- FERPA-aware AI infrastructure
- consent systems

- escalation boundaries
- AI recommendation limitations
- wellness protections
- minors protections
- transparency requirements

This document should be treated as:

- a master AI governance infrastructure document
- a platform-wide AI behavioral policy architecture
- a dependency for all AI-enabled AcademIQ systems

Including:

- Emma Infrastructure
- Discovery Engine Infrastructure
- Messaging Infrastructure
- ONE Thread Infrastructure
- Trust & Safety Infrastructure
- Institution Ecosystem Infrastructure
- Human Feedback & AI Reinforcement Infrastructure
- Verification & Credential Infrastructure

2. Core Philosophy of AI Governance

AcademIQ is NOT building AI systems to:

- replace educators
- replace counselors
- replace institutions
- replace human relationships
- replace human judgment

AcademIQ AI exists to:

- support educational continuity
- improve support accessibility
- strengthen educational workflows
- improve organization
- improve reinforcement continuity

- assist educational collaboration
- intelligently route support systems

The goal is NOT:

- autonomous educational decision-making
- unrestricted AI intervention
- AI authority over institutions
- AI authority over disciplinary systems

The goal IS:

- human-centered AI assistance
- institution-safe AI systems
- continuity-aware educational intelligence
- ethically governed educational orchestration

3. Foundational AI Governance Principles

All AI systems inside AcademIQ must follow these principles.

Human-Centered Governance

Humans remain the final authority.

AI should:

- assist
- organize
- recommend
- summarize
- reinforce
- support

AI should NOT:

- independently discipline users
- independently diagnose users
- independently determine academic outcomes
- independently override institutions

Recommendation-Only Principle

AI systems should primarily function as:

- recommendation engines
- support assistants
- continuity systems
- organizational systems

Critical decisions should require:

- human oversight
- institution review
- moderator review
- instructor review
- counselor review

Explainability Principle

AI behavior should remain:

- understandable
- auditable
- reviewable
- institution-visible

Users and institutions should understand:

- why recommendations occur
- why escalations occur
- why moderation actions occur
- why support routing occurs

Consent-Driven AI Participation

Users must maintain control over:

- personalization
- AI memory

- AI participation
- recommendations
- AI continuity systems

4. AI System Classification Infrastructure

AcademIQ AI systems should be classified by operational sensitivity.

Tier 1 — Low-Risk AI Systems

Examples:

- summaries
- scheduling reminders
- thread organization
- educational recommendations
- study planning

These systems may operate with lower oversight.

Tier 2 — Moderate-Risk AI Systems

Examples:

- reinforcement systems
- educational continuity recommendations
- support routing
- burnout detection signals
- engagement analysis

These systems require:

- explainability
- auditability
- moderation visibility

Tier 3 — High-Risk AI Systems

Examples:

- wellness-related routing
- crisis escalation
- academic integrity workflows
- minors-related systems
- institution-sensitive workflows

These systems require:

- human oversight
- institution governance
- moderation visibility
- escalation controls
- strict audit logging

5. Emma Behavioral Governance

Emma must follow strict behavioral boundaries.

Emma Allowed Behaviors

Examples:

- educational organization
- study support
- continuity assistance
- support recommendations
- productivity reinforcement
- educational summaries
- scheduling assistance

Emma Restricted Behaviors

Emma should NOT:

- diagnose mental health conditions
- provide therapy
- replace counselors
- provide legal advice
- provide medical advice
- independently discipline users
- manipulate users emotionally
- encourage dependency

Emma Escalation Responsibilities

Emma should:

- encourage human support systems
- recommend counselors
- recommend instructors
- escalate emergencies appropriately
- encourage institution-approved pathways

6. Wellness & Emotional Support Governance

AcademIQ includes:

- wellness support systems
- Real Talk infrastructure
- counselor ecosystems
- emotional support pathways

This creates elevated governance requirements.

AI Wellness Boundaries

AI systems should:

- encourage support
- recommend human professionals
- route users appropriately
- recognize elevated-risk interactions

AI systems should NOT:

- provide therapy
- diagnose conditions
- simulate licensed care
- replace counselors
- create emotional dependency

Crisis Escalation Infrastructure

Potential high-risk situations may trigger:

- moderation review
- trust & safety escalation
- institution escalation
- counselor escalation
- emergency intervention workflows

Subject to:

- institution policies
- age protections
- regional compliance laws
- privacy governance

7. Minors Protection Infrastructure

AcademIQ must support elevated protections for minors.

Minor Safety Protections

Examples:

- restricted communication pathways
- moderation escalation
- institution visibility
- age-aware AI behavior
- restricted wellness interactions
- controlled recommendations

Minor AI Restrictions

AI systems interacting with minors should:

- avoid manipulative behavior
- avoid emotional dependency systems
- avoid unsafe guidance
- escalate elevated-risk situations
- remain institution-safe

8. Academic Integrity Governance

AI systems must maintain strict boundaries around:

- assignment support
- academic integrity
- assessment workflows
- plagiarism-related systems

AI Integrity Restrictions

AI systems should NOT:

- complete assignments dishonestly
- bypass academic policies
- facilitate cheating
- undermine institution integrity rules

Integrity Workflow Governance

Human oversight should remain required for:

- disciplinary review
- integrity escalations
- academic misconduct decisions
- institutional enforcement

9. Reinforcement Governance Infrastructure

Reinforcement systems must remain ethically governed.

Allowed Reinforcement Behaviors

Examples:

- productivity encouragement
- study reminders
- accountability encouragement
- continuity reinforcement
- educational motivation

Restricted Reinforcement Behaviors

AI systems should NOT:

- exploit emotional vulnerability
- manipulate engagement psychologically
- optimize addiction-style engagement
- pressure unhealthy participation

10. AI Recommendation Governance

Recommendations should remain:

- educationally relevant
- support-focused
- explainable
- institution-safe
- moderation-aware

Recommendation Restrictions

Recommendations should NOT:

- expose restricted users
- violate institution boundaries
- surface unsafe content
- bypass moderation restrictions
- violate user privacy settings

11. Human Oversight Infrastructure

High-sensitivity AI workflows must support human oversight.

Human Review Required Examples

Examples:

- moderation escalations
- academic integrity escalations
- crisis situations

- institution disciplinary workflows
- suspicious AI recommendations

Human Override Systems

Humans must be able to:

- override AI actions
- disable AI recommendations
- review AI decisions
- audit AI behavior
- restrict AI participation

12. Institution AI Governance Infrastructure

Institutions must maintain governance authority over AI systems.

Institution Controls

Institutions may configure:

- AI participation permissions
- moderation policies
- escalation pathways
- AI visibility
- recommendation restrictions
- communication boundaries
- wellness routing permissions

Institution AI Visibility

Institutions should maintain visibility into:

- AI escalations
- AI moderation assistance
- AI support routing
- AI institutional workflows

Subject to:

- privacy controls
- FERPA protections
- governance restrictions

13. AI Moderation Governance

AI may assist moderation systems.

However:
AI should not function as the sole authority for severe enforcement actions.

AI Moderation Assistance Examples

Examples:

- spam detection
- harmful content flagging
- abuse detection assistance
- escalation prioritization
- moderation queue assistance

Human Moderation Requirements

Human review should remain required for:

- permanent restrictions
- institution escalations
- severe disciplinary actions
- crisis-related actions

14. Consent Infrastructure

Users must explicitly control:

- AI participation
- personalization
- AI memory
- continuity systems
- recommendation systems

Consent Categories

Examples:

- Emma continuity
- personalization systems
- reinforcement systems
- educational recommendations
- memory participation

Opt-Out Systems

Users should support:

- partial AI participation
- recommendation opt-outs
- memory restrictions
- personalization restrictions

15. Transparency & Explainability Infrastructure

AI systems must remain:

- auditable
- explainable
- institution-reviewable
- moderation-visible

Explainability Examples

Examples:

- why recommendations occurred
- why escalations occurred
- why support routing occurred
- why moderation flags occurred

16. Auditability Infrastructure

All sensitive AI interactions should support:

- audit logs
- moderation traceability
- institution traceability
- escalation history

Audit Log Examples

Examples:

- AI escalations
- moderation recommendations
- institutional routing
- recommendation history
- AI-generated summaries

17. FERPA / Privacy / Governance Requirements

All AI systems must maintain:

- FERPA protections
- educational privacy protections
- institution governance
- data minimization
- consent systems
- explainability
- auditability

Data Governance Requirements

AI systems should support:

- anonymization
- retention controls
- secure storage
- institution boundaries
- access restrictions

18. AI Provider Governance

AcademIQ should remain:
AI-provider agnostic.

AI systems should operate through:
Unified AI Service Layer.

AI Provider Requirements

The platform should support:

- provider abstraction
- routing systems
- fallback systems
- model switching
- institution-specific AI controls

Approved AI Provider Examples

Examples:

- DeepSeek
- Gemini
- OpenAI
- future educational models
- institution-hosted models

19. Analytics & Governance Monitoring

The platform should monitor:

- AI recommendation quality
- escalation effectiveness
- moderation effectiveness
- support routing outcomes
- AI safety metrics
- reinforcement effectiveness

Institutional Governance Analytics

Institutions may view:

- AI usage metrics
- escalation metrics

- support routing trends
- moderation assistance metrics

Subject to:

- FERPA protections
- permissions
- governance controls

20. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of AI interactions
- distributed orchestration
- scalable audit systems
- provider abstraction
- institution-level governance
- moderation-aware AI infrastructure
- low-latency AI routing

21. Future Expansion Architecture

Potential future capabilities:

- institution-hosted AI governance
- sovereign educational AI systems
- advanced explainability systems
- AI safety scoring
- adaptive AI governance systems
- region-specific governance controls
- institution-specific AI copilots

22. Final Engineering Recommendations

AI systems inside AcademIQ should be treated as:

- educational support infrastructure
- continuity-aware intelligence systems
- institution-governed assistance systems

- human-centered orchestration systems
- not autonomous authority systems

The goal is to optimize:

- educational continuity
- support accessibility
- institution trust
- ethical AI participation
- safe educational intelligence
- moderation-aware orchestration

The dev team should prioritize:

- human oversight systems
- provider-agnostic orchestration
- explainability
- auditability
- FERPA-aware AI behavior
- institution governance controls
- moderation-aware AI infrastructure
- ethical reinforcement systems

Because AI governance becomes one of the most critical trust and safety layers powering the entire AcademIQ ecosystem.

Moderation Escalation & Enforcement Infra System

AcademIQ Moderation Escalation & Enforcement Infrastructure System

DevSpec Document

1. Introduction

This document defines the operational moderation architecture, escalation systems, enforcement workflows, and institutional moderation infrastructure for the AcademIQ ecosystem.

AcademIQ includes:

- threads
- messaging
- study circles
- study sessions
- institutional communication
- Real Talk
- educational collaboration
- wellness-related interaction
- minors participation
- AI-assisted systems

This creates the need for:

- scalable moderation
- institution-safe enforcement
- escalation workflows
- trust systems
- operational governance
- moderation transparency
- educationally appropriate enforcement systems

This document defines:

- moderation architecture
- escalation chains
- moderation queues
- institution moderation systems
- enforcement lifecycle systems
- trust scoring enforcement
- temporary restriction systems
- appeals systems

- emergency escalation systems
- AI-assisted moderation governance
- moderation analytics
- auditability infrastructure

This document should be treated as:

- the operational moderation engine of AcademIQ
- a platform-wide enforcement infrastructure document
- a dependency for all collaborative and communication systems

Including:

- ONE Thread Infrastructure
- Messaging Infrastructure
- Study Circle Infrastructure
- Institution Ecosystem Infrastructure
- AI Boundary Governance Infrastructure
- Trust & Safety Infrastructure
- Verification & Credential Infrastructure
- Discovery Engine Infrastructure

2. Core Philosophy of Moderation

AcademIQ moderation should NOT operate like:

- entertainment social media moderation
- engagement-driven enforcement
- opaque automated punishment systems
- overly punitive environments

Moderation should optimize:

- educational safety
- institutional trust
- healthy collaboration
- support accessibility
- continuity protection
- educational appropriateness
- transparent governance

The goal is NOT:

- excessive punishment
- algorithmic over-enforcement
- fear-based moderation

The goal IS:

- safe educational ecosystems
- transparent enforcement
- institution-aware governance
- scalable support protection
- continuity-aware moderation

3. What the Moderation System Actually Is

The Moderation Infrastructure is:

an operational governance and enforcement system designed to maintain safe, productive, institution-aware educational collaboration across the AcademIQ ecosystem.

The system coordinates:

- moderation review
- enforcement workflows
- escalation chains
- institution governance
- trust scoring
- appeals systems
- AI-assisted moderation
- emergency intervention
- communication restrictions
- continuity protection

4. Core Objectives

The moderation system must support:

- scalable moderation workflows
- institution-aware enforcement

- transparent escalation systems
- educationally appropriate moderation
- trust scoring enforcement
- appeals infrastructure
- emergency escalation
- moderation auditability
- AI-assisted moderation
- scalable governance systems

5. Moderation Architecture Overview

The moderation system should support:

- AI-assisted moderation
- human moderation
- institution moderation
- emergency moderation
- trust & safety escalation
- role-aware moderation
- educational governance

Moderation Sources

Moderation may originate from:

- user reports
- AI detection systems
- institution escalation
- automated safety systems
- emergency workflows
- moderator review

6. Moderation Queue Infrastructure

The system should support structured moderation queues.

Queue Categories

General Moderation Queue

Examples:

- spam
- harassment
- inappropriate behavior
- impersonation

Institution Moderation Queue

Examples:

- institution policy violations
- academic misconduct
- classroom disruption
- institution communication abuse

Wellness & Crisis Queue

Examples:

- self-harm concerns
- threats
- crisis escalation
- dangerous behavior

High-Risk Queue

Examples:

- minors-related safety issues
- severe harassment
- predatory behavior
- emergency escalation

7. Escalation Chain Infrastructure

Moderation systems should support structured escalation chains.

Escalation Levels

Level 1 — Automated Flagging

AI-assisted identification.

Level 2 — Human Moderator Review

Operational moderation review.

Level 3 — Institution Review

Institution-specific governance review.

Level 4 — Trust & Safety Escalation

Severe platform-wide safety review.

Level 5 — Emergency Escalation

Critical intervention workflows.

8. Enforcement Lifecycle Infrastructure

Enforcement should follow structured lifecycle systems.

Enforcement Lifecycle Stages

Warning Stage

Educational guidance and warnings.

Restriction Stage

Limited communication or participation restrictions.

Temporary Suspension Stage

Temporary ecosystem restrictions.

Institution Escalation Stage

Institution-level governance review.

Permanent Enforcement Stage

Reserved for severe violations.

9. Temporary Restriction Infrastructure

The system should support temporary restrictions.

Restriction Examples

Examples:

- messaging restrictions
- thread participation restrictions
- study circle restrictions
- upload restrictions
- institution communication restrictions

Restriction Goals

Restrictions should:

- protect users
- reduce escalation risk
- preserve educational continuity where possible
- remain proportionate

10. Institution Moderation Infrastructure

Institutions must maintain governance authority.

Institution Moderation Features

Examples:

- institution moderators
- department moderators
- cohort moderation
- institution escalation workflows
- institution enforcement policies

Institution Authority Examples

Institutions may:

- restrict participation
- review reports
- escalate users
- enforce institution policies

- review communication systems

11. Trust Scoring Enforcement Infrastructure

Moderation should integrate deeply with:
Trust & Safety Infrastructure.

Trust Signals

Examples:

- report history
- moderation history
- verification status
- institution affiliation
- behavioral patterns
- repeated violations

Trust-Based Enforcement

Trust scoring may influence:

- messaging permissions
- thread visibility
- Discovery Engine ranking
- support participation
- moderation prioritization

12. Reporting Infrastructure

Users must support:

- reporting
- flagging
- escalation requests
- safety concerns

Report Categories

Examples:

- harassment
- bullying
- threats
- inappropriate behavior
- spam
- impersonation
- academic misconduct
- unsafe wellness behavior

Reporting Locations

Users may report:

- threads
- messages
- study circles
- study sessions
- resources
- profiles
- comments
- institutional communication

13. ONE Thread Moderation Infrastructure

Threads should integrate deeply with moderation systems.

Thread Enforcement Features

Examples:

- thread muting
- read-only mode
- participant removal
- thread restriction
- moderation review
- institution escalation

14. Messaging Moderation Infrastructure

Messaging systems must support:

- abuse detection

- harassment detection
- spam filtering
- predatory behavior detection
- moderation escalation

Messaging Enforcement Examples

Examples:

- temporary messaging suspension
- attachment restrictions
- communication throttling
- institution review

15. Study Circle & Study Session Moderation

Study Circles and Study Sessions should support:

- participant moderation
- session reporting
- moderator controls
- emergency intervention

Moderation Features

Examples:

- participant removal
- session restrictions
- session reporting
- institution intervention
- recording review

16. Emergency Escalation Infrastructure

The system should support emergency intervention workflows.

Emergency Examples

Examples:

- threats
- predatory behavior
- crisis situations
- dangerous conduct
- severe harassment
- minors-related safety risks

Emergency Routing

Escalations may route to:

- trust & safety teams
- institution leadership
- counselors
- emergency moderation teams

Subject to:

- institution policies
- privacy laws
- regional compliance

17. AI-Assisted Moderation Governance

AI may assist moderation systems.

However:
AI should not independently perform severe enforcement actions without oversight.

AI Moderation Assistance Examples

Examples:

- spam detection
- harmful language detection
- escalation prioritization
- moderation queue organization
- behavioral pattern analysis

Human Oversight Requirements

Human review should remain required for:

- permanent suspensions
- institution escalations
- emergency actions
- severe moderation actions

18. Appeals Infrastructure

Users should support moderation appeals.

Appeals Features

Examples:

- enforcement review requests
- institution appeals
- moderation clarification requests
- escalation review

Appeals Workflow

Appeals may route to:

- moderation review
- institution review
- trust & safety review
- governance review

19. Moderation Transparency Infrastructure

Moderation systems should remain:

- explainable
- auditable
- institution-visible
- operationally transparent

Transparency Features

Examples:

- enforcement explanations
- moderation reasoning
- escalation visibility
- moderation history

20. FERPA / Privacy / Governance Requirements

Moderation systems must maintain:

- FERPA protections
- educational privacy protections
- institution governance
- auditability
- consent-aware enforcement

Required Governance Features

Audit Logs

The system should maintain:

- moderation actions
- escalation history
- enforcement history
- institution actions
- appeal history

Institution Governance

Institutions may enforce:

- institution policies
- communication restrictions
- participation restrictions
- moderation workflows

Visibility Restrictions

Moderation visibility should remain:

- role-aware
- institution-aware
- privacy-aware

21. Analytics Infrastructure

The system should track:

- moderation trends
- enforcement effectiveness
- escalation frequency
- trust scoring patterns
- appeals trends
- institution moderation metrics

Institutional Analytics

Institutions may view:

- moderation activity
- communication trends
- safety trends
- escalation metrics
- enforcement statistics

Subject to:

- permissions
- FERPA protections
- governance controls

22. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of moderation events
- distributed moderation systems
- scalable queues

- real-time escalation systems
- AI-assisted moderation
- institution-level governance
- high concurrency

23. Future Expansion Architecture

Potential future capabilities:

- predictive moderation systems
- adaptive trust scoring
- multilingual moderation
- institution AI moderation copilots
- advanced behavioral risk detection
- region-specific moderation governance
- educational safety scoring systems

24. Final Engineering Recommendations

The Moderation Escalation & Enforcement system should be treated as:

- an educational governance infrastructure system
- a continuity protection layer
- a trust and safety operational engine
- not a generic social moderation system

The goal is to optimize:

- educational safety
- institutional trust
- healthy collaboration
- transparent governance
- scalable moderation
- continuity-aware enforcement

The dev team should prioritize:

- structured escalation chains
- institution governance integration
- moderation transparency

- AI-assisted moderation with human oversight
- scalable moderation queues
- appeals infrastructure
- FERPA-aware enforcement
- educationally appropriate moderation systems

Because Moderation & Enforcement Infrastructure becomes one of the most critical operational trust systems powering the entire AcademIQ ecosystem.

Analytics & Educational Intelligence Infra System

AcademIQ Analytics & Educational Intelligence Infrastructure System

DevSpec Document

1. Introduction

This document defines the infrastructure architecture for the AcademIQ Analytics & Educational Intelligence system.

The Analytics & Educational Intelligence Infrastructure is one of the foundational intelligence systems of the AcademIQ ecosystem.

This infrastructure powers:

- engagement analytics
- educational continuity analytics
- retention analytics
- institutional dashboards
- support effectiveness analytics
- reinforcement intelligence
- educational intelligence systems
- predictive educational insights
- continuity mapping
- institutional reporting
- AI-assisted educational intelligence
- support outcome analysis

This document defines:

- analytics architecture
- educational intelligence systems
- institutional dashboards
- continuity analytics
- engagement analytics
- reinforcement analytics
- predictive systems
- support outcome systems
- AI-assisted educational insights
- analytics governance
- FERPA-aware analytics infrastructure
- institution intelligence systems

This document should be treated as:

- a master analytics infrastructure document
- an educational intelligence systems reference
- a dependency for multiple AcademIQ systems

Including:

- Discovery Engine Infrastructure
- Emma Infrastructure
- ONE Thread Infrastructure
- Institution Ecosystem Infrastructure
- Scheduling Infrastructure
- Study Circle Infrastructure
- Human Feedback & AI Reinforcement Infrastructure
- Moderation & Enforcement Infrastructure
- Verification & Credential Infrastructure

2. Core Philosophy of Educational Intelligence

Traditional analytics systems are often:

- engagement-only focused
- disconnected from educational outcomes
- continuity-unaware
- institutionally fragmented

AcademIQ analytics are fundamentally different.

The Analytics & Educational Intelligence system should optimize:

- educational continuity
- support effectiveness
- healthy engagement
- institutional insight
- educational collaboration
- academic support accessibility
- reinforcement effectiveness
- educational outcomes

The goal is NOT:

- addictive engagement optimization
- surveillance-driven analytics
- exploitative behavioral tracking

The goal IS:

- educational intelligence
- continuity insight
- support optimization
- institutional visibility
- educational outcome improvement

3. What the Analytics System Actually Is

The Analytics & Educational Intelligence Infrastructure is:

an educational intelligence and continuity-aware analytics system that evaluates educational interaction, support systems, collaboration patterns, and institutional engagement across the AcademIQ ecosystem.

The system analyzes:

- educational participation
- continuity behavior
- support engagement
- reinforcement effectiveness
- collaboration systems
- educational pathways
- institutional interaction
- scheduling participation
- study behavior

4. Core Objectives

The Analytics system must support:

- educational intelligence
- continuity analytics
- engagement analytics
- support effectiveness analysis
- institutional dashboards
- predictive educational systems

- reinforcement analytics
- scalable analytics infrastructure
- FERPA-aware reporting
- institution governance

5. Analytics Infrastructure Overview

The platform should support:

- user analytics
- institutional analytics
- cohort analytics
- engagement analytics
- continuity analytics
- support analytics
- moderation analytics
- reinforcement analytics
- collaboration analytics
- scheduling analytics

Analytics Categories

Engagement Analytics

Examples:

- thread participation
- study circle participation
- session attendance
- messaging activity

Continuity Analytics

Examples:

- recurring participation
- support continuity
- tutoring continuity
- accountability consistency

Institutional Analytics

Examples:

- cohort engagement
- retention trends
- support participation
- academic collaboration

Reinforcement Analytics

Examples:

- reminder effectiveness
- accountability participation
- productivity reinforcement
- support engagement trends

6. Educational Intelligence Architecture

Educational intelligence systems should evaluate:

- educational continuity
- engagement quality
- support accessibility
- collaboration effectiveness
- participation consistency
- educational progression

Educational Intelligence Layers

Interaction Intelligence Layer

Tracks educational interaction patterns.

Continuity Intelligence Layer

Tracks recurring educational continuity.

Support Intelligence Layer

Tracks support effectiveness and engagement.

Institutional Intelligence Layer

Tracks institution-wide educational behavior.

Predictive Intelligence Layer

Identifies potential educational trends and risks.

7. Continuity Analytics Infrastructure

Continuity is one of the foundational intelligence systems inside AcademIQ.

Continuity Metrics

Examples:

- recurring study participation
- tutoring consistency
- support continuity
- accountability consistency
- onboarding retention
- educational relationship persistence

Continuity Goals

The system should identify:

- continuity gaps
- support disengagement
- collaboration decline
- educational inconsistency

8. Engagement Analytics Infrastructure

Engagement analytics should focus on:

- productive educational interaction
- healthy participation
- educational collaboration
- support engagement

Engagement Signals

Examples:

- thread participation
- study circle attendance
- tutoring participation
- resource usage
- office hours attendance
- support interaction

Engagement Restrictions

Analytics should NOT optimize:

- addiction-style engagement
- unhealthy usage patterns
- exploitative interaction systems

9. Support Effectiveness Infrastructure

The platform should evaluate:

- support accessibility
- tutoring effectiveness
- counselor engagement
- study support participation
- educational reinforcement effectiveness

Support Metrics

Examples:

- tutoring continuity
- office hours participation
- support thread engagement
- accountability effectiveness
- study session participation

10. Reinforcement Intelligence Infrastructure

The system should integrate deeply with:
Human Feedback & AI Reinforcement Infrastructure.

Reinforcement Analytics

Examples:

- reminder effectiveness
- productivity reinforcement engagement
- accountability consistency
- educational motivation trends

Reinforcement Goals

The system should identify:

- effective reinforcement systems
- support drop-off patterns
- engagement improvements
- accountability success patterns

11. Predictive Educational Intelligence

Predictive systems should remain:

- assistive
- explainable
- institution-aware
- ethically governed

Predictive Examples

Examples:

- disengagement risk
- continuity gaps
- support accessibility risks
- burnout indicators
- participation decline

Predictive Restrictions

Predictive systems should NOT:

- autonomously discipline users
- create harmful profiling systems
- replace human educational judgment
- operate without explainability

12. Institutional Dashboard Infrastructure

Institutions should support advanced educational dashboards.

Dashboard Categories

Student Engagement Dashboards

Examples:

- participation metrics
- support participation
- continuity trends

Cohort Dashboards

Examples:

- cohort retention
- study participation
- collaboration metrics

Department Dashboards

Examples:

- office hours engagement
- tutoring utilization
- support engagement

Institution-Wide Dashboards

Examples:

- educational engagement
- continuity metrics
- institutional participation
- support system effectiveness

13. Discovery Engine Analytics Integration

Analytics should integrate deeply with:
Discovery Engine Infrastructure.

Discovery Intelligence Examples

Examples:

- recommendation effectiveness
- support matching success
- educational relevance scoring
- collaboration engagement

14. Emma Analytics Integration

Emma should integrate deeply with:
Analytics & Educational Intelligence Infrastructure.

Emma Intelligence Examples

Examples:

- reinforcement effectiveness
- scheduling participation
- educational continuity trends
- support engagement
- recommendation quality

15. ONE Thread Analytics Integration

Threads should support:

- participation analytics
- continuity tracking
- educational engagement metrics
- collaboration analysis

Thread Metrics

Examples:

- thread participation
- recurring engagement
- support continuity
- educational collaboration quality

16. Study Circle & Study Session Analytics

Study systems should support:

- attendance analytics
- collaboration metrics
- accountability effectiveness
- continuity participation

Study Metrics

Examples:

- recurring attendance
- study consistency
- collaboration participation
- support engagement

17. Scheduling Analytics Infrastructure

Scheduling systems should support:

- attendance trends
- office hours utilization
- tutoring scheduling trends
- accountability scheduling participation

Scheduling Metrics

Examples:

- no-show rates
- recurring attendance
- support accessibility
- session participation

18. Moderation Analytics Infrastructure

Moderation systems should support:

- moderation trends
- escalation frequency
- trust scoring trends
- institution safety metrics

Moderation Metrics

Examples:

- report frequency
- moderation escalation rates
- enforcement trends
- communication safety metrics

19. Verification Analytics Infrastructure

Verification systems should support:

- verification participation
- credential legitimacy metrics
- institution verification rates
- trust scoring patterns

20. AI-Assisted Analytics Infrastructure

AI systems may assist with:

- educational insight generation
- engagement trend analysis
- support effectiveness analysis
- continuity mapping
- predictive educational recommendations

However:
all AI analytics systems should remain explainable and governance-aware.

21. FERPA / Privacy / Governance Requirements

Analytics systems must maintain:

- FERPA protections
- educational privacy protections
- anonymization support
- institution governance
- explainability
- auditability

Data Governance Requirements

Analytics systems should support:

- anonymized reporting
- role-aware visibility
- institution-aware access
- data minimization
- retention controls

Consent Controls

Users should control:

- analytics participation
- personalization systems
- educational recommendations
- AI participation

22. Auditability Infrastructure

Analytics systems should maintain:

- analytics generation logs
- AI insight logs
- recommendation visibility
- institution access logs

Audit Examples

Examples:

- predictive recommendation logs
- dashboard access history
- institution analytics access
- educational intelligence reports

23. Scalability & Infrastructure Requirements

Infrastructure must support:

- millions of analytics events
- distributed analytics systems
- scalable dashboards
- AI-assisted intelligence generation
- institution-level isolation
- real-time reporting
- high concurrency

24. Future Expansion Architecture

Potential future capabilities:

- adaptive educational intelligence
- predictive retention systems
- institution AI analytics copilots
- burnout-aware educational insights
- multilingual analytics systems
- educational success mapping
- advanced learning graph intelligence

25. Final Engineering Recommendations

The Analytics & Educational Intelligence system should be treated as:

- a continuity-aware educational intelligence system
- an institutional insight infrastructure layer
- a support optimization system
- not a generic engagement analytics platform

The goal is to optimize:

- educational continuity
- support accessibility
- healthy engagement
- institutional insight
- educational collaboration
- educational outcomes
- continuity-aware educational intelligence

The dev team should prioritize:

- continuity analytics
- institutional dashboards
- reinforcement intelligence
- predictive systems with explainability
- FERPA-aware reporting
- AI-assisted educational intelligence
- scalable analytics infrastructure
- institution governance controls

Because Analytics & Educational Intelligence Infrastructure becomes one of the foundational intelligence systems powering the entire AcademIQ ecosystem.